Probe-Guided Rebalancing of Persistent Subscription Connections with Greedy and Deep Q-Learning Control

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Transport load balancers cannot observe the application-level signals that affect persistent subscription delivery. These include the subscriptions on each connection, backlog per subscription, client acknowledgement capacity, and pod reliability under subscription pressure. As a result, balanced connection counts can still hide large differences in subscription counts and queue backlog. We model client-side rebalancing as a sequence of decisions based on partial observations. We present a probe-guided rebalancing framework for persistent subscription transports, implemented and tested with STOMP. A client combines telemetry from active persistent connections, local queue and throughput counters, and one subscription-free connection that the existing load balancer assigns to a random pod. The framework separates policy selection from receipt-gated execution. It supports deterministic, pressure-aware greedy control and Deep Q-Learning (DQL) over bounded actions. Both use subscription count and backlog to assess a candidate; neither needs full cluster visibility. The DQL policy additionally learns the tradeoff among backlog, imbalance, throughput, and intervention cost. Execution guards block infeasible actions. When migration is selected, the probe is promoted through a receipt-gated standby, quiesce, drain, activate, detach, and old-socket retirement protocol.

In a locked scale-out simulator experiment, 20,000 subscriptions carry 775 million queued events on ten connections established on one Pod. Across 20 paired seeds, adding three Pods with unchanged connection placement yields 58,000 events/s. Probe-guided DQL and guarded-greedy migration yield 219,762 and 221,260 events/s, respectively. DQL reduces normalized backlog area by 12.30% relative to unchanged placement (paired difference -0.117504; 95% confidence interval $[-0.121210, -0.113345]$). The comparison establishes the value of connection reassociation under the stated service model. In a separate mixed-workload holdout, DQL improves mean composite reward by 0.0318 and reduces subscription-count variance by 90.0% relative to the two-dimensional heuristic, with a 4.5% throughput tradeoff. Oracle Cloud experiments establish execution of the mechanism: ten connection replacements reassociate 4,000 of 20,000 subscriptions, and a loaded scale-out trial completes three selective handoffs while retaining the other sessions and confirming all 8,000 observed delivery ACKs. The framework therefore offers both a transparent greedy policy and a learned multi-objective policy, with their benefits evaluated separately from the shared handoff mechanism.

CCS Concepts: • **Computing methodologies** → **Reinforcement learning**; • **Computer systems organization** → **Cloud computing**; • **Software and its engineering** → *Message passing*; *Publish-subscribe / event-based architectures.*

Additional Key Words and Phrases: persistent connections, subscription rebalancing, deep Q-learning, partial observability, probe promotion, greedy control, messaging systems, autonomic control

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1 INTRODUCTION

Distributed messaging systems support cloud-native and event-driven applications. Real-time analytics, notifications, workflow engines, and microservices need reliable event delivery as workloads change. Many deployments run several subscriber-service instances behind a transport-layer load balancer. Clients keep persistent STOMP/TCP, raw TCP, or WebSocket-style connections to these services.

Load balancing is therefore an important design concern. Poor distribution leaves some subscriber instances with growing backlogs and others with spare capacity. Clients then experience uneven delivery delays. Cloud resource-management reviews describe machine-learning methods for changing demand and capacity [30]. Deep-reinforcement-learning reviews also discuss adaptive policies for different resource types and competing objectives [50]. Conventional load balancers, however, usually distribute connections through policies such as round-robin or least-connections.

Those policies are simple and scalable, but connection count alone does not describe subscription-heavy messaging. A persistent connection may carry a few lightweight subscriptions or hundreds of high-rate subscriptions with growing queues. The load balancer may count both cases as one connection. The subscriber pod handles very different workloads in these cases. Its load depends on subscription count, backlog per subscription, and the client's message-processing and acknowledgement rates.

Subscriber-pod scale-out exposes this mismatch directly. If ten persistent connections are established while only one pod exists, new pods increase available processing capacity but do not change those established endpoints. The client can use the new capacity by promoting a probe that reaches a suitable pod and reassociating subscription-bearing connections to it. This scale-out recovery is the central experiment in Section 7.1. The corresponding live handoff mechanism is examined during loaded Oracle Cloud cluster scale-out in Section 6.5.

This mismatch creates two related problems. First, a client may reach a pod that already manages too many subscriptions, even when its connection count is acceptable. Second, the client may place too many subscriptions on one of its own connections. This can create local backlog even when the pod has enough capacity overall. A fixed connection pool cannot increase parallelism as subscription counts grow or event rates rise. Opening more connections without moving subscriptions, however, can leave the workload unevenly distributed.

Deep reinforcement learning has been used for hierarchical cloud scheduling [51], multidimensional data-center allocation [45], and distributed cloud-edge offloading [37]. Decentralized bandit methods select service instances from client QoS feedback [14] or coordinate edge offloading through server rewards [48]. BLOC also shows how client-side load balancing can use overload feedback without a central metadata store [3]. These studies provide examples of adaptive control with limited information. Our focus is how the client changes persistent subscription connections. It retains a probed socket, promotes it only when the required checks pass, and transfers subscription ownership while preserving the durable queues.

Server placement, connection scaling, and subscription assignment depend on each other. Placement determines which pod a connection reaches. Scaling determines how many connections the client can use in parallel. Subscription assignment determines how evenly the work is spread across those connections. A practical controller must coordinate these decisions using only the information available to the client at runtime.

We propose an adaptive framework with coordinated control loops for persistent subscription messaging. It applies to long-lived STOMP/TCP, raw TCP, or WebSocket connections that carry multiple subscriptions and periodically report application-level statistics. The framework has three control loops at different system layers:

(1) A policy layer that chooses bounded actions from active-connection and probe statistics. We evaluate deterministic greedy rules and DQL as alternative selectors under partial observability.
(2) A client-side adaptive parallel connection scaling loop that adjusts the number of persistent transport connections based on total subscription count and per-subscription backlog characteristics.
(3) A connection-level load rebalancing loop that minimizes both subscription-count variance and per-subscription-backlog variance across scaled connections.

Together, these loops let the client choose and carry out rebalancing actions without full cluster visibility. Both policy families use the client's own queues, active-connection statistics, and one retained probe assigned by the load balancer. The greedy policy responds to measured pressure and imbalance. DQL learns action values for the same operational objectives. The design keeps the existing transport load balancer.

Contributions

The key contributions of this work are:

- A client-side connection-replacement mechanism for persistent subscription transports. The client retains an authenticated, load-balancer-assigned probe and conditionally promotes that same socket, preserving the observed endpoint without requiring direct routing to a chosen Pod.
- A policy-independent separation of observation, action selection, and guarded execution. A pressure-aware greedy policy and a partially observable DQL policy use client-local queues and sampled Pod statistics. The experiments distinguish the benefit of the common mechanism from the choice of selector.
- Subscription- and backlog-aware connection scaling and two-dimensional connection rebalancing. These mechanisms distinguish adding transport parallelism, redistributing subscriptions across existing connections, and replacing a connection's delivery endpoint.
- An implemented receipt-gated handoff that retains subscription identity and durable queues. Standby target bindings, source quiescence, a fresh zero-in-flight check, target activation, and source retirement make migration an explicit protocol rather than an instantaneous reassignment.
- Complementary deployment and policy evidence: a 20,000-subscription Oracle Cloud cluster connection-replacement experiment, an isolated loaded scale-out trial with selective DQL handoffs, and controlled simulator comparisons and ablations. The evaluation separates executable handoff behavior, the benefit of rebalancing, and the contribution of learned action selection.

The contribution is not a new Q-learning update or a new greedy optimizer. It is their integration with retained-probe selection and subscription-aware connection replacement, supported by separate mechanism and policy tests. Greedy control provides a transparent reference and a usable control policy; DQL provides a learned choice among competing operational objectives. The reported experiments evaluate them separately. They do not use a DQL model to tune the scripted comparator or claim an evaluated hybrid policy.

2 BACKGROUND AND RELATED WORK

2.1 Persistent Messaging and Stream Processing

We consider persistent subscription transports in which a client keeps a long-lived connection to a subscriber service and multiplexes multiple subscriptions over that connection. The abstraction covers STOMP-over-TCP, custom TCP subscription streams, and WebSocket-based messaging. In each case, the load created by one connection is not captured

by the connection count alone; it also depends on how many subscriptions are multiplexed over the connection and how much backlog has accumulated behind those subscriptions.

Modern event-driven platforms and stream-processing engines face related elasticity problems. Dynamic stream redirection can improve runtime elasticity in distributed stream-processing systems [43]. Cardellini et al. survey runtime-adaptation strategies and organize them by their control objectives and mechanisms [5]. Recent adaptive join-processing work also shows that workload redistribution can improve latency and throughput under changing stream rates [44]. These studies motivate adaptive runtime control. Their actions change operator placement, stream routing, or stateful processing inside the stream runtime. They do not address the client-side persistent-transport problem in which subscription count and per-subscription backlog must be balanced across a client's own long-lived connections.

Fragkoulis et al. distinguish elasticity, reconfiguration, state management, and fault tolerance in the evolution of stream-processing systems [19]. This distinction matters here because the durable queue stays in place: migration changes its subscription's delivery owner, not the queue's stored contents. Transactional stream processing also separates processing guarantees from ordinary stream execution [49]. Accordingly, a completed connection replacement is not, by itself, evidence of end-to-end exactly-once processing.

2.2 Cloud, Microservice, and Load-Balancing Literature

Recent cloud resource-management studies emphasize that adaptive schedulers must operate under workload uncertainty, heterogeneous resources, and competing objectives such as latency, throughput, energy, and utilization [30]. Reviews of DRL-based cloud control identify the same practical issues that matter here: reward design, non-stationary demand, and operational deployment [50]. DRL has been used for algorithm selection in hierarchical cloud scheduling [51], feedback-control-based cloud resource allocation [10], multi-objective virtual-machine placement [7], and multidimensional allocation in distributed data centers [45]. These works show how learned control can account for delayed effects and competing objectives. However, their controlled objects are usually tasks, virtual machines, jobs, or data-center resources. Our controlled objects are lower in the messaging stack: subscriptions, persistent client connections, and observed subscriber pods.

Cloud scheduling offers several distinct approaches, not all of them learning-based. Belgacem et al. use a spacing multi-objective antlion algorithm to optimize makespan and virtual-machine cost [2]. Ding et al. instead use Q-learning for energy-efficient task scheduling [16], while Chen et al. use actor–critic DRL for resource allocation in cloud data centers [11]. Collaborative scheduling addresses heterogeneous workflows [8]; workload-time windows provide temporal context for cloud-service allocation [9]. Li et al. apply DRL to energy-aware Spark scheduling [32]. Sanjalawe et al. provide a comprehensive review of AI-driven cloud job scheduling [40].

The controlled workload also matters. Dong et al. use DQN for manufacturing task scheduling [17], and Chen et al. study attention-based scheduling across manufacturing cloud and edge resources [12]. Cheng et al. introduce preemption into cost-aware cloud scheduling [13]; Liu et al. consider inter-region bandwidth and traffic costs for data-intensive jobs [33]. These papers supply scheduling context. Our problem instead requires subscription reassociation while retaining queue identity and delivery ownership across persistent connections.

Edge research is relevant here in two narrower respects: decentralized observations and explicit tradeoffs among resource costs. Iftikhar et al. provide a taxonomy of AI-based fog and edge computing [26], while Hortelano et al. survey RL-based computation offloading [23]. A concrete decentralized example is Nieto et al.'s per-device actor–critic offloading policy, which trades battery-related QoE against task-latency requirements [37]. Hou et al. explicitly formulate

distributed offloading with partial global information [24]. Partial observability is therefore already addressed in earlier work. It is not, by itself, a new contribution of our controller.

Other offloading papers inform the choice of cost terms, rather than the transport handoff mechanism. Xu et al. optimize cloud-edge resource rental and allocation costs [47]. Huang et al. combine computation offloading with resource allocation in wireless-powered edge networks [25]. Pan et al. model dependent tasks and train an offloading scheduler using a modified PPO method [38]. Jayanetti et al. consider energy and execution time for precedence-constrained tasks [28]. Mitsis et al. incorporate price and risk into offloading decisions [35], whereas Cui et al. optimize caching and request routing for latency-sensitive content distribution [15]. Our migration and connection costs likewise make operational tradeoffs explicit, but their definitions and the subscription-ownership contract are specific to persistent messaging.

Microservice autoscaling provides a closer infrastructure comparison. Abdel Khaleq and Ra use reinforcement learning to control horizontal scaling from resource demand [1]. HephaestusForge learns microservice placement across Kubernetes clusters with multiple deployment objectives [41]. These approaches change service capacity or placement. Our mechanism addresses the complementary client-side step: existing subscription-bearing sockets must reach the added capacity, and candidate promotion must account for both subscription and backlog pressure.

2.3 Autonomic and Learning-Based Control

Gheibi et al. organize machine-learning support for self-adaptation around monitor–analyze–plan–execute feedback loops, including policy updates and resource management [21]. Our controller follows that established pattern; DQL alone is not the differentiating mechanism. The industrial survey by Weyns et al. also examines the practical motivations, risks, and difficulties of deploying self-adaptation [46]. This motivates making the observation boundary and the concrete connection-handoff procedure explicit.

Recent DRL literature is also clear about the risks of applying learning methods directly to production systems. Partial observability, non-stationary workloads, reward design, and safe online adaptation remain difficult [18]. Cloud DRL surveys make the same point for operational cloud control [50]. The closer load-balancing comparisons already include decentralized operation. The QEdgeProxy preprint describes multi-player bandits and per-client QoS feedback for selecting service instances [14]. Yahya et al. combine mean-field bandits with server-reward adjustment for decentralized task offloading [48]. BLOC uses overload feedback and dynamic capacity estimation without a centralized metadata store [3]. Selvakumar et al. use predicted waiting times and service-chain communication costs to prioritize requests and select instances [42]. These mechanisms provide direct comparisons for decision inputs and adaptive selection. Our narrower distinction is conditional reuse of a retained probe socket together with a concrete subscription-ownership handoff.

Negi et al. classify soft-computing approaches to VM- and physical-machine-level load balancing [36]. In our messaging setting, a bounded action-template design limits the corresponding control space: the DQL agent chooses among deployable rebalancing actions rather than directly searching the full combinatorial subscription-assignment space.

Metzger et al. use feature models to guide exploration through large or evolving adaptation spaces [34]. Our bounded templates reduce a different search space: the possible subscription moves and connection replacements available to one client. Recent RS-DRL work instead changes reward shaping during replay to manage uncertainty in self-adaptation [29]. We retain a conventional DQL learner and specialize the observations, feasible actions, and handoff semantics to persistent messaging.

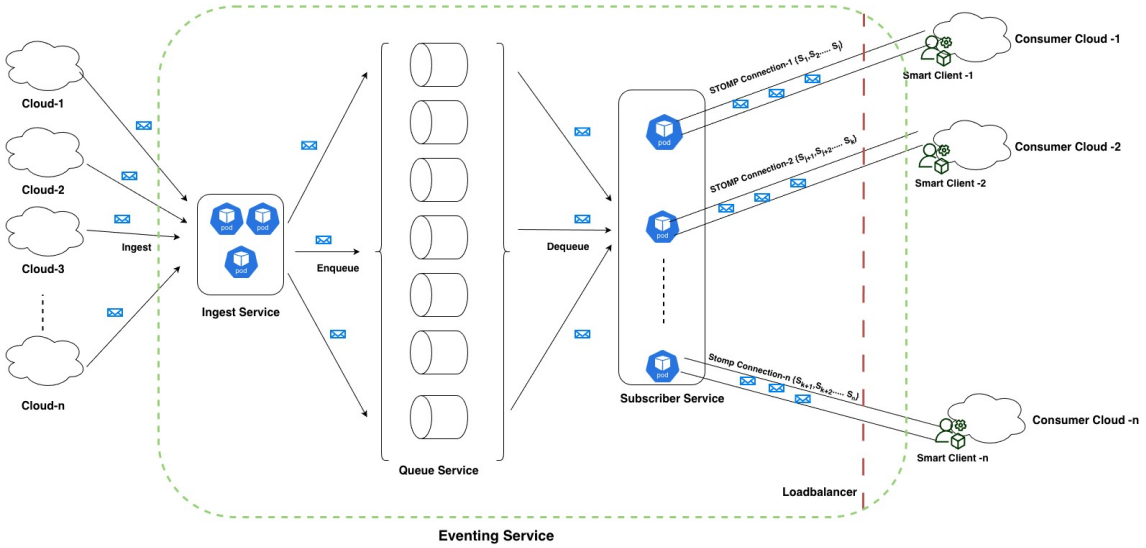


Fig. 1. System architecture with the DQL policy option shown. Producers send events through ingest services to durable subscription queues. Subscriber Pods deliver events over persistent connections. A client retains a load-balancer-assigned probe and promotes that same socket only after policy selection and execution checks. Greedy selection uses the same observation and actuation interfaces.

2.4 Research Gap and Novelty

The comparison above identifies a specific gap in how actions are chosen and carried out. Decentralized load balancing already exists. BLOC, QEdgeProxy, and decentralized offloading already address decisions without a central full-state controller. Our mechanism combines four requirements. It treats pod subscription count and backlog as separate pressure signals. It tracks the client's subscription-to-connection assignments. It promotes the same authenticated probe socket only after checking that candidate. Finally, it uses receipt-gated delivery handoff while durable queues stay in place. These requirements define the comparison in Table 1.

The proposed mechanism connects these concerns through an endpoint-preserving probe workflow. The load balancer selects a reachable Pod when the probe is opened; the selected policy then evaluates whether to use that already-established connection. Promotion is conditional, not an automatic consequence of probing. An admissible migration reuses the observed socket and transfers subscription delivery ownership under the handoff contract, while the durable queues remain stationary. This differs from choosing a Pod by name and opening another load-balanced socket, whose endpoint could differ from the one evaluated. Local subscription and backlog measurements also support scaling and redistribution across existing connections. The contribution is this partially observable control and execution mechanism, evaluated with established greedy and Q-learning methods. Their comparison is specific to the measured workloads. Table 1 summarizes the positioning of the proposed approach relative to representative literature.

3 SYSTEM MODEL

Figure 1 illustrates the overall architecture of the event-driven messaging system considered in this work. The system consists of four major components: (1) event producers, (2) ingest and queue services, (3) subscriber services, and (4) distributed consumer clients connected via persistent subscription transport connections.

Table 1. Comparison of Representative Work and the Proposed Framework

Work	Main focus	Observed information	Controlled object	How this work differs
BLOC [3]	Request balancing and overload control	Overload feedback; no central metadata store	Request admission and server selection	Adds retained-socket promotion and subscription ownership transfer
QEdgeProxy [14]	QoS-aware service selection	Client QoS outcomes; decentralized bandits	Service-instance selection	Preserves the probed endpoint during connection replacement
Adaptive load balancing [42]	Service-chain latency	Predicted waiting times; inter-service costs	Request priorities and instance selection	Reassociates existing subscription-bearing connections
Decentralized offloading [48]	Task distribution among edge servers	Unknown network information; adjusted server rewards	Offloading population control	Reassociates persistent subscription groups
Per-device DRL [37]	Energy- and latency-aware offloading	Device, server, channel, and task state	Local, MEC, or cloud task execution	Uses subscription counts and queued-event pressure
Partial-observation DRL [24]	QoE-aware distributed offloading	Partial observations; device QoE preferences	IoT task-offloading decisions	Evaluates a retained probe and transfers subscriptions
Dr-Stream [43]	Stream redirection and operator elasticity	Stream-runtime measurements	Operator instances and stream state	Keeps durable queues stationary during transport handoff
Aj-Stream [44]	Skew-aware stream joining	Predicted key frequencies; instance load	Key grouping and operator scaling	Moves subscription ownership while queues stay in place
RL autoscaling [1]	Microservice scaling thresholds	Resource demand and response-time QoS	Available service capacity	Moves existing sticky sockets to available capacity
HephaestusForge [41]	Multi-cluster microservice placement	Network latency; resource capacity; deployment cost	Pod-replica placement across clusters	Connects existing subscriptions to available Pod capacity
Feature-model-guided RL [34]	Exploration of adaptation choices	Feature models; configuration changes; rewards	Runtime feature configurations	Uses bounded connection actions with receipt-gated handoff
RS-DRL [29]	Reward-shaped self-adaptation	Runtime feedback; replayed transitions	Self-adaptive IoT configurations	Specializes actions to guarded transport and subscription handoff

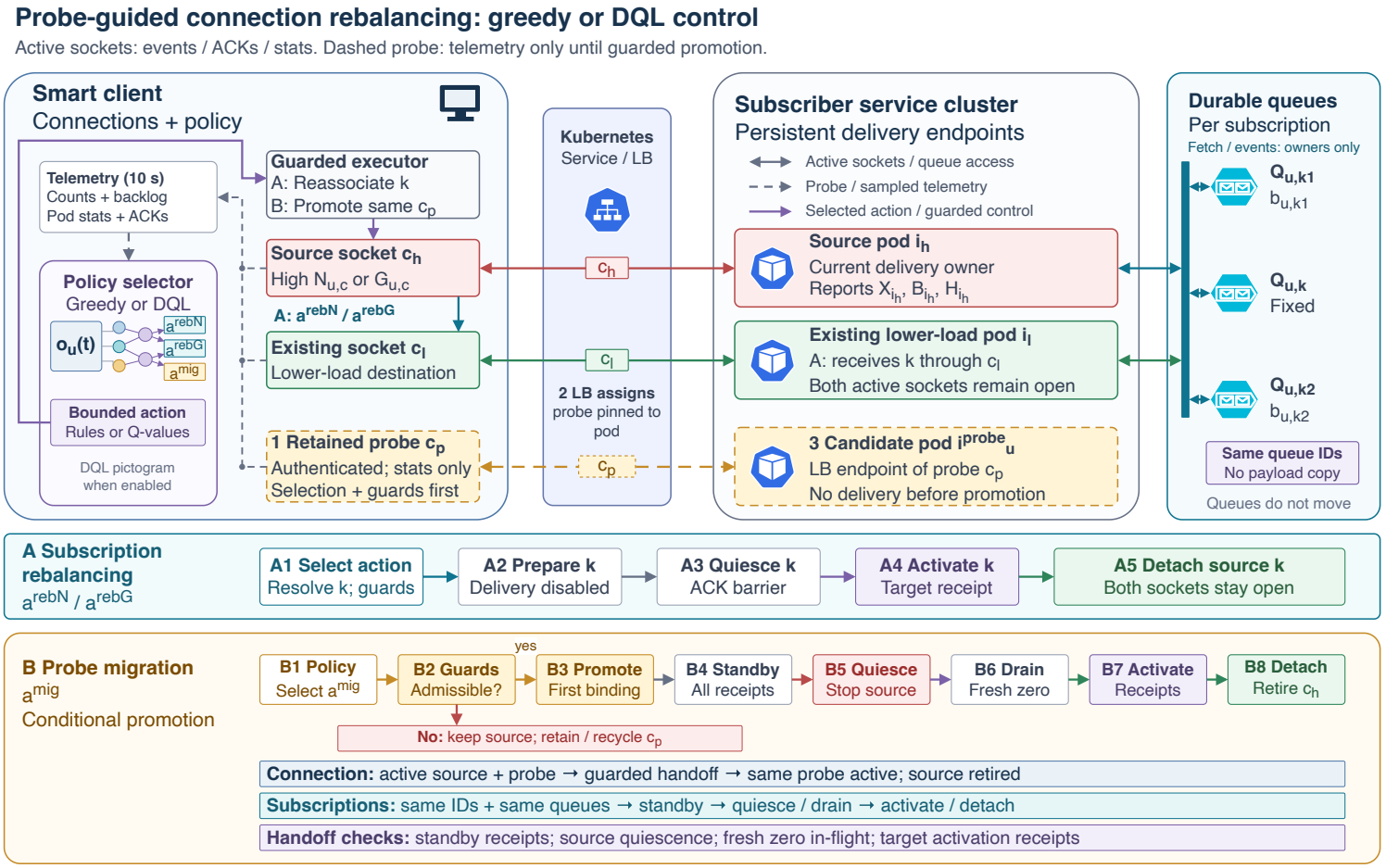


Fig. 2. Shared probe-guided architecture with greedy or DQL selection. Path A rebalances subscriptions between existing sockets. Path B promotes the retained probe through its first standby binding, waits for all standby receipts, quiesces and drains the source, activates the target, detaches the source bindings, and retires the old socket. Unselected sockets and durable queues remain in place. The DQL pictogram denotes the optional learned selector, not an additional handoff service.

3.1 Event Producers and Ingest Service

Event producers run in independent cloud environments, shown as Cloud-1 through Cloud-n in Figure 1. They generate events asynchronously and send them to the ingest service, which is the system's entry point. The ingest service is implemented as a horizontally scalable set of service instances deployed as pods. Each ingest instance processes incoming events and enqueues them into the distributed queue service. The ingest service ensures reliable message intake and provides elasticity to handle fluctuating workloads.

3.2 Queue Service

The queue service acts as an intermediate buffering layer between event producers and subscribers. It provides durable storage and ensures reliable delivery of messages. Messages received from the ingest service are placed into queues, where they remain until consumed by subscriber services.The queue service supports parallel queues and can scale as event volume grows. Buffering lets producers and consumers work independently, without matching their processing rates.

3.3 Subscriber Service

The subscriber service is responsible for dequeuing messages from the queue service and delivering them to consumer clients. Similar to the ingest service, the subscriber service is deployed as multiple horizontally scalable instances. Each subscriber service instance maintains persistent transport connections with multiple consumer clients. These connections may use STOMP-over-TCP, raw TCP, WebSocket, or another long-lived subscription-bearing transport. They allow real-time event delivery and enable clients to subscribe to specific event streams. Subscriber service instances perform the following key functions:

- Dequeue messages from the queue service
- Maintain subscription state for connected clients
- Deliver messages to subscribed clients
- Track subscription load, backlog, and delivery rate

The subscriber service instances are deployed behind a load balancer, which distributes incoming client connection requests across available subscriber instances.

3.4 Load Balancer

The load balancer serves as the initial connection routing layer between consumer clients and subscriber service instances. It distributes client connections across available subscriber instances using traditional transport-level algorithms such as round-robin or least-connections.

However, the load balancer operates without visibility into application-level metrics such as subscription count, message backlog, or processing load. As a result, connection assignments may not reflect the true load distribution across subscriber instances. This limitation motivates the need for adaptive client-driven connection placement mechanisms.

3.5 Consumer Clients and Persistent Transport Connections

Consumer clients, referred to as Smart Clients in Figure 1, are deployed across multiple consumer cloud environments. These clients establish persistent subscription transport connections to subscriber service instances.

Each persistent transport connection can support multiple subscriptions. For client u, let $\mathcal{K}_u(t)$ be its subscription set and $C_u(t)$ its active delivery-connection set. The subscriptions assigned to connection $c \in C_u(t)$ are

$$\mathcal{K}_{u,c}(t) = \{k \in \mathcal{K}_u(t) : z_{u,k,c}(t) = 1\}. \tag{1}$$

Here $z_{u,k,c}(t)$ is the binary assignment indicator defined in Section 4. Each subscription corresponds to a specific event stream or topic; k indexes a subscription, not a Pod.

The subscriber service maintains subscription state and delivers messages to clients based on these subscriptions. Because subscription counts and event rates may vary significantly across connections, the load imposed by each client connection is highly variable.

3.6 Connection Placement and Rebalancing Model

Let the set of subscriber service instances be denoted as:

$$\mathcal{S} = \{1, 2, \ldots, n\}. \tag{2}$$

Each subscriber instance exposes the following dynamic statistics to active connections and probe connections:

$$y_i(t) = \big(p_i, X_i(t), B_i(t), H_i\big). \tag{3}$$

where:

- p_i is the pod name
- $X_i(t)$ is the pod-level subscription count
- $B_i(t)$ is the cumulative backlog over all subscription queues served by the pod
- H_i is the configured maximum processing threshold, used only as a capacity-normalization signal

These statistics are exposed to clients through protocol-specific statistics frames, headers, or control messages. Actual drain behavior is measured from the client side using delivered, processed, and acknowledged message counts, because the effective drain rate also depends on client processing capacity.

Decentralized monitoring research similarly treats telemetry quality and monitoring overhead as coupled concerns. DEMon, for example, uses adaptive gossip-based monitoring in volatile edge environments [27]. Here the telemetry scope is deliberately smaller: active sockets and a retained probe provide sampled Pod observations, not a complete cluster view.

Clients maintain persistent connections and periodically evaluate alternative subscriber instances using a learning-based placement and rebalancing strategy. Based on observed load characteristics, clients may rebalance subscriptions across their own connections, scale connection parallelism, or migrate subscriptions to alternative subscriber instances.This adaptive mechanism enables the system to dynamically rebalance load across client connections and subscriber service instances while operating only on client-visible statistics.

3.7 Retained-Probe Acquisition and Receipt-Gated Handoff

Figure 2 makes the selected-Pod mechanism explicit. The client does not choose a Pod name and then attempt to bypass the Kubernetes Service. It opens an authenticated, statistics-only probe through the same transport load balancer used by ordinary connections. The load balancer selects one ready subscriber Pod when that TCP/WebSocket connection is established, after which the socket remains endpoint-bound to that Pod. The STOMP implementation identifies

the authenticated session and its Pod in telemetry. A probe initially has no subscription bindings and does not fetch events. Promotion changes the role of that existing session; it does not create a second connection whose load-balanced destination might differ.

The selector evaluates a candidate that is already connected. Fresh and sufficiently complete telemetry, a distinct source and target Pod, capacity, cooldown, and handoff checks determine whether migration is admissible. The DQL selector also applies its action-value hysteresis test. A rejected or unselected probe remains statistics-only until refreshed or recycled. Thus, probing does not imply migration, and the client never moves an established TCP socket between endpoints. It replaces one delivery socket with the already-connected probe and reassociates the source subscriptions. In Figure 2, c_p labels the retained probe, whose endpoint is $i_u^{\text{probe}}(t)$. The source and existing target Pod indices are $i_h = i_{u,c_h}(t)$ and $i_l = i_{u,c_l}(t)$. Thus i_h and i_l identify Pods, while $P_i(t)$ denotes a pressure score.

Queued events remain in the same durable subscription queues $Q_{u,k}(t)$. The client retains each subscription identifier, listener, message permits, and negative-acknowledgement visibility setting. For the selected subscription set $\mathcal{K}_{\text{move}}$, the executed protocol is:

(1) Create each target binding in *standby*, waiting for its STOMP receipt. The first subscription request carries the probe-promotion flag. The server checks the authenticated session and its active-connection cap, then changes the probe's role. Standby bindings do not deliver events. The source remains active while these target bindings are prepared.
(2) After all standby receipts, add the promoted socket to the client pool and clear its probe role. Send a receipt-gated quiesce request for each source binding. Quiescence stops new delivery while existing in-flight messages remain on the source acknowledgement path.
(3) Record a barrier timestamp after the quiesce requests. Wait for a newer source telemetry frame that reports every selected subscription as quiesced with zero in-flight messages. A zero from an older frame cannot satisfy the barrier. Drain time is observed, not estimated from H_i.
(4) Activate the target bindings one at a time, waiting for their receipts. The source remains quiesced during this phase.
(5) Unsubscribe the source bindings with receipt checks. After all selected bindings are detached, remove and disconnect the old socket for a full-connection migration. Replenish the subscription-free probe asynchronously. Unselected active sockets remain connected.

Let s_b be the barrier timestamp, s_f the timestamp of a received source frame, and $I_{c_h,k}(s_f)$ its reported in-flight count. The target-activation precondition is

$$s_f \geq s_b \quad \wedge \quad \bigwedge_{k \in \mathcal{K}_{\text{move}}} \left[\text{state}_{c_h,k}(s_f) = \text{quiesced} \wedge I_{c_h,k}(s_f) = 0 \right]. \tag{4}$$

This precondition describes the implemented planned handoff, assuming timely, accurate telemetry and servers that honor quiescence. It is not a distributed ownership lease or an epoch-fencing guarantee during partitions. On an exception, the client attempts to quiesce and close the target before restoring the source; unsuccessful cleanup is logged rather than counted as a completed handoff. The protocol uses authenticated session state and receipts, not a separate signed promotion token or an atomic cross-Pod ownership commit. Receipt completion records server acknowledgement of the corresponding operation; it is not by itself proof of an end-to-end exactly-once event census. The delivery model remains unordered and permits redelivery after an unconfirmed acknowledgement. The live results report observed deliveries, ACK receipts, and unresolved records separately.

3.8 System Characteristics

The system exhibits the following important properties:

- **Persistent connections.** Clients maintain long-lived STOMP-over-TCP, raw TCP, WebSocket, or equivalent subscription-bearing connections.
- **Subscription-Based Load:** Server load is determined primarily by subscription count and event rate, rather than connection count.
- **Dynamic Workload:** Event rates and subscription distributions change over time.
- **Distributed Architecture:** Subscriber services operate independently and scale horizontally.
- **Limited Load Balancer Visibility:** Load balancers lack application-level visibility into subscription load.

These characteristics make adaptive, client-driven connection placement particularly suitable for achieving optimal load balancing.

4 PROBLEM FORMULATION

The main quantities are client subscriptions $\mathcal{K}_u$, active sockets C_u, per-subscription backlog $b_{u,k}$, and the corresponding connection loads $(N_{u,c}, G_{u,c})$. Pod-wide quantities (X_i, B_i) include all clients served by Pod i. Appendix A collects the full symbol reference; definitions also appear at their first use below.

We model the system at discrete control epochs $t = 0, 1, \ldots, T$. The interval between adjacent epochs is Δ seconds: the next epoch is $t + 1$, while its physical time is $(t + 1)\Delta$. Live-event timestamps and durations are reported separately in seconds. Let $\mathcal{S} = \{1, \ldots, n\}$ be the set of subscriber pods and $\mathcal{U} = \{1, \ldots, m\}$ be the set of clients. Client u maintains a subscription set $\mathcal{K}_u(t)$ and a set of persistent transport connections $\mathcal{C}_u(t)$, with $1 \le |\mathcal{C}_u(t)| \le C_{\max}$. Indices i, u, c, k refer to Pods, clients, connections, and subscriptions, respectively. Calligraphic symbols denote sets; their cardinalities are counts. In the single-client scaling rules, $C(t) = |\mathcal{C}_u(t)|$ and $N_{\text{sub}}(t) = |\mathcal{K}_u(t)|$ are count shorthands with the fixed client index suppressed. They are never used as sets. Figure labels and algorithm steps may omit (t) when all quantities refer to the same decision epoch.

4.1 Per-Subscription Queues

Each subscription has a queue that stores messages waiting to be delivered and processed. Let $Q_{u,k}(t)$ be the queue of subscription $k \in \mathcal{K}_u(t)$, and let

$$b_{u,k}(t) = |Q_{u,k}(t)| \tag{5}$$

be the backlog for that subscription. This is the basic backlog measure used by the client. At the pod level, backlog is a server-owned aggregate. Let $\mathcal{R}_i(t)$ denote the set of subscription queues currently served by pod i, and let $\tilde{b}_{i,r}(t)$ be the backlog of server-side subscription queue $r \in \mathcal{R}_i(t)$. If queue r corresponds to client u and subscription k, then $\tilde{b}_{i,r}(t)$ is the server-side counterpart of $b_{u,k}(t)$; however, $\mathcal{R}_i(t)$ also contains queues belonging to other clients and is maintained by the pod, not by any single client.

Let $z_{u,k,c}(t) \in \{0, 1\}$ indicate whether subscription k of client u is assigned to connection $c \in \mathcal{C}_u(t)$. Feasible assignment requires

$$\sum_{c \in \mathcal{C}_u(t)} z_{u,k,c}(t) = 1, \qquad \forall u,\ k \in \mathcal{K}_u(t). \tag{6}$$

Let $i_{u,c}(t) \in \mathcal{S}$ denote the pod index reached by connection c. This endpoint mapping differs from the Pod name p_i and from the selected DQL action $a_u(t)$. The number of subscriptions and cumulative subscription backlog on connection c

are

$$N_{u,c}(t) = \sum_{k \in \mathcal{K}_u(t)} z_{u,k,c}(t), \tag{7}$$

$$G_{u,c}(t) = \sum_{k \in \mathcal{K}_u(t)} b_{u,k}(t) z_{u,k,c}(t). \tag{8}$$

The first term captures subscription-management pressure on the connection; the second captures queue pressure induced by the subscriptions mapped to that connection.

4.2 Pod Statistics and Partial Observability

Each pod exposes a statistics tuple through active connections and probe connections:

$$y_i(t) = \big(p_i,\ X_i(t),\ B_i(t),\ H_i\big), \tag{9}$$

Here p_i is the pod name, and $X_i(t)$ is its subscription count. The value $B_i(t)$ is the cumulative backlog across all subscription queues served by pod i. The value H_i is the configured maximum processing threshold. Using the server-side queue set $\mathcal{R}_i(t)$, the pod-level subscription count and cumulative pod backlog are

$$X_i(t) = |\mathcal{R}_i(t)|, \qquad B_i(t) = \sum_{r \in \mathcal{R}_i(t)} \tilde{b}_{i,r}(t). \tag{10}$$

Thus $B_i(t)$ is not defined by summing over the connection set $C_u(t)$ of a particular client. A client's connections determine which pod statistics it can observe and which subscriptions it can move, but the pod backlog itself is the cumulative backlog over all subscription queues served by that pod. These pod-level values include traffic from many clients and are not directly computable by a single client unless reported by the pod.

At each statistics interval Δ (10 seconds in the target deployment), every active connection receives the tuple of its current pod. In addition, client u maintains one authenticated probe connection whose endpoint is selected by the load balancer and receives its tuple. The client retains the probe socket and its endpoint identity until promotion, expiry, or recycling; it does not resample a different pod merely by reading the next observation. Therefore the client observes only a subset $O_u(t) \subseteq \mathcal{S}$ of pod states:

$$O_u(t) = \{i_{u,c}(t) : c \in C_u(t)\} \cup \{i_u^{\text{probe}}(t)\}. \tag{11}$$

The client stores the newest tuple for each observed pod in a local cache. This makes the placement problem partially observable: global cluster variance is available to an offline evaluator or simulator, but not to an individual client controller. In particular, an admissible migration target must be represented by a currently retained probe socket; the client cannot replace the sampled candidate with an unobserved globally optimal pod.

Assumption 1 (Available deployment signals). *For every pod i observed by client u, where $i \in O_u(t)$, the protocol supplies $(p_i, X_i(t), B_i(t), H_i)$ in a statistics frame, header, or control message. Client u can also measure its own per-subscription backlogs $b_{u,k}(t)$, delivered message count, processed message count, acknowledged message count, active connection count, and migration count over each interval Δ.*

Assumption 2 (Transport abstraction). *The underlying transport is required only to maintain long-lived client–pod connections, multiplex one or more subscriptions over a connection, support subscription migration or reassignment,*

and expose periodic statistics through headers, frames, or control messages. STOMP-over-TCP, custom TCP streams, and WebSocket-based messaging are concrete examples of this abstraction.

Remark 1. *The pod threshold H_i is not treated as an actual drain rate. The true drain rate depends on the queue subsystem, pod scheduling, network transport, connection behavior, client processing capacity, and acknowledgement rate. Hence $B_i(t)/(H_i\Delta + \varepsilon_0)$ is used only as a dimensionless pod-side pressure ratio. Here B_i counts messages, H_i has units of events per second, and ε_0 has units of messages. The ratio is not an estimate of the actual drain time.*

4.3 Local Imbalance Measures

For client u, define the mean subscription count and mean backlog load across its own connections:

$$\bar{N}_u(t) = \frac{1}{|C_u(t)|} \sum_{c \in C_u(t)} N_{u,c}(t), \tag{12}$$

$$\bar{G}_u(t) = \frac{1}{|C_u(t)|} \sum_{c \in C_u(t)} G_{u,c}(t). \tag{13}$$

The corresponding variances are

$$V_u^N(t) = \frac{1}{|C_u(t)|} \sum_{c \in C_u(t)} \big(N_{u,c}(t) - \bar{N}_u(t)\big)^2, \tag{14}$$

$$V_u^G(t) = \frac{1}{|C_u(t)|} \sum_{c \in C_u(t)} \big(G_{u,c}(t) - \bar{G}_u(t)\big)^2. \tag{15}$$

The first variance captures subscription-management skew, while the second captures backlog skew. The local rebalancing objective is therefore

$$V_u(t) = \eta_N \widehat{V}_u^N(t) + \eta_G \widehat{V}_u^G(t), \qquad \eta_N + \eta_G = 1, \tag{16}$$

where hats denote normalized quantities. This formulation is necessary because two connections can have equal backlog but very different subscription counts, or equal subscription counts but very different backlog. The normalized terms place the two variances on a common scale before they are combined. In the evaluation setup, we use positive reference values V_{ref}^N and V_{ref}^G and compute

$$\widehat{V}_u^N(t) = \min\left\{\frac{V_u^N(t)}{V_{\text{ref}}^N}, 1\right\},$$

$$\widehat{V}_u^G(t) = \min\left\{\frac{V_u^G(t)}{V_{\text{ref}}^G}, 1\right\}.$$

Thus $V_u(t)$ is a dimensionless connection-imbalance score in which the subscription-count component and the backlog-load component are comparable. The subscription-count reference V_{ref}^N is chosen from an operational tolerance rather than learned by DQL. Since $\sqrt{V_u^N(t)}$ is the standard deviation of subscription counts across client u's active connections, a tolerable subscription-count standard deviation σ_{tol}^N defines

$$V_{\text{ref}}^N = (\sigma_{\text{tol}}^N)^2.$$

For example, if the operator allows roughly a Δ_N-subscription spread between a heavy and a light connection, then $\sigma_{\text{tol}}^N \approx \Delta_N/2$ is a practical calibration. If the subscription-density threshold is Θ_N, a default choice is $\sigma_{\text{tol}}^N = \rho_N \Theta_N$ with $\rho_N \in [0.1, 0.25]$; smaller values increase sensitivity to subscription-count skew. The backlog-load reference V_{ref}^G is

chosen in the same way, but in backlog units. Since $\sqrt{V_u^G(t)}$ is the standard deviation of cumulative connection backlog $G_{u,c}(t)$ across client u's active connections, a tolerable backlog-load standard deviation σ_{tol}^G defines

$$V_{\text{ref}}^G = (\sigma_{\text{tol}}^G)^2.$$

If the operator allows roughly a Δ_G-message backlog spread between a heavy and a light connection, then $\sigma_{\text{tol}}^G \approx \Delta_G/2$ is the direct calibration. When the tolerance is latency-driven, Δ_G is derived from the client-observed processing capability rather than from pod capacity alone. For example, if $\widehat{r}_u^{\text{ack}}$ is a conservative ACK or processing rate measured by client u and τ_G is the acceptable additional drain-time skew, then $\Delta_G = \widehat{r}_u^{\text{ack}}\tau_G$ is a defensible choice. If no latency SLO is available, a practical default is $\sigma_{\text{tol}}^G = \rho_G \beta \Theta_N$ with $\rho_G \in [0.1, 0.25]$, where $\beta\Theta_N$ is the nominal backlog budget of a connection carrying Θ_N subscriptions. The weights $\eta_N, \eta_G \geq 0$ determine how much the controller prioritizes subscription-management fairness versus backlog-load fairness. For example, larger η_N makes the controller more sensitive to connections carrying too many subscriptions, while larger η_G makes it more sensitive to connections accumulating more queued messages. The constraint $\eta_N + \eta_G = 1$ makes $V_u(t)$ a convex combination rather than an unbounded sum.

4.4 Pod Pressure and Deployable Reward

For an observed pod $i \in O_u(t)$, define normalized pressure terms

$$P_i^X(t) = \min\left\{\frac{X_i(t)}{X_{\text{safe}}}, 1\right\}, \tag{17}$$

$$P_i^B(t) = \min\left\{\frac{B_i(t)}{B_{\text{ref}}}, 1\right\}, \tag{18}$$

$$P_i^H(t) = \min\left\{\frac{B_i(t)}{H_i\Delta + \varepsilon_0}, 1\right\}, \qquad \varepsilon_0 > 0. \tag{19}$$

Here X_{safe} is the subscription count at which pod reliability begins to degrade. The value X_{hard} is the hard reference count at which we treat the pod as severely overloaded by subscription-management work. The value B_{ref} is the backlog reference used for normalization. The condition $X_{\text{hard}} > X_{\text{safe}}$ is required. Since high subscription count has been observed to make connections unreliable, we also define a nonlinear reliability-risk penalty

$$P_i^R(t) = \left[\frac{X_i(t) - X_{\text{safe}}}{X_{\text{hard}} - X_{\text{safe}}}\right]_+^2, \qquad [x]_+ = \max\{x, 0\}. \tag{20}$$

The pod pressure score is

$$\begin{aligned} P_i(t) =& \omega_X P_i^X(t) + \omega_B P_i^B(t) \\ &+ \omega_H P_i^H(t) + \omega_R P_i^R(t), \\ &\sum_{q \in \{X,B,H,R\}} \omega_q = 1, \qquad \omega_q \geq 0. \end{aligned} \tag{21}$$

Here $P_i^X(t)$ measures normalized subscription-count pressure. The term $P_i^B(t)$ measures normalized cumulative-backlog pressure. The term $P_i^H(t)$ measures backlog pressure relative to the processing threshold. Finally, $P_i^R(t)$ measures nonlinear reliability risk and becomes active only when $X_i(t) > X_{\text{safe}}$. The coefficients $\omega_X, \omega_B, \omega_H, \omega_R$ specify how much importance the policy assigns to each pressure component. For example, if subscription-heavy pods are known to become unreliable before backlog becomes large, then ω_X and ω_R receive greater weight than ω_B and ω_H. If queue

accumulation is the dominant failure mode, ω_B and ω_H receive greater weight. This score keeps subscription count and backlog count as distinct load dimensions.

Client u also computes local backlog and backlog-growth pressure:

$$B_u(t) = \sum_{k \in \mathcal{K}_u(t)} b_{u,k}(t), \tag{22}$$

$$\mathrm{Grow}_u(t) = \left[B_u(t) - B_u(t-1)\right]_+, \qquad t \geq 1. \tag{23}$$

At initialization, $\mathrm{Grow}_u(0) = 0$. Let $T_u(t)$ be the effective client-side throughput in events per second. The rates $T_u^{\mathrm{del}}(t)$, $T_u^{\mathrm{proc}}(t)$, and $T_u^{\mathrm{ack}}(t)$ are the delivered, processed, and acknowledged counts in the measured interval, each divided by its duration Δ:

$$T_u(t) = \min\{T_u^{\mathrm{del}}(t), T_u^{\mathrm{proc}}(t), T_u^{\mathrm{ack}}(t)\}. \tag{24}$$

For an action that selects or keeps pod i, the deployable reward is

$$\begin{aligned} R_u(t) = &- \lambda_1 \widehat{B}_u(t) - \lambda_2 \widehat{\mathrm{Grow}}_u(t) - \lambda_3 P_i(t) - \lambda_4 V_u(t) \\ &- \lambda_5 M_u(t) - \lambda_6 \widehat{C}_u(t) + \lambda_7 \widehat{T}_u(t), \end{aligned} \tag{25}$$

where $M_u(t)$ is the migration indicator or normalized migration count, $\widehat{C}_u(t) = |C_u(t)|/C_{\max}$, and $\lambda_\ell \geq 0$ with $\sum_{\ell=1}^{7} \lambda_\ell = 1$. The hats on backlog, backlog growth, and throughput denote their normalized versions under the stated experiment's scaling convention, not different physical quantities. This reward is computable by the client from local measurements and observed pod tuples.

4.5 DQL State, Action, and Value Function

The client-side controller is modeled as a partially observable sequential decision problem. At each control epoch, client u constructs an observation vector

$$o_u(t) = \left[o_u^B(t), o_u^N(t), o_u^G(t), o_u^C(t), o_u^P(t), o_u^D(t)\right], \tag{26}$$

where $o_u^B(t)$ contains normalized local backlog and backlog growth. The branch $o_u^N(t)$ describes subscription-count imbalance using terms such as $N_{u,c}(t)$ and $V_u^N(t)$. The branch $o_u^G(t)$ describes backlog imbalance using terms such as $G_{u,c}(t)$ and $V_u^G(t)$. The branch $o_u^C(t)$ contains connection count and effective throughput. The branch $o_u^P(t)$ contains pressure, subscription count, and cumulative backlog for the current and probe pods. Finally, $o_u^D(t)$ contains cooldown and statistics-staleness features. For current-pod features, $c^\star$ denotes the connection being evaluated. All quantities are normalized before they enter the Q-network.

The DQL action space is defined using bounded action templates rather than all possible subscription assignments, which keeps the decision problem tractable:

$$\mathcal{A} = \{a^{\mathrm{stay}}, a^{\mathrm{reb}N}, a^{\mathrm{reb}G}, a^{\mathrm{scale}+}, a^{\mathrm{scale}-}, a^{\mathrm{mig}}\}. \tag{27}$$

Here a^{stay} makes no change. The action $a^{\mathrm{reb}N}$ moves one subscription from the connection with the most subscriptions to the one with the fewest. The action $a^{\mathrm{reb}G}$ moves the subscription with the largest predicted decrease in backlog variance. The action $a^{\mathrm{scale}+}$ opens one new connection and, when allowed, makes bounded subscription moves toward the target Θ_N. The action $a^{\mathrm{scale}-}$ closes a subscription-free connection only if the remaining connections meet the required capacity and observed backlog is zero. The action a^{mig} requests promotion of the retained probe. It transfers the

evaluated connection or bounded subscription group only if the admissibility and hysteresis conditions pass. The DQL network selects the action template; it does not choose an arbitrary Pod or directly change a subscription assignment. The deterministic executor computes the concrete source connection, target connection, and subscription using the load-aware rules in Algorithms 1 and 2, and may reject an inapplicable template as a guarded no-op.

The scale-up action is one logical option even though its guarded executor can contain several reassociations. After the new socket is ready, the executor repeatedly applies the same receipt-gated $a^{\mathrm{reb}N}$ primitive while $\max_c N_{u,c}(t) > \Theta_N$ and an eligible move exists. It records the opened session, completed move count, final maximum count, elapsed time, and whether the target was restored. If opening fails, the unused socket is removed. If a later reassociation fails, the completed receipt-gated moves remain valid and the option reports failure rather than claiming an atomic rollback of the whole sequence. New checkpoints carry an action-contract version. Full-action execution rejects the older open-only meaning of $a^{\mathrm{scale+}}$. An explicit rebalancing-only allowlist may load a legacy checkpoint because it disables both scaling actions. This is the profile used in the October loaded cluster trial. Table 12 separates the evaluated contracts; the legacy checkpoint is not evidence for learned cap-restoring scale-up.

DQL uses two action-value networks with the same architecture but different roles. The online network $Q_\theta(o, a)$ is the trainable network used for current action evaluation and epsilon-greedy action selection. The target network $Q_{\theta^-}(o, a)$ is a delayed copy of the online network. It is not used for selecting the live action and is not updated by every gradient step; it is used only to compute a stable Bellman target for the next observation. For a transition $\xi_t = (o_u(t), a_u(t), R_u(t), o_u(t+1))$ sampled from replay buffer $\mathcal{D}$, the target is

$$y_t = R_u(t) + \gamma \max_{a' \in \mathcal{A}_u^{\mathrm{adm}}(t+1)} Q_{\theta^-}(o_u(t+1), a'), \tag{28}$$

where $\gamma \in [0, 1)$ is the discount factor and θ^- denotes the target network parameters. The set $\mathcal{A}_u^{\mathrm{adm}}(t+1) \subseteq \mathcal{A}$ contains the actions admitted by the next observation's execution guards; an infeasible action cannot supply the bootstrap value. The equation applies to non-terminal transitions. At a true terminal state, $y_t = R_u(t)$. An interrupted observation window is not assigned an artificial terminal reward. The online network parameters θ are trained by minimizing

$$\mathcal{L}(\theta) = \mathbb{E}_{\xi_t \sim \mathcal{D}} \left[(y_t - Q_\theta(o_u(t), a_u(t)))^2 \right]. \tag{29}$$

Every K_{sync} training updates, the target network is synchronized by copying the online parameters, $\theta^- \leftarrow \theta$. This separation prevents the same rapidly changing network from defining both the prediction $Q_\theta(o_u(t), a_u(t))$ and the target y_t. The parameters θ and θ^- contain all network weights and biases; Θ_N is instead a subscription-count threshold. The learning rate α controls the update $\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}(\theta)$ and is distinct from the imbalance weights η_N, η_G. This formulation makes rebalancing a sequential control problem: a migration, new connection, or subscription move can reduce future backlog but may incur immediate migration or connection cost.

4.5.1 Persistent Learning and Frozen Evaluation. Learning is retained between runs through two checkpoint types. A research checkpoint stores both networks' weights and biases, replay transitions and their next-state action masks, the random-generator state, the exploration rate, the training-update count, and the learner configuration. Restoring this checkpoint continues the learner from its saved state rather than initializing new weights. Simulator continuation starts fresh workload episodes; the checkpoint does not restore an interrupted Kubernetes deployment or simulator trajectory. Checkpoints also retain the action-contract version so that replay from incompatible action meanings is not silently combined.

Each eligible transition is recorded before its model update, using the action actually applied and the subsequent observed reward. A masked decision or shadow recommendation is not evidence that a migration occurred; shadow execution contributes only its applied stay action. Stale or incomplete telemetry interrupts the learning trajectory instead of joining observations across the missing interval. An append-only transition record and checkpoint hashes preserve a traceable history of later training.

A deployment checkpoint is a separate frozen copy: exploration is disabled, network updates are disabled, and replay is not modified during evaluation. The evaluation records observations and outcomes without turning them into training data for that comparison. Further development uses separately designated training data and produces a new candidate, which requires its own validation before deployment. Thus, retained learning avoids restarting from zero without making a fixed-policy experiment adapt to its test outcomes.

4.6 Global Evaluation Objective

The controller does not require cluster-wide variance. However, simulation and offline validation can report global metrics. A finite-horizon evaluation cost is

$$J_T = \sum_{t=0}^{T} \Big[\Lambda_B \sum_{i \in \mathcal{S}} B_i(t) + \Lambda_X \Phi_X(t) + \Lambda_C \sum_{u \in \mathcal{U}} V_u(t) + \Lambda_M M(t) \Big]. \tag{30}$$

where $\Phi_X(t)$ is pod subscription-count variance and $M(t)$ is total migration count. This objective is used to evaluate system-level behavior, not as information assumed available to every client.

The quantity J_T is an offline finite-horizon cost: lower values indicate better system-level behavior over the evaluation horizon $0, \ldots, T$. The first term penalizes total pod backlog, where $B_i(t)$ is the cumulative backlog over all subscription queues served by pod i. The second term penalizes uneven subscription placement across pods. Specifically, if

$$\bar{X}(t) = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} X_i(t),$$

$$\Phi_X(t) = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \big(X_i(t) - \bar{X}(t)\big)^2,$$

then $\Phi_X(t)$ is high when some pods carry many more subscriptions than others. The third term aggregates client-side connection imbalance, using $V_u(t)$ from (16); it measures whether a client's own persistent connections are uneven in subscription count or backlog load. The fourth term penalizes operational churn through $M(t)$, the number of migration or reconnect operations performed during epoch t. The coefficients $\Lambda_B, \Lambda_X, \Lambda_C, \Lambda_M \geq 0$ determine the relative importance of backlog reduction, pod-level subscription balance, client-side connection balance, and migration stability. Normalization before weighting places the components on comparable scales, preventing a large-scale metric from dominating J_T merely because of its units.

4.7 Rebalancing Properties

Proposition 1 (Subscription-density bound). *If* $1 \leq |\mathcal{K}_u(t)| \leq \Theta_N C_{\max}$ *and the scaling rule sets*

$$|C_u(t)| = \left\lceil \frac{|\mathcal{K}_u(t)|}{\Theta_N} \right\rceil, \tag{31}$$

then the mean subscription density satisfies $\bar{N}_u(t) \le \Theta_N$.

Proof. By definition, $\bar{N}_u(t) = |\mathcal{K}_u(t)|/|C_u(t)|$. Since $|C_u(t)| \ge |\mathcal{K}_u(t)|/\Theta_N$, it follows that $\bar{N}_u(t) \le \Theta_N$. □

Proposition 2 (Backlog-variance reduction). *For client* u, *let* $c_h, c_l \in C_u(t)$ *be two active persistent connections, where* c_h *is the source connection from which subscription* k *is removed and* c_l *is the target connection to which subscription* k *is assigned. Let* $G_{u,c_h}(t)$ *and* $G_{u,c_l}(t)$ *be the cumulative backlog loads on these two connections before the move, and let* $b_{u,k}(t) > 0$ *be the backlog of the moved subscription queue. If*

$$G_{u,c_h}(t) - G_{u,c_l}(t) > b_{u,k}(t), \tag{32}$$

then moving subscription k *from* c_h *to* c_l *strictly decreases the connection-level backlog variance* $V_u^G(t)$.

The condition states that the source connection is heavier than the target connection by more than the backlog carried by the moved subscription. Hence the move reduces their squared-deviation sum. It can cross the equal-load point and still reduce variance: loads $(10, 0)$ become $(2, 8)$ after a move of size 8. Avoiding that crossing would require the stronger condition $2b_{u,k}(t) \le G_{u,c_h}(t) - G_{u,c_l}(t)$, which is not needed by the proposition.

Proof. The move preserves total backlog and therefore preserves $\bar{G}_u(t)$. Let $d_h = G_{u,c_h}(t) - \bar{G}_u(t)$ and $d_l = G_{u,c_l}(t) - \bar{G}_u(t)$. Only two squared-deviation terms change. With $b = b_{u,k}(t)$, the unnormalized variance change is

$$\begin{aligned}&(d_h - b)^2 + (d_l + b)^2 - d_h^2 - d_l^2\\ &\quad = 2b\big(b - (G_{u,c_h}(t) - G_{u,c_l}(t))\big) < 0.\end{aligned} \tag{33}$$

Thus $V_u^G(t)$ decreases. □

Remark 2. *The placement loop optimizes a client-observed reward rather than an unobserved global cluster objective. It is a decentralized control mechanism operating under partial observability: active connections and probe connections provide fresh but incomplete pod statistics, and the client optimizes its own reward using these observations.*

5 POLICY SELECTION AND GUARDED REBALANCING

The framework places a policy selector above three concrete runtime mechanisms: pod migration, client-side connection scaling, and subscription redistribution across active connections. These mechanisms solve different parts of the same load-distribution problem. Migration changes where a connection terminates, scaling changes how much parallelism the client maintains, and rebalancing changes how subscriptions use that parallelism. Keeping the mechanisms separate makes each action deployable, while the selected policy decides which action is appropriate at a given control epoch.

The separation between reward and feasibility is deliberate. Safe-RL research distinguishes performance optimization from constraint satisfaction [22]; a migration penalty alone cannot enforce ownership or readiness. Krasowski et al. distinguish action masking, replacement, and projection as intervention mechanisms [31]. Our masks and execution-time checks restrict the deployable actions, but do not confer the formal safety guarantees of a verified environment model.

5.1 Greedy and Learned Policy Roles

Both policy families use the client-visible measurements and deterministic execution checks defined above. In the scale-out study, the guarded-greedy selector requests a^{mig} whenever that action is admissible and otherwise stays. Its

guard assesses projected source and target pressure after moving the selected subscription group, so a lightly subscribed but heavily backlogged candidate need not be accepted. The source-selection and pressure checks are shared with DQL; they are not learned from DQL's outcomes.

The mixed-workload study uses a different deterministic comparator, labelled *Heuristic*: it applies threshold rules for subscription-count and backlog imbalance as well as migration. The independent *Scripted* timing controls use pressure-unaware schedules and are sensitivity controls, not substitutes for the stronger guarded-greedy comparison. Table 2 keeps these identities distinct.

Table 2. Policy roles and the comparison each supports. Greedy and DQL share the rebalancing mechanism; the studies retain their original policy labels.

Policy	Decision rule	Evidence role
Guarded-greedy	Promote an admissible pressure-improving probe	Strong non-learning scale-out control
Heuristic	Threshold rules over count, backlog, and placement	Non-learning multi-objective comparator
Frozen DQL	Masked maximum learned action value; no evaluation updates	Learned multi-objective selector
Scripted schedules	Fixed or jittered timing; random source and target	Pressure-unaware sensitivity controls
Static / scale-out only	Retain established connection assignment	No-rebalancing reference

The shared contribution is probe-guided actuation, not the invention of either greedy search or Q-learning. The experiments compare policy choices without feeding DQL's holdout decisions back into the scripted rules. Learning traces can inform separate development work, but automatic DQL-to-script tuning is not part of the evaluated system.

The layer is designed to:

- Dynamically scale the number of parallel persistent transport connections
- Improve parallel processing for high-backlog subscriptions
- Maintain balanced subscription distribution across connections
- Prevent overload of individual connections

5.2 Adaptive Parallel Connection Scaling - Algorithm

5.2.1 Mathematical Model. Unlike server-level placement, which changes the pod reached by a connection, connection scaling changes the amount of parallelism available to a client. In subscription-driven messaging systems, too few persistent connections can force many high-backlog subscriptions through the same processing path, increasing latency and delaying acknowledgements.

The algorithms use the same backlog $b_{u,k}(t)$ and connection subscription count $N_{u,c}(t)$ as the formal model. For compactness, the single-client scaling counts below suppress the fixed client index u.

Let the client-side subscription population be

$$N_{\text{sub}}(t) = |\mathcal{K}_u(t)|, \tag{34}$$

where $\mathcal{K}_u(t)$ is the set of active subscriptions maintained by client u at time t. Thus $N_{\text{sub}}(t)$ counts subscriptions, not messages.

Let the current connection count be

$$C(t) = |\mathcal{C}_u(t)|, \tag{35}$$

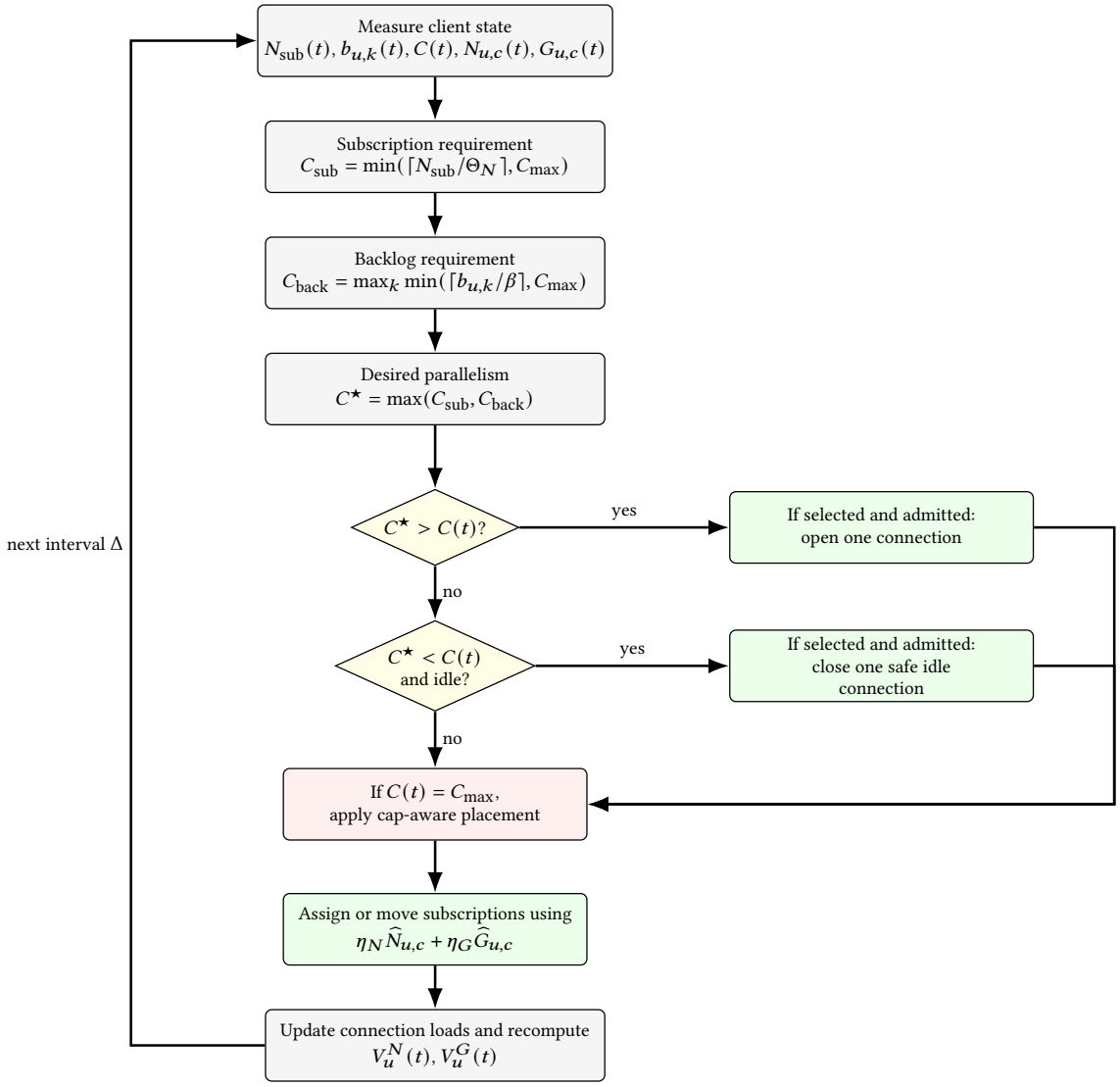


Fig. 3. Adaptive connection-scaling workflow. The client first computes the connection requirement induced by subscription density and the requirement induced by per-subscription backlog. The larger requirement determines the desired number of persistent connections, bounded by C_{max}. If the cap is reached, the client assigns or moves subscriptions using a two-dimensional score that combines subscription-count pressure and backlog pressure. One DQL a^{scale+} decision creates at most one connection and then restores the subscription target through bounded receipt-gated reassociation.

where $\mathcal{C}_u(t)$ is the set of active persistent transport connections used by client u for message delivery at time t. Hence $C(t)$ is the current number of usable STOMP/TCP/WebSocket-style delivery connections before the scaling decision is applied. The retained statistics-only probe is not counted in $C(t)$ unless it is promoted into a persistent delivery connection.

The desired count $C^{\star}(t)$ can differ from the amount executed in one DQL decision. Algorithm 1 computes the desired parallelism, whereas one $a^{\text{scale+}}$ option adds at most one connection. It then moves whole subscriptions until no active connection exceeds Θ_N, or until no eligible move remains. A subscription has exactly one active delivery binding outside the receipt-gated handoff interval; this mechanism does not split one subscription queue across several connections. Using the requirements defined below, the workflow abbreviates $C_{\text{back}}(t) = \max_{k \in \mathcal{K}_u(t)} C_k(t)$ and $C^{\star}(t) = \max\{C_{\text{sub}}(t), C_{\text{back}}(t)\}$ for a nonempty subscription set.

We define:

- Θ_N : maximum desired subscription count per connection
- β : backlog threshold
- $C_{\max}$: maximum allowable connections

The complete notation reference is collected in Appendix A.

5.2.2 Subscription-Based Scaling. As the number of subscriptions increases, each connection must handle more message streams. To prevent overload, the desired number of connections is defined as:

$$C_{\text{sub}}(t) = \min\left(\left\lceil \frac{N_{\text{sub}}(t)}{\Theta_N} \right\rceil, C_{\max}\right) \tag{36}$$

Here Θ_N is a configured subscription-density threshold: it is the maximum desired number of subscriptions assigned to one persistent connection under normal operating conditions. It is not learned by DQL and is not the DQL parameter vector θ used in $Q_\theta(o, a)$.

If:

$$C_{\text{sub}}(t) > C(t)$$

the desired count exceeds the current count. The policy may request one admissible scale-up action; the sizing calculation itself opens no sockets.

The count rule bounds mean subscription density when sufficient capacity is available. A per-connection bound additionally requires the reassociations to complete; the executor records whether that target was restored.

The sizing rule and the reassociation executor therefore have separate roles: one requests enough sockets, and the other redistributes eligible subscriptions.

5.2.3 Max-Cap Distribution Strategy. When:

$$C(t) = C_{\max}$$

further scaling is restricted. In this scenario, newly arriving subscriptions are assigned using load-aware distribution.

Let:

$$N_{u,c_j}(t) \tag{37}$$

denote the number of subscriptions assigned to connection c_j.

New subscriptions are assigned to the connection with the lowest combined subscription-count and backlog pressure:

$$c^{\text{place}} = \arg\min_{c_j \in C_u(t)} \left(\eta_N \widehat{N}_{u,c_j}(t) + \eta_G \widehat{G}_{u,c_j}(t)\right) \tag{38}$$

This greedy balancing strategy avoids assigning new subscriptions to a connection that is already heavy either in subscription count or cumulative subscription backlog. Here c^{place} is the selected placement connection, while $c^{\star}$ is the connection evaluated for migration. The hats on N_{u,c_j} and G_{u,c_j} denote normalized loads, not the normalized variances $\widehat{V}_u^N$ and $\widehat{V}_u^G$.

5.2.4 Backlog-Based Scaling. Subscription backlog signals queued work, but its cause may lie in Pod service, client acknowledgement capacity, or shared queue bandwidth. Scaling is useful when additional connections can reach otherwise unused service capacity.

For subscription k, let backlog be:

$$b_{u,k}(t) \tag{39}$$

If:

$$b_{u,k}(t) > \beta$$

the configured backlog-pressure threshold has been crossed.

The required number of connections is:

$$C_k(t) = \min\left(\left\lceil \frac{b_{u,k}(t)}{\beta} \right\rceil, C_{\max}\right) \tag{40}$$

If:

$$C_k(t) > C(t)$$

backlog suggests additional parallelism. The selector still decides whether to execute a bounded scaling action under the current guards.

The additional capacity can then receive eligible whole subscriptions from loaded connections. Subscription k itself remains active on exactly one connection; its queue is not partitioned by this mechanism. Thus backlog-based scaling can improve parallelism across a set of subscriptions, but it cannot parallelize the delivery of one indivisible subscription queue by itself.

5.2.5 Scaling Objective. The objective of adaptive parallel connection scaling is to minimize backlog while respecting connection constraints:

$$\min \sum_{k \in \mathcal{K}_u(t)} b_{u,k}(t) \tag{41}$$

subject to:

$$C(t) \le C_{\max} \tag{42}$$

This represents a constrained parallel resource allocation problem.

Algorithm 1: Desired Connection Count (Sizing Only)

Input : Subscription set $\mathcal{K}_u(t)$, backlogs $b_{u,k}(t)$; thresholds Θ_N, β; hard cap $C_{\max}$
Output : Desired count $C^\star(t)$; no connection mutation

$N_{\text{sub}}(t) \leftarrow |\mathcal{K}_u(t)|$;
$C_{\text{sub}}(t) \leftarrow \min\{\lceil N_{\text{sub}}(t)/\Theta_N \rceil, C_{\max}\}$;
$C_{\text{back}}(t) \leftarrow 0$;
foreach $k \in \mathcal{K}_u(t)$ **do**
    $C_k(t) \leftarrow \min\{\lceil b_{u,k}(t)/\beta \rceil, C_{\max}\}$;
    $C_{\text{back}}(t) \leftarrow \max\{C_{\text{back}}(t), C_k(t)\}$;
$C^\star(t) \leftarrow \max\{C_{\text{sub}}(t), C_{\text{back}}(t)\}$;
return $C^\star(t)$ as decision context for the selector;

5.3 Connection-Level Load Rebalancing - Algorithm

5.3.1 Motivation. Adaptive connection scaling increases processing capacity as subscription count or backlog grows. Scaling alone, however, does not guarantee an even workload. New subscriptions, changing backlogs, and scaling decisions made at different times can leave some connections with far more subscriptions than others.

Such imbalance leads to:

- Uneven CPU utilization
- Increased processing latency on overloaded connections
- Underutilized parallel capacity on lightly loaded connections

Therefore, a dedicated rebalancing mechanism is required to maintain fairness and stability across scaled connections.

5.3.2 Load Distribution Model. Connection load is represented by two variables. For connection c, $N_{u,c}(t)$ is the number of subscriptions assigned to the connection, and $G_{u,c}(t)$ is the cumulative backlog of those subscriptions. The rebalancing controller therefore monitors both $V_u^N(t)$ and $V_u^G(t)$ as defined in the problem formulation.

High $V_u^N(t)$ means that subscription-management work is skewed across the client's persistent connections. High $V_u^G(t)$ means that queue pressure is skewed even if subscription counts look balanced. Either condition can degrade delivery stability.

5.3.3 Rebalancing Objective. The local objective of connection-level rebalancing is:

$$\min \left(\eta_N \widehat{V}_u^N(t) + \eta_G \widehat{V}_u^G(t) \right) \tag{43}$$

subject to:

$$1 \le |C_u(t)| \le C_{\max}. \tag{44}$$

This objective keeps both subscription-count skew and backlog skew under control.

5.3.4 Guarded Connection-Level Rebalancing Strategy. The two rebalancing templates have separate admissibility conditions:

$$\widehat{V}_u^N(t) > \gamma_N \quad \text{or} \quad \widehat{V}_u^G(t) > \gamma_G. \tag{45}$$

The DQL policy, rather than Algorithm 2, decides whether the current observation calls for $a^{\mathrm{reb}N}$, $a^{\mathrm{reb}G}$, or another action. If it selects $a^{\mathrm{reb}N}$, the executor moves one eligible subscription from the connection with largest $N_{u,c}(t)$ to the connection with smallest $N_{u,c}(t)$. If it selects $a^{\mathrm{reb}G}$, the executor chooses the subscription whose reassociation from a high-backlog connection to a low-backlog connection produces the largest predicted decrease in $V_u^G(t)$. The corresponding threshold is then applied as a deterministic deployment guard. At most one bounded reassociation is executed in a control interval. This limits repeated back-and-forth moves and links the next reward to a clear action.

Algorithm 2: Guarded Executor for DQL-Selected Connection-Level Rebalancing

Input : Selected template $a_u(t) \in \{a^{\mathrm{reb}N}, a^{\mathrm{reb}G}\}$; connections $C_u(t)$; subscription counts $N_{u,c}$; connection backlogs $G_{u,c}$; thresholds γ_N, γ_G

Output : Updated assignment, or the unchanged assignment when the selected template is inadmissible

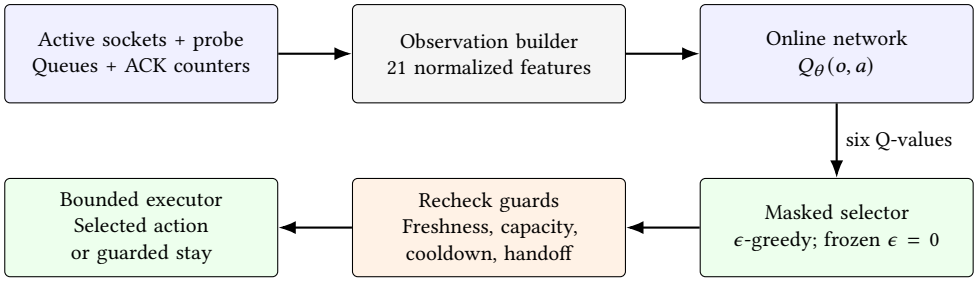
Compute $\widehat{V}_u^N(t)$ and $\widehat{V}_u^G(t)$;
if $a_u(t) = a^{\mathrm{reb}N}$ **then**
  if $\widehat{V}_u^N(t) \le \gamma_N$ **then**
    return unchanged assignment ; `// Guarded no-op`
  end if
  $c_h \leftarrow \arg\max_{c \in C_u(t)} N_{u,c}(t)$;
  $c_l \leftarrow \arg\min_{c \in C_u(t)} N_{u,c}(t)$;
  Select an eligible movable subscription k from c_h using a stable tie-break;
else
  if $\widehat{V}_u^G(t) \le \gamma_G$ **then**
    return unchanged assignment ; `// Guarded no-op`
  end if
  $c_h \leftarrow \arg\max_{c \in C_u(t)} G_{u,c}(t)$;
  $c_l \leftarrow \arg\min_{c \in C_u(t)} G_{u,c}(t)$;
  Select eligible k on c_h maximizing the predicted decrease in $V_u^G(t)$;
end if
if $c_h = c_l$ *or no eligible* k *exists* **then**
  return unchanged assignment ; `// Guarded no-op`
end if
Reassociate k from c_h to c_l using the guarded executor;
Update $N_{u,c_h}, N_{u,c_l}, G_{u,c_h}, G_{u,c_l}$;
return updated assignment;

Algorithm 2 carries out the selected action; it is not a second policy. In particular, it never compares $\widehat{V}_u^N(t)$ with $\widehat{V}_u^G(t)$ to decide which objective will dominate. These normalized quantities can have different observed distributions even though both are bounded. The selected DQL template is the stored action, while the reward and next observation reflect the executor outcome, including a guarded no-op. The learned value therefore accounts for whether an action can be executed, without introducing a second rebalancing policy.

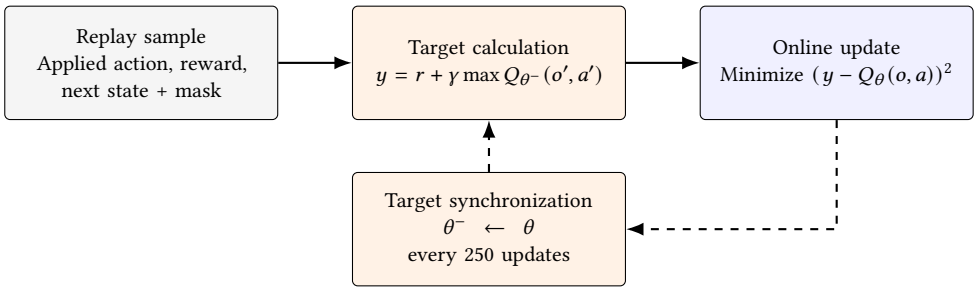
5.4 DQL-Based Rebalancing and Placement Algorithm

This section presents the client-driven DQL controller under the observability constraints described above. Each client keeps a statistics cache with two sources. Active connections report their current pod's tuple every $\Delta = 10$ seconds. An authenticated, statistics-only probe provides the other source. The load balancer assigns its endpoint when the socket is established. The client retains that socket and candidate-pod identity until promotion, expiry, or recycling while

periodically obtaining the tuple in (9). The controller then uses the observation vector in (26) to select one of the action templates in (27).

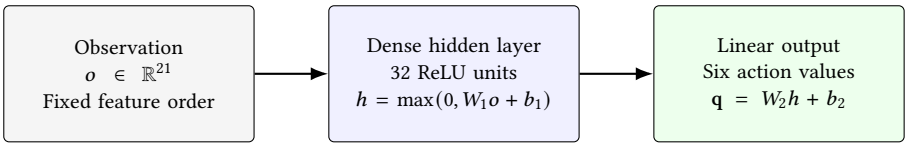


(a) Runtime decision path. The target network is not on this path.



(b) Training path for the evaluated v3 learner. Frozen evaluation disables updates.

Fig. 4. DQL selection and learning are separate from the shared guarded executor. Solid arrows show runtime or training data flow; dashed arrows copy parameters to the target network. Only the online network ranks live actions. The target network stabilizes the next-state value used during training.



$$W_1 : 32 \times 21, \quad b_1 : 32, \quad W_2 : 6 \times 32, \quad b_2 : 6$$

Fig. 5. The evaluated Q-network: 21 input features, one fully connected 32-unit ReLU layer, and six linear outputs. Each arrow denotes a dense matrix operation, not a sparse connection between individual neurons. There are no separate learned feature encoders. The online and target networks have this same architecture and different parameter-update roles.

In Figure 5, o abbreviates $o_u(t)$ and $\theta = (W_1, b_1, W_2, b_2)$. There are 902 trainable scalar parameters in each network. The output order is $(a^{\text{stay}}, a^{\text{reb}N}, a^{\text{reb}G}, a^{\text{scale}+}, a^{\text{scale}-}, a^{\text{mig}})$. Grouping features by meaning in (26) does not introduce additional neural layers. Table 3 gives the evaluated v3 configuration; Appendix B gives the exact input order and normalization.

Table 3. Configuration of the v3 DQL policy used in the reported holdout, scale-out simulator, and October live trial. Later learner variants are reported separately in Appendix C.

Setting	Value
Network; optimizer	Dense 21–32–6; ReLU hidden, linear output; mini-batch gradient descent
Discount; learning rate	$\gamma = 0.99$; $\alpha = 0.001$
Replay; batch; target copy	Uniform replay, capacity 5,000; batch 16; copy every 250 updates
Bootstrap; loss	Target-network maximum over admitted actions; squared TD error
Training budget	10,000 transitions per candidate; 9,985 updates after batch warm-up
Training exploration	$\epsilon_0 = 1$, multiplier $\exp(\log(0.05)/10{,}000)$; floor 0.05
Reward weights $(\lambda_1, \ldots, \lambda_7)$	$(0.20, 0.15, 0.20, 0.15, 0.10, 0.05, 0.15)$
Pressure weights $(\omega_X, \omega_B, \omega_H, \omega_R)$	$(0.30, 0.25, 0.20, 0.25)$
Imbalance weights	$\eta_N = \eta_G = 0.5$
Frozen evaluation	$\epsilon = 0$; no model or replay updates; checkpoint hash retained

5.4.1 Reward-Based Learning. For every observed pod $i \in O_u(t)$, the client computes pod pressure $P_i(t)$ from (21). This pressure explicitly penalizes both pod subscription count and pod cumulative backlog. The subscription-count term captures the observed reliability degradation of pods with excessive subscription-management load, while the backlog term captures queue accumulation. The threshold H_i is used only as a capacity-normalized pressure term; it is not interpreted as the actual client-side drain rate.

After action execution, the client computes reward $R_u(t)$ using (25). This reward includes local backlog, backlog growth, local connection imbalance, migration cost, connection cost, pod pressure, and effective client-side throughput. Hence the learned policy is trained using information that the client can actually observe at runtime.

5.4.2 Exploration, Hysteresis, and Migration. Let $\mathcal{A}_u^{\mathrm{adm}}(t) \subseteq \mathcal{A}$ contain only the templates that pass the current freshness, capacity, cooldown, and handoff guards. The DQL policy uses epsilon-greedy exploration over this masked set:

$$a_u(t) = \begin{cases} \mathrm{Uniform}(\mathcal{A}_u^{\mathrm{adm}}(t)), & \text{w.p. } \epsilon, \\ \arg\max_{a \in \mathcal{A}_u^{\mathrm{adm}}(t)} Q_\theta(o_u(t), a), & \text{w.p. } 1-\epsilon. \end{cases} \tag{46}$$

The exploration probability decreases during training. Frozen validation and deployment policies use $\epsilon = 0$. This prevents unreviewed exploratory changes during evaluation. Online exploration would require a separate safety study. Migration is executed only when the selected action is a^{mig} and the retained probe candidate improves the estimated value by a hysteresis threshold:

$$Q_\theta(o_u(t), a^{\mathrm{mig}}) > Q_\theta(o_u(t), a^{\mathrm{stay}}) + \delta. \tag{47}$$

This threshold prevents oscillatory migrations caused by small or stale statistical differences. It does not choose the probe endpoint: the Kubernetes load balancer already fixed that endpoint when the probe socket was opened. Passing this value test is necessary but not sufficient. The executor also checks telemetry freshness, target capacity, source–target separation, cooldown, and the absence of a conflicting handoff. As a result, most probes can remain telemetry-only and may expire or be recycled without carrying a subscription.

Algorithm 3: DQL-Based Client-Side Persistent Transport Rebalancing

Input : Replay buffer $\mathcal{D}$; online network Q_θ; target network Q_{θ^-}
Observed pod set $O_u(t)$ and its cached tuples $y_i(t)$; retained authenticated probe and endpoint $i_u^{\text{probe}}(t)$
Exploration probability ϵ; discount factor γ; migration threshold δ; target Θ_N; hard cap $C_{\max}$
Output : Adaptive rebalancing, scaling, and migration decisions

$t \leftarrow 0$; use the active delivery pool and one authenticated probe;
Load the declared checkpoint; initialize networks/replay only for new training;
while *client is active* **do**
  Update the pod cache from active and retained-probe telemetry;
  Measure per-subscription backlogs, connection loads, throughput, and acknowledgements;
  Construct observation $o_u(t)$ using (26);
  Construct the masked admissible set $\mathcal{A}_u^{\text{adm}}(t)$ from freshness, capacity, cooldown, and handoff guards;
  $a_u(t) \leftarrow \epsilon$-greedy$(Q_\theta, o_u(t), \mathcal{A}_u^{\text{adm}}(t))$;
  if $a_u(t) \in \{a^{\text{reb}N}, a^{\text{reb}G}\}$ **then**
    Instantiate the selected template using Algorithm 2;
  else
    if $a_u(t) = a^{\text{mig}}$ **then**
      if *probe and handoff guards pass, including the hysteresis test* **then**
        Promote the same retained probe and execute the receipt-gated handoff in Section 3.7;
        Open a new statistics-only probe through the load balancer after handoff;
      else
        Record a guarded no-op; preserve source ownership and probe role;
      end if
    else
      if $a_u(t) = a^{\text{scale+}}$ **then**
        if *capacity and subscription-demand guards still pass* **then**
          Open one connection; apply receipt-gated $a^{\text{reb}N}$ moves until $\max_{c \in C_u(t)} N_{u,c}(t) \le \Theta_N$ or no move exists;
          Record move count and target-restored status;
        else
          Record a guarded no-op;
        end if
      else
        if $a_u(t) = a^{\text{scale}-}$ **then**
          Close one safe idle connection after rechecking its guard;
        else
          Preserve the current topology for a^{stay};
        end if
      end if
    end if
  end if
  Observe reward $R_u(t)$ from (25) and next observation $o_u(t+1)$;
  Log the applied action, reward, next observation, and guard outcome;
  if *training is enabled and the transition is valid* **then**
    Store the applied-action transition and next-state mask in $\mathcal{D}$;
    Update from a mini-batch using (29); synchronize the target when due;
  end if
  Wait Δ; $t \leftarrow t + 1$;
end while
return policy and recorded action outcomes;

$$\begin{aligned}\text{computeReward}(i,u,t) = R_u(t) = &-\lambda_1 \widehat{B}_u - \lambda_2 \widehat{\text{Grow}}_u \\ &- \lambda_3 P_i - \lambda_4 V_u \\ &- \lambda_5 M_u - \lambda_6 \widehat{C}_u + \lambda_7 \widehat{T}_u \end{aligned} \tag{48}$$

The epsilon-greedy rule is therefore used as the exploration mechanism inside DQL, not as the main learning model. Unlike a stateless bandit controller, DQL can learn delayed effects. Scale-up with subscription restoration has an immediate connection and reassociation cost but may later reduce backlog growth. Migration may have an immediate penalty but later avoid an unreliable pod with too many subscriptions. A rebalancing move may help only after queues have had time to drain.

5.5 Hierarchical Optimization Structure

The resulting control structure has two complementary levels:

(1) **Server-Level Optimization:** The DQL controller selects migration only when the observed pod statistics and local client state suggest that moving to a candidate pod is likely to improve long-term reward after migration cost.
(2) **Connection-Level Optimization:** The same policy decides when to rebalance subscriptions, increase or decrease connection parallelism, or leave the current assignment unchanged.

This hierarchical design enables simultaneous optimization of:

- Inter-server load distribution
- Intra-client parallelism
- Subscription fairness
- Backlog reduction

The learned component does not create arbitrary subscription assignments. It chooses bounded action templates that the client can execute and check against deployment guards.

6 ORACLE CLOUD CLUSTER DEPLOYMENT EXPERIMENT

We deployed the connection-replacement mechanism in an Oracle Cloud cluster running Kubernetes in a development environment and first exercised it with 20,000 registered subscriptions. This initial experiment measures live probe promotion, subscription reassociation, handoff timing, and post-migration topology. The controlled scale-out and policy comparisons in Section 7 use a simulator and are reported separately. Sections 6.4 and 6.5 report separate isolated trials, including selective DQL migration during delivery and a one-to-four-Pod scale-out sequence.

6.1 Deployment and Execution

The deployment contained ten subscriber-service pods. Ten client processes each maintained five active persistent connections, with 400 subscriptions per connection, and one additional subscription-free probe. The inventory had four event-type groups, each with 5,000 subscriptions. Every client loaded the same frozen DQL checkpoint, with exploration disabled and no training updates during evaluation. The deployed action profile admitted rebalancing and migration but masked connection scale-up and scale-down.

Table 4. Measured Oracle Cloud cluster deployment results. Handoff and drain statistics summarize ten executed connection migrations. Drain wait is part of handoff time, not an additional independent interval.

Measurement	Observed value
Subscriber-service pods / client processes	10 / 10
Registered and attached subscriptions	20,000
Active persistent connections / probes	50 / 10
Executed full-connection migrations	10
Subscriptions moved per migration / total	400 / 4,000
Handoff time: minimum / median / maximum (s)	7.898 / 14.498 / 101.563
Drain wait: minimum / median / maximum (s)	1.264 / 7.347 / 92.450
Exact final topologies after 1,800 s	10/10
Recorded deliveries / processed and acknowledged	114,986 / 113,149
Processing failures / ACK failures / NACK failures	0 / 0 / 0
Recorded duplicate candidates	0
Independent pod samples / collection errors	310 / 0
Pod restart increase during observation	0

Each client could execute at most one guarded connection migration. A selected probe was promoted only after telemetry and execution checks passed. The executor then reassociated the source connection's 400 subscriptions, drained in-flight delivery, retired the source socket, and opened a fresh subscription-free probe. Subscription receipts and server-reported counts were checked against the client assignment. After execution, each process held its topology for 1,800 seconds and checked it again before shutdown. The statistical unit is one deployment run, containing ten client migrations.

6.2 Observed Handoff and Topology

Table 4 shows that all ten clients replaced one active connection and moved its full subscription set. In every case, the destination was the retained probe, the other four active connections remained in place, and the replacement probe had no subscriptions. The final checks confirmed all 2,000 subscriptions and five active connections in each client after the 30-minute hold. Thus, 4,000 subscriptions changed their transport association while the complete 20,000-subscription client population remained accounted for.

Handoff duration ranged from 7.898 to 101.563 seconds, with a median of 14.498 seconds. The corresponding source-drain waits ranged from 1.264 to 92.450 seconds, with a median of 7.347 seconds. The range is important: replacement time includes waiting for outstanding work, not just opening a socket. This run deliberately used eight processing workers and a two-second processing delay per event in each client. Those settings exercise handoff under a bounded client drain rate. The endpoint is handoff behavior under this processing limit, rather than maximum Pod throughput.

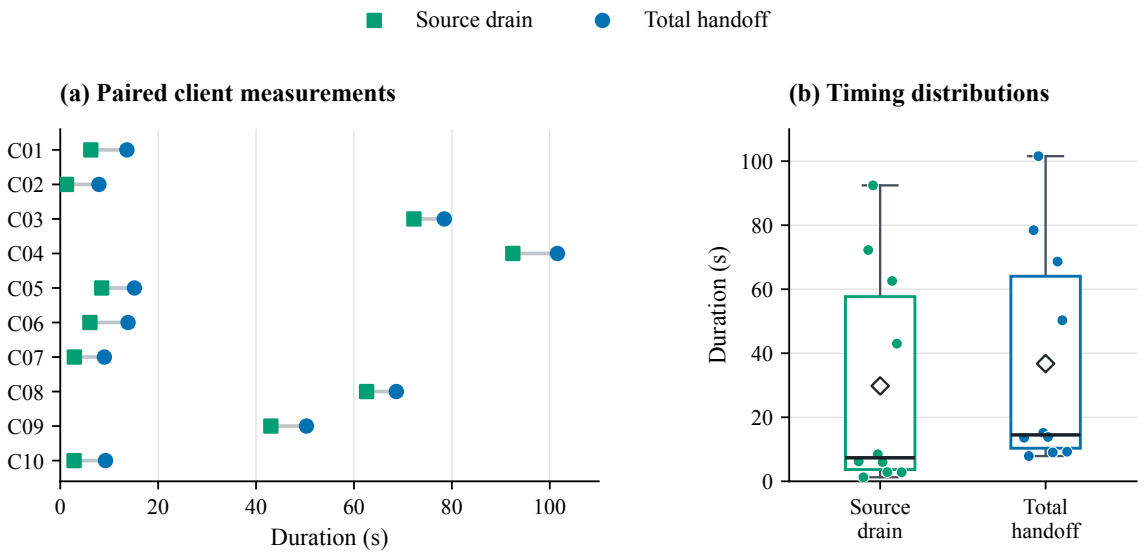


Fig. 6. Measured Oracle Cloud cluster migration timing. Left: paired source-drain and total-handoff durations for each of the ten client migrations. Right: quartile boxes with median lines, mean diamonds, and all ten observations; whiskers extend to observations within 1.5 interquartile ranges. These are ten migrations in one cluster run, not ten independent cluster trials.

Figure 6 retains the pairing that a summary median cannot show. For client u, the interval between its two markers is $T_u^{\text{handoff}} - T_u^{\text{drain}}$, the measured non-drain remainder of the handoff; it is not an estimate of network latency alone. The box extends from the sample's first to third quartile, so the interquartile range describes the central half of the timings. All points, including the long-duration observations, remain visible. Longer handoffs coincide with longer source drains, consistent with the receipt-gated mechanism waiting for outstanding work. These are within-run timing distributions; their scope is the observed deployment, without a between-deployment confidence interval.

An independent collector retained 31 observations of each of the ten pods over approximately 2,422 seconds. All pods remained Ready, with unchanged pod, image, and process identities and no restart increase. These observations establish operational continuity over this observation period. Long-term availability is outside the measurement window.

6.3 Event Accounting and Interpretation

The clients recorded 114,986 deliveries and 113,149 processed and acknowledged events, with no recorded processing, acknowledgement, or negative- acknowledgement failures. The final difference is 1,837 deliveries whose eventual status is not resolved by these counters. Zero duplicate candidates describes the configured observation window, rather than an exactly-once delivery guarantee. The run consumed existing queued work without a matched static-policy workload. Its contribution is measured probe-based connection replacement with 20,000 subscriptions: subscription reassociation, source retirement, and the connection layout after the hold period. Its endpoints are handoff execution and topology retention, not a comparative backlog-drain rate.

The private evidence archive retains the frozen-policy and inventory hashes, per-client decision and topology records, delivery counters, and independent pod observations. A recursive manifest covers 134 files; aggregate values in Table 4 come from the active-execution gate and pod-observation summaries, not from simulator output.

6.4 Concentrated Placement and Startup Under Load

We conducted a separate Oracle Cloud cluster experiment with an isolated subscriber Deployment and Service to examine what happens when persistent TCP connections remain on one Pod. In an initial 4,000-subscription static calibration, ten delivery connections and two probes terminated on one Pod. Adding three Ready Pods did not relocate these sockets: the source retained all twelve established connections, while each added Pod had none. The clients nevertheless received and acknowledged events while subscription activation was still progressing. Full activation took approximately 1,398.4 seconds; the completed run reconciled 137,008 observed deliveries with confirmed ACKs. This shows the difference between having server capacity and reaching it through existing persistent connections.

A later frozen-DQL test used 2,000 other subscriptions, divided equally among four event-type groups. One client attached five connections, each carrying 400 subscriptions, to the single source Pod. Three destination Pods became Ready before standby bindings were activated. The client used 32 processing workers without an artificial processing delay, ten message permits per subscription, and a window of 32 outstanding activation requests. Exact lifecycle records used batched SQLite transactions with FULL synchronization and a mandatory flush at measurement and closure boundaries. The earlier calibration used a different subscription population and recorder settings. The two timing records are therefore reported as separate experiments rather than a matched static-policy comparison.

Table 5. Measured isolated Oracle Cloud cluster connection-rebalancing qualification. These values describe one live run, distinct from the earlier 20,000-subscription deployment.

Measurement	Observed value
Available Pods / subscribed Pods before and after	4 / 1 and 3
Subscriptions / active connections / probe	2,000 / 5 / 1
Time to 50% / 90% / full activation (s)	238.066 / 388.139 / 426.305
Committed connection migrations / subscriptions moved	3 / 1,200
Source subscriptions before / after	2,000 / 800
Final subscribed-Pod populations	800, 800, 400
Handoff times, in execution order (s)	10.878, 10.992, 10.648
Source-drain waits, in execution order (s)	6.069, 6.174, 6.848
Delivered / processed / receipt-confirmed ACKs	73,639 / 73,639 / 73,639
Unresolved observed attempts at closure	0
Recorded ACK failures / duplicate candidates	0 / 0

The frozen controller committed three complete connection replacements, each moving 400 subscriptions through a retained probe. The source population fell from 2,000 to 800, a reduction of $(2{,}000 - 800)/2{,}000 = 60\%$. Two destination Pods acquired 400 and 800 subscriptions. The active-connection count remained five, and all 2,000 subscriptions were confirmed in the final topology after the two-minute hold. The migrations therefore changed which Pods served these subscriptions without increasing the client's active-connection count.

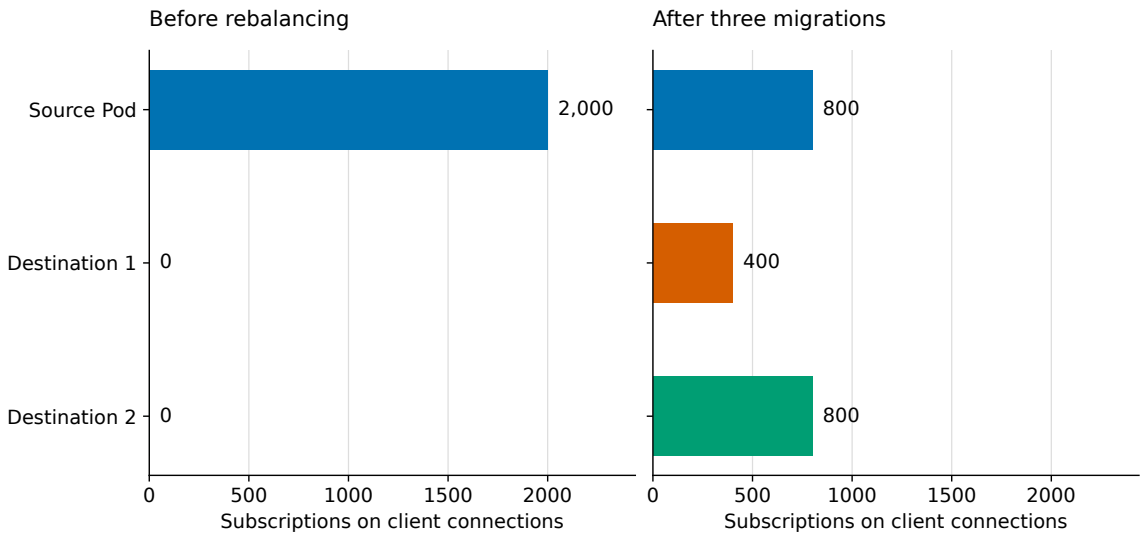


Fig. 7. Verified client subscription placement before and after three frozen-DQL connection migrations in the Oracle Cloud cluster. Bars aggregate the client's receipt-confirmed bindings by serving Pod. Initial zeros mean no bindings belonging to this client, not a measurement of every cluster subscription. Both panels account for the same 2,000 subscriptions and five connections.

Figure 7 shows the main operational effect: newly available capacity becomes reachable by moving subscription-bearing connections. It reports verified connection endpoints, rather than estimating changes between observations. A probe was first resampled because it did not offer an admissible improvement. Eleven later decision intervals selected stay while backlog telemetry was incomplete; migration resumed when the observation checks passed. The online and target network parameters remained identical to the loaded checkpoint, while valid transitions were retained.

6.4.1 Activation and Delivery Coexistence. For subscription k, let t_k^{send} be activation submission time and t_k^{ready} its successful receipt time, relative to the start of activation. At elapsed time s in seconds, the cumulative completion count is $A^{\text{act}}(s) = \sum_{k=1}^{2000} \mathbf{1}\{t_k^{\text{ready}} \le s\}$, and $2000 - A^{\text{act}}(s)$ is the pending count. This separates attachment from confirmed activation and makes startup delay visible even while events flow. The 50th, 90th and 100th completion percentiles were 238.066, 388.139 and 426.305 seconds, respectively. No activation retry was needed in this run.

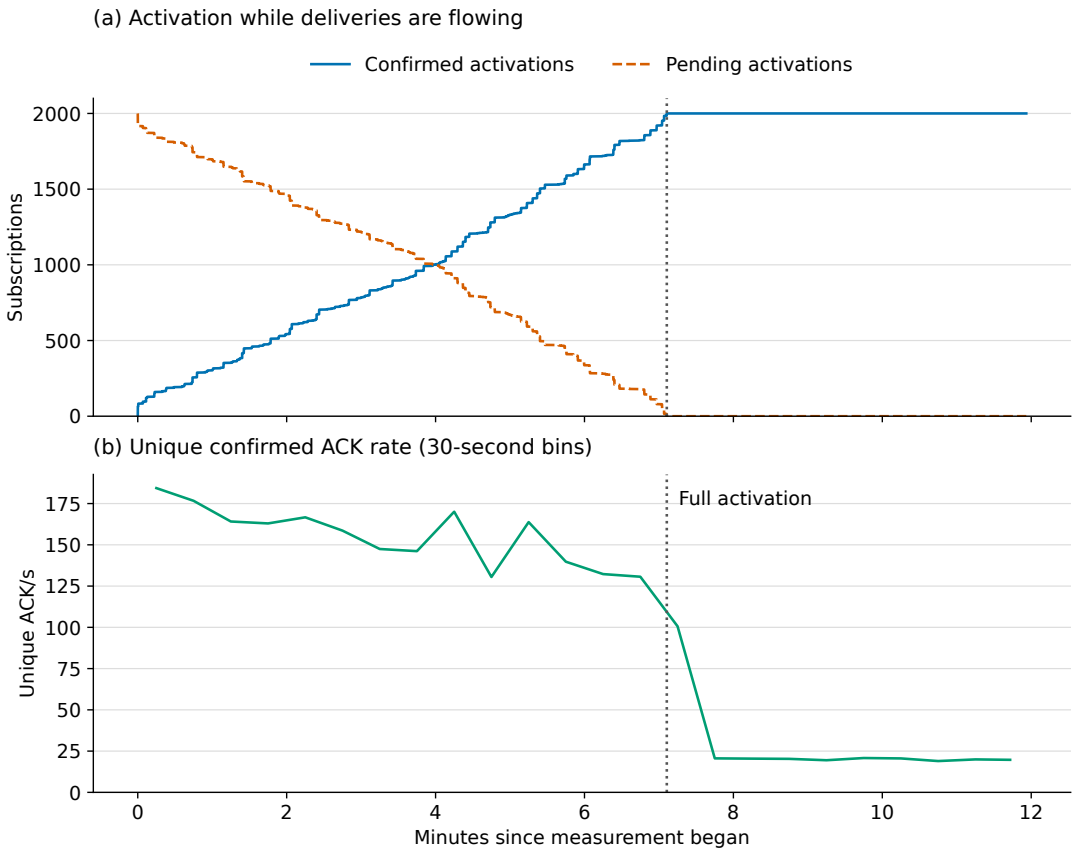


Fig. 8. Activation and acknowledged delivery in the isolated Oracle Cloud cluster run. Top: confirmed and pending activation counts. Bottom: distinct observed subscription–message pairs whose first successful ACK receipt occurs in each 30-second bin, divided by that bin's actual duration. The dotted line marks full activation. DQL decisions start after this line.

Figure 8 retains the startup interval rather than treating it as discarded warm-up. Of 73,587 distinct first ACKs inside the 716.017-second measurement window, 66,400 occurred before full activation. A further 52 attempts completed during shutdown, bringing final reconciled deliveries to 73,639. The rate curve describes consumption of existing queued work. It describes startup and workload drainage rather than constant-offered-load capacity. DQL begins after full activation, so these measurements establish delivery during startup followed by redistribution. Activation time precedes the policy intervention.

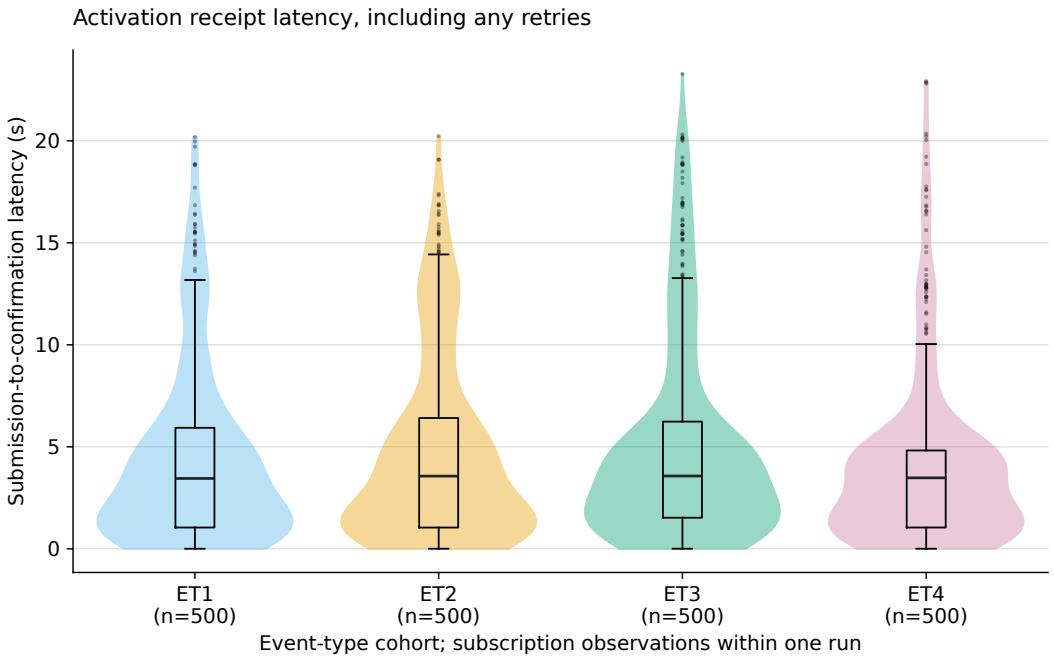


Fig. 9. Within-run activation latency $t_k^{\text{act}} = t_k^{\text{ready}} - t_k^{\text{send}}$ for 500 subscriptions in each event-type cohort. Violin widths show smoothed empirical density; boxes show the interquartile range and median. Whiskers use 1.5 interquartile ranges, with observations beyond them retained as points. These are subscription observations in one run, not independent cluster trials.

Figure 9 distinguishes an individual request's latency from the time to activate the complete population. Cohort medians were 3.443, 3.560, 3.567 and 3.476 seconds; nearest-rank 99th percentiles were 18.805, 16.907, 20.072 and 20.038 seconds. The subscriptions share upstream services and scheduling conditions, so their observations are dependent. The distributions describe within-run variation, without estimating a between-policy confidence interval. A concurrent thread snapshot found fourteen subscription handlers waiting for upstream transition confirmation. This locates the observed startup bottleneck in upstream confirmation waits; the snapshot does not establish CPU saturation.

The complete ledger covers observed delivery attempts, with publisher-to-Queue event counts outside its scope. This qualification establishes receipt-confirmed activation under delivery load, probe-based redistribution with a 60% reduction in source subscriptions, and reconciled closure. The recorded delivery rates include workload depletion and startup, so the experiment reports redistribution rather than an isolated throughput effect.

6.5 Loaded DQL Migration During Pod Scale-Out

We next tested selective connection replacement during event delivery in a separate Oracle Cloud cluster run. One client attached 1,000 registered subscriptions to four persistent connections on one subscriber Pod, with 250 subscriptions per connection. The isolated deployment then grew from one to four Ready Pods while the same client process remained running. The second Pod became Ready before standby activation was released; the third and fourth Pods were added during observation. The transport load balancer assigned each empty probe to a Pod. The client used the retained socket only when DQL selected migration and the execution guards passed.

The frozen checkpoint had 9,985 training updates and used $\epsilon = 0$. The trial allowed three executed actions in a configured 600-second window, with a 60-attempt probe-discovery budget. Telemetry had to satisfy the 180-second age limit and 0.8 minimum backlog coverage. Each Pod requested 1.5 CPU cores and 4 GiB memory, with limits of three cores and 6 GiB. The client used 32 workers and a fixed one-second processing delay to keep work outstanding during handoff. This setting tests acknowledgement-aware migration under a bounded client drain rate, rather than maximum Pod capacity.

Two bounded batches each published four synthetic source events. Each event matched 5,000 registered subscriptions, including the client's 1,000-subscription cohort. Thus, publication exposed the queue to 40,000 copies while the measured cohort accounted for 8,000 delivery attempts. Capacity was checked before each batch against the existing 400 MB limit, retaining an 80 MB reserve and prior unreconciled publication costs. No queue purge or quota increase was used.

Table 6. Executed frozen-DQL handoffs in the loaded Oracle Cloud scale-out run. P1 is the initial Pod; P2 and P3 are added Pods. Drain time is included in total handoff time. These are three actions within one run.

Action	Path	Subscriptions	Other sessions	Drain (s)	Total (s)
H1	P1 to P2	250	3 retained	93.862	97.555
H2	P1 to P2	250	3 retained	8.339	10.881
H3	P2 to P3	250	3 retained	76.166	79.657

Table 6 records actual policy execution, not just a migration recommendation. The three observed differences between the migration and stay Q-values were 1.7126, 1.2919 and 1.8201. Each migration was admissible and selected by the frozen network. Before-and-after session maps show that every handoff preserved the other three active connections. Two original sessions remained unchanged throughout the run. The mechanism therefore replaced selected connections without disconnecting the whole pool.

Figure 10(a) separates moved subscriptions from handoff operations. H3 moves the group already transferred by H1, so the three actions comprise 750 subscription-handoff operations involving 500 distinct subscriptions. The final distribution across the four available Pods is 500, 250, 250 and zero. The maximum client population on one Pod falls from 1,000 to 500, a 50% reduction, without increasing the four active connections. The last Pod remains a probe candidate after the three-action budget is reached. Discovering it does not automatically promote it.

Figure 10(b) uses the confirmed ACK receipt timestamp of each recorded delivery. Its cumulative total reaches 8,000, and each of the 1,000 registered subscription hashes has eight acknowledged delivery records. The ledger contains zero recorded ACK failures, processing failures, duplicate candidates or unresolved deliveries at clean closure. Long drain intervals remain visible: source retirement waits for outstanding work instead of discarding it to shorten the measured handoff. The configured window contains 60 decision records; its actual measurement boundaries span 609.716 seconds, including boundary handling.

The identity check uses STOMP message identifiers and confirms observed delivery/ACK accounting. Source CloudEvent identities and individual backend queue deletions are outside that ledger's scope. A separate unloaded 1,000-subscription run executed two handoffs in 5.244 and 5.172 seconds; those timings are retained separately because the processing conditions differ. The online and target weights remained identical to the input checkpoint in both runs. The loaded run saved 39 valid experience transitions without applying new training updates. The complete private archive verifies 256 files, and the manuscript figure uses identifier-free aggregates derived from those records. The isolated deployment was restored to one Ready Pod; all 13 shared-resource isolation checks passed.

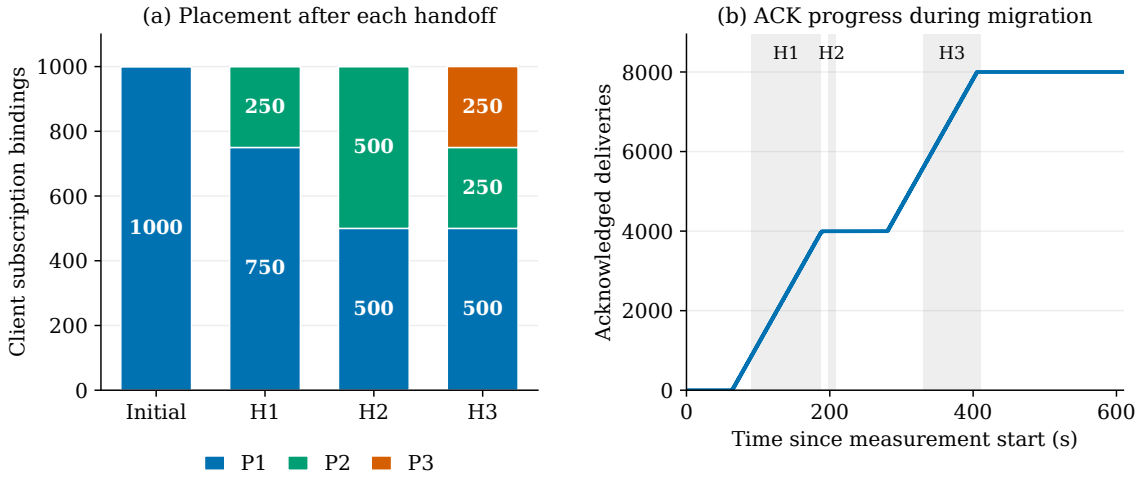


Fig. 10. Measured loaded DQL migration in the Oracle Cloud cluster. (a) Receipt-confirmed client bindings at the initial snapshot and after H1–H3. Each bar accounts for all 1,000 subscriptions; the four active connections each retain 250 subscriptions. P4 has no client bindings and is observed only through a probe. (b) Cumulative ACK receipts, aggregated in one-second bins. Shaded spans show the measured complete handoff intervals, including source drain. The curve describes this bounded workload, not a matched between-policy throughput comparison.

6.6 TTL-Based Reconnection Under Delivery Load

Time-to-live (TTL) reconnection is an existing alternative to client-directed connection replacement in the subscriber service. It was disabled in the preceding Oracle Cloud cluster experiments and enabled in the isolated recovery trials reported below. The deployment team reported that enabling age-based disconnection under event load caused widespread delivery interruptions and slow resubscription: delivery restarted on early-restored subscriptions while the remaining subscriptions were still being attached. These operational reports provide context rather than a frequency or severity estimate. The matched-preload trial in Section 6.6.5 measures expiry-related interruptions and ACK completion using server-correlated records.

6.6.1 Implemented Expiry and Recovery Paths. The server enables expiry only when the configured connection TTL is positive. For a connection established at time t_c^{open}, its nominal expiry is

$$t_c^{\text{exp}} = t_c^{\text{open}} + T_{\text{TTL}} + 60J_c, \qquad J_c \in \{1, \ldots, 60\},$$

where T_{TTL} is in seconds and J_c is the implementation's uniformly sampled integer jitter in minutes. A scheduled scan runs with a 60-second fixed delay. An expired connection is submitted to a cleanup executor, its server-side subscription associations are removed, and its socket is closed. Scheduling and cleanup add delay beyond t_c^{exp}; the scan interval is not a hard upper bound on closure latency. Several expired connections can be selected in one scan, although the random jitter reduces deliberate synchronization. The implementation does not imply that every client connection expires simultaneously.

The Python client reconnects the affected socket through the transport load balancer and resends the subscriptions retained for that socket. Recovery applies to the affected connection. It does not automatically restart the whole client

pool. Each subscription retains its previous delivery state. Consequently, an active subscription requests active delivery immediately on resubscription; there is no whole-connection barrier that first restores every subscription and then releases their events together. Transport reconnection can therefore finish well before subscription restoration.

This sequence explains the two distinct concerns in the operational report. First, the expired socket cannot deliver while it is being replaced, and overlapping recovery intervals can interrupt a large fraction of the pool. Second, early-restored subscriptions can resume event delivery while other `SUBSCRIBE` operations are still outstanding. Delivery and restoration then overlap in the same service. The server's cleanup also traverses shared subscription state under a write lock. These operations can compete for shared resources. Their effects on throughput and latency depend on workload and cannot be inferred from connection age alone.

Table 7. Implemented TTL reconnection versus guarded DQL migration. This compares control mechanisms, not measured performance rankings.

Property	Existing TTL strategy	Guarded DQL migration
Trigger	Age, jitter and periodic scan	Observed pressure, Q-values and feasibility guards
Destination	Reconnect assigned by load balancer	Existing load-balancer-assigned probe evaluated before promotion
Replacement	Cleanup and close before reconnect	Target standby before source quiescence and drain
Subscription restart	Retained active subscriptions resume individually	Receipt-gated handoff and controlled target activation
In-flight work	No DQL drain barrier in TTL closure path	Source drain checked before retirement
Scope	Every connection eligible by expiry time	Bounded selected connection; other sockets retained

Table 7 identifies the intended advantage of probe-guided migration: destination evaluation and subscription handoff happen before the old socket is retired. TTL can also reach a newly added Pod, but selects by connection age rather than the destination's subscription or backlog pressure. DQL evaluates that pressure and controls source retirement; its handoff duration still includes source drain and target activation, so delivery can pause during the transfer.

6.6.2 Recovery Endpoints and Verification. Let d_c be the observed disconnect time, h_c the replacement `CONNECTED` time, and r_c the time when all retained subscriptions have successful replacement-session receipts. The two recovery endpoints are

$$T_c^{\text{conn}} = h_c - d_c, \qquad T_c^{\text{restore}} = r_c - d_c.$$

If recovery is incomplete at the next disconnect or the end of observation, we record only the elapsed waiting time. This is a right-censored observation, not a successful fast reconnect. Let s denote elapsed trial time in seconds, and let $I_c(s)$ equal one while connection c is disconnected or its retained bindings remain unrestored, and zero otherwise. With C_0 denoting the active connection pool at the start of a TTL trial, the affected fraction is

$$F(s) = \frac{1}{|C_0|} \sum_{c \in C_0} I_c(s).$$

The peak of F and its time integral distinguish a brief single-socket outage from widespread or prolonged recovery. Delivery-rate degradation on sockets that remain connected is measured separately; it is not counted as a physical

disconnect. We also measure unique receipt-confirmed ACK rate, gaps between deliveries, pending subscription count, and receipt latency per subscription. These show the effects on both event delivery and subscription operations. Events delivered before r_c directly establish overlap between delivery and incomplete restoration, but overlap alone does not prove CPU or network saturation.

The recovery instrumentation requests receipts for reconnect subscriptions without serializing their dispatch. It preserves pending caller Futures, distinguishes successive sessions, and stores timestamps alongside the exact delivery ledger. A common instrumentation contract defines the measurements across comparison arms. Client disconnects are attributed to TTL only when matched to server expiry records; unrelated network failures remain separate.

The implementation-level checks exercise the disabled TTL branch, expired connection closure, and selection of expired rather than unexpired sockets in a common scan. Client checks cover receipt ownership, unchanged delivery states, interrupted recovery, and censored timing analysis. These checks validate implementation behavior. The live comparisons reported below measure expiry-related interruptions, receipt completion, and recovery timing using the native jitter. This separates observed behavior from an artificial simultaneous-disconnection scenario.

6.6.3 Recovery Observed Within the Existing Storage Limit. The isolated recovery trial retained the installed 400,000,000-byte Queue limit and consumed existing experimental events without a storage increase or new publication. Four clients attached 1,250 subscriptions each to one subscriber Pod, giving 5,000 receipt-confirmed standby subscriptions across 20 delivery connections and four probes. Each delivery connection held 250 subscriptions. Activation then released delivery with a bounded window of 16 outstanding requests per client. Native TTL used a configured value of 60 seconds with the original 1–60-minute jitter and scheduled scan; no simultaneous-disconnection rule was introduced.

A sealed client record captures a complete recovery of 250 retained subscriptions. The replacement socket reached `CONNECTED` in 0.018 seconds, whereas receipt-confirmed restoration took 104.174 seconds. During that restoration interval, 82 delivery attempts arrived on the replacement session. Thus, transport recovery did not mean that the subscription population was ready, and delivery demonstrably overlapped with restoration. The corresponding server expiry record was not retained. We therefore report this as recovery during a TTL-enabled trial, not as a disconnect independently confirmed to have been caused by TTL. A second client record does match an explicit server TTL-close request by session identity and records the subsequent disconnect; this partial run confirms that native expiry occurred, but it is not a completed performance trial.

The TTL-enabled execution ended during activation after a receipt timeout. One client sealed its evidence; the supervisor's earlier cleanup deadline interrupted the other three before sealing. These records are retained as diagnostics rather than treated as four successful repetitions. ACK errors also occurred on connections without a recorded disconnect, so they are not all assigned to TTL. The observed recovery sequence supports distinguishing socket readiness, subscription readiness and delivery activity; it does not by itself estimate a TTL-versus-DQL throughput effect.

After restoring TTL to its disabled state, a functional control used the same 5,000-subscription population and existing events. Its four verified client records contain 117,269 delivery attempts, 110,411 receipt-confirmed ACKs and 6,858 unresolved attempts. No active-connection interruption was recorded, but activation ended after an explicit upstream state-transition failure exhausted its retry budget; 2,929 bindings had confirmed active state in the last progress records. This distinguishes the expiry/recovery mechanism from an activation bottleneck that also occurs without TTL. Because the first execution had already consumed part of the workload, the two runs do not provide a matched-backlog

throughput comparison. Both preserve the frozen model without additional training. The isolated Deployment was restored, and all thirteen shared-resource isolation checks passed.

6.6.4 Native TTL During Two Delivery Bursts. Before the preload trial, a separate live pair tested native TTL against a static TTL-disabled control on one isolated Oracle Cloud subscriber Pod. Both arms used 5,000 subscriptions, five clients, 20 occupied connections and five empty probes, with zero artificial client processing delay. Each started from an empty Queue and received two bursts of three accepted source events, giving 30,000 recorded delivery attempts per arm. Both used the same server image and 900-second post-activation observation. Native TTL retained its original jitter. The control replayed the accepted burst schedule: its first requests were 0.462 and 0.578 seconds later than the native relative targets of 26.182 and 319.207 seconds. Queue-statistics budget failures stopped other publication blocks before any event attempt; those failures were retained.

Table 8. Oracle Cloud two-burst TTL comparison, separate from the subsequent preload pair. Both arms use static placement on one Pod; TTL off is not a learned DQL intervention. Latency statistics include confirmed ACKs only.

Measurement	Native TTL	Static TTL off
Delivery attempts	30,000	30,000
Receipt-confirmed ACKs	29,631	30,000
Unresolved ACK receipts	369	0
Server-matched TTL interruptions	5	0
Confirmed ACK latency: median (s)	19.439	21.407
Confirmed ACK latency: 95th percentile (s)	238.392	55.826
Clients with receipt-complete shutdown	2 of 5	5 of 5

Table 8 shows the tail that a median alone would miss: native TTL's confirmed-ACK p95 was 4.27 times the control's, although its median was 1.968 seconds lower. Five interruptions matched server expiry records. Transport reconnection took 17–24 ms, whereas restoring all 250 bindings on each affected connection took 0.504–0.973 seconds. One recovery received 22 delivery callbacks before all binding receipts arrived. This directly demonstrates delivery overlapping resubscription within the observed short recovery windows; prolonged stalls and whole-pool expiry were not observed. All 369 unresolved attempts belonged to expired sessions; three clients reached the 120-second receipt-drain deadline. Missing receipts are not counted as lost Queue events. All ten sealed client bundles were recovered and verified. This single sequential pair measures TTL-related disruption and confirmed-receipt latency under static placement. It estimates neither repeated-run capacity nor a DQL-specific effect.

6.6.5 Matched-Preload Comparison in the Oracle Cloud Cluster. A subsequent isolated trial compared native TTL with the frozen DQL controller under the same initial queued workload. Each arm used the same 5,000 registered ET1 subscriptions: five clients held 1,000 subscriptions each on four occupied connections, with 250 subscriptions per connection and one empty probe per client. All 20 occupied connections initially terminated on one subscriber Pod. Three identical Pods became Ready before activation. Each Pod had a limit of three CPU cores and 6 GiB memory, and both arms used the same pinned server image. The installed Queue limit remained 400,000,000 bytes; shared subscriber resources were not changed.

All bindings were attached in standby before six source events were published through the authenticated ingest path. A pre-activation Queue snapshot verified 30,000 visible copies and zero in-flight copies in each arm. The recorded

delivery ledger subsequently contained six attempts for every subscription in the same 5,000-subscription cohort. Native TTL retained the 60-second base, 1–60-minute random jitter and scheduled scan. The DQL arm disabled TTL and loaded unchanged online and target networks from the checkpoint with 9,985 training updates; exploration and further learning were disabled. Observation continued for 900 seconds after activation, with activation itself also recorded.

The original workload included a six-event refill at five minutes. Its first Queue-statistics budget check returned HTTP 429 before any event was attempted. Before the DQL arm started, the comparison was amended to use the six-event preload without refill. Three DQL preload budget checks also failed before publication; cooldown and fresh checks admitted the remaining events while all bindings stayed in standby. The initial copy counts therefore matched, but the setup proceeded at different rates. Queue occupancy was approximately 20.3 MB in both arms, not 400 MB. This sequential pair measures fixed-preload drainage. Sustained arrivals and subscriber saturation are outside that workload.

6.6.6 Receipt Completion and Connection Recovery. Table 9 reports the observed outcomes. Let D_{obs} be the number of recorded delivery attempts and A_{conf} the number with a confirmed ACK receipt. Receipt completion is $A_{\text{conf}}/D_{\text{obs}}$; the remainder is unresolved, not classified as event loss. Both arms recorded 30,000 attempts. Native TTL confirmed 26,831 ACKs (89.44%), compared with 30,000 (100%) in the TTL-disabled arm, a difference of 10.56 percentage points in this pair. The latter arm is labelled *guard-held DQL*: its controller ran, but telemetry guards held every decision at *stay*, as explained below.

Table 9. Measured Oracle Cloud outcomes for the matched 5,000-subscription preload. The DQL arm had TTL disabled and executed no migration. Latencies are conditional on confirmed ACKs; unresolved receipts are reported separately.

Measurement	Native TTL	Guard-held DQL
Delivery attempts	30,000	30,000
Receipt-confirmed ACKs	26,831	30,000
Unresolved ACK receipts	3,169	0
Receipt completion (%)	89.44	100.00
Server-matched TTL interruptions	12	0
Distinct occupied sockets interrupted	11 of 20	0 of 20
Peak simultaneously recovering sockets	3 of 20	0 of 20
Incomplete-restoration exposure (connection-s)	10.983	0
Confirmed ACK latency: median (s)	69.862	69.836
Confirmed ACK latency: 95th percentile (s)	251.546	249.874
Executed DQL migrations	Not applicable	0

All twelve native interruptions matched server expiry records by session identity and disconnect time. They affected eleven distinct occupied sockets, with at most four interrupted within a rolling minute. At most three sockets were concurrently unrestored, giving $\max_s F(s) = 3/20 = 0.15$ and $\int \sum_c I_c(s)\, ds = 10.983$ connection-seconds. Complete binding restoration took 0.492–1.835 seconds. These observations show overlapping recovery periods affecting part of the pool, rather than a whole-pool outage. All 3,169 unresolved native attempts belonged to sessions with matched expiries, but a missing ACK receipt does not reveal whether the underlying Queue deletion succeeded.

Figure 11 retains the unresolved attempts in the final level rather than rescaling each curve using only successful attempts. Time starts when all five client observation windows overlap within each arm. Those common windows lasted 931.347 and 937.743 seconds, respectively. Their mean confirmed-ACK rates were 28.809 and 31.992/s, including idle time; these are whole-window completion rates, not estimates of maximum Pod bandwidth. Four native clients

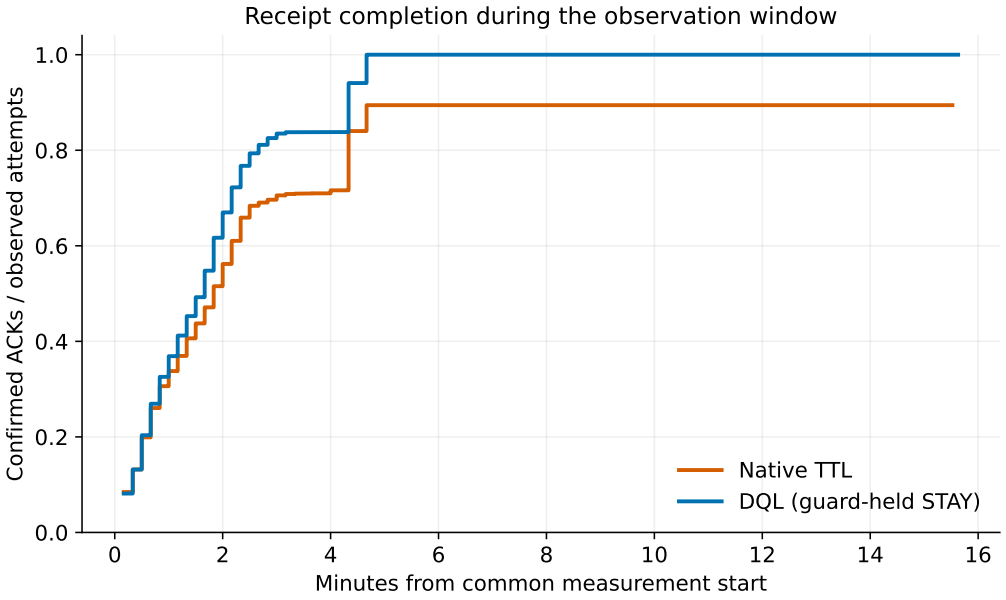


Fig. 11. Receipt completion in the Oracle Cloud matched-preload pair. Each curve accumulates confirmed ACKs in ten-second bins and divides by all 30,000 observed attempts in its arm. Unresolved attempts remain in the denominator. DQL denotes the guard-held, TTL-disabled execution, not a completed migration comparison.

reached the 120-second final ACK-drain deadline. We retain their incomplete drains and the coordinator's nonzero exit status as outcomes. All ten client evidence bundles were sealed and verified; the DQL coordinator completed cleanly.

Figure 12 separates receipt completion from conditional latency. The confirmed medians differ by only 0.026 seconds, and the 95th percentiles are also close. Native unresolved receipts are absent from this latency distribution but remain explicit in Table 9. Violin width describes within-run latency variation. The five clients and 30,000 attempts belong to one pair rather than independent trials, so between-run uncertainty is not estimated. Unlike the earlier 104.174-second diagnostic, this pair recorded no new delivery callbacks during subscription-restoration windows. It captures a different recovery condition from resubscription under concurrent delivery.

6.6.7 Telemetry Freshness and Action Eligibility. The DQL trace contains 450 decision intervals, all held at *stay* by `invalid-backlog-telemetry`. No migration executed, and all 5,000 bindings remained on the source Pod despite four Pods being Ready. Statistics collection paced individual subscription requests at 40 ms for a 5,000-subscription sweep. Even before network delay or retries, the sweep required at least $5{,}000 \times 0.040 = 200$ seconds, exceeding the 180-second oldest-sample deadline. The minimum coverage across the Pod and occupied-connection components ranged from 0.560 to 0.808. Only ten observations met every 80% coverage threshold; their oldest constituent samples were 183.127–185.974 seconds old and failed freshness. A recently received telemetry frame can therefore contain stale Queue observations. Queue-statistics rate limits further constrained coverage.

This finding separates two operational properties. Native expiry produced measurable interruptions and receipt uncertainty; the TTL-disabled, guard-held arm preserved its occupied sockets and completed all observed ACKs. Freshness checks held migration inactive, so the measured benefit concerns connection retention and receipt completion,

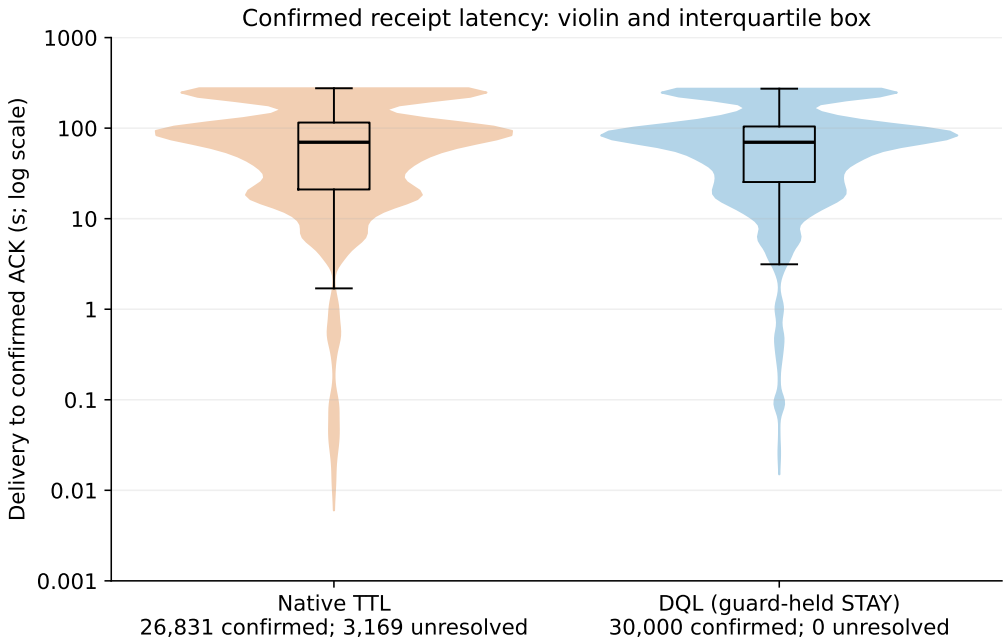


Fig. 12. Confirmed delivery-to-ACK latency in the Oracle Cloud preload pair. Violin density is estimated on $\log_{10}$ latency; boxes show its median and interquartile range, with 1.5-IQR whiskers. Outlier markers are suppressed, not removed from the observations. Counts exclude unresolved receipts.

rather than throughput from added Pod capacity. The checkpoint and guards stayed fixed throughout the trial. Cleanup verified empty Queue and dead-letter Queue inventories, restored isolated TTL to zero and one Ready Pod, and passed all thirteen shared-resource isolation checks.

7 EXPERIMENTAL METHOD AND LOCKED SIMULATOR EVIDENCE

We evaluate recovery after subscriber-pod scale-out using 20,000 subscriptions, ten persistent connections, and 775 million queued events. The primary comparison measures the use of added pod capacity with guarded-greedy and DQL-guided migration against retaining the initial connection placement. Supporting studies examine multi-objective control, action selection, and adverse probe promotion.

7.1 Recovery After Subscriber-Pod Scale-Out

The primary recovery experiment models the capacity increase from one subscriber pod to four. Ten persistent TCP connections are established on the original pod, carrying 20,000 subscriptions. At the measurement boundary, three additional ready pods provide processing capacity, while the ten established connections retain their original endpoints. The comparison therefore measures how effectively the client uses the added capacity through probe promotion and subscription reassociation.

Four equal subscription groups begin with 100,000, 50,000, 5,000, and zero queued events per subscription, respectively. The initial backlog is $B(0) = 775{,}000{,}000$ events. Round-robin assignment gives each connection 2,000 subscriptions and

Table 10. Persistent-connection recovery after scaling from one to four subscriber Pods. Twenty paired seeds each start with 20,000 subscriptions and 775 million queued events on ten TCP connections pinned to Pod 0. Lower backlog measures are better.

Policy	Backlog AUC	Final backlog (%)	Throughput (k/s)	Active Pods	Migrations
Frozen DQL	0.8376	65.97	219.8	4.00	7.75
Kubernetes scale-out only	0.9551	91.02	58.0	1.00	0.00
Scripted probe migration	0.8359	65.74	221.3	4.00	7.75
DQL vs. scale-out only: 12.30% lower backlog AUC; 278.9% higher mean throughput.					

77.5 million queued events. Consequently, $V_u^N(0) = V_u^G(0) = 0$: the initial bottleneck is concentrated pod placement. This setup isolates whole-connection migration from pre-existing imbalance between the client's connections.

The model assigns each pod a processing threshold of 100,000 events/s and the client a base acknowledgement capacity of 250,000 events/s. Its bounded connection-parallelism factor is three at ten connections. Service is limited by both client capacity and each connection's share of its pod capacity. The simulator's pressure-dependent service factor is $\max\{0.35, 1 - 0.42P_i^{\mathrm{sim}}\}$, where $P_i^{\mathrm{sim}} \in [0, 1]$ is the simulator's clipped load score, distinct from the deployable pressure in (21). These numerical rates define the simulator's service assumptions rather than measured Pod benchmarks. Pressure is sampled before the action and refreshed after the interval. For the clipped score, this service multiplier lies in $[0.58, 1]$: the model represents pressure-degraded service, not complete Pod service collapse. Arrival rates and processing rates are converted to event counts using the declared $\Delta = 10$ seconds. The drain phase introduces no new arrivals and lasts 120 intervals, or 20 minutes.

We compare *Frozen DQL*, *Kubernetes scale-out only*, and *Scripted probe migration*. All three receive the same capacity increase. The second policy retains the existing connection assignment. The third promotes every candidate admitted by the migration guard and evaluates the framework's deterministic guarded-greedy policy. DQL uses the previously frozen checkpoint without exploration or training. Adaptive policies can replace at most 12 non-improving probes, and actions have a three-interval cooldown. The scripted control uses no learned Q-values: it shares the pressure-aware admission rule and source selection with DQL. We therefore distinguish this *guarded-greedy* policy from the independent ad-hoc schedules below. Its mean throughput is 221,260 events/s, compared with 219,762 for DQL. These are two effective selectors for the shared rebalancing mechanism, not evidence that Q-learning is necessary to use added capacity. The original registered primary contrast remains DQL versus unchanged placement; this broader interpretation does not change that protocol or relabel a post-hoc contrast as confirmatory. An exploratory paired reanalysis of these same twenty seeds gives a greedy-minus-DQL mean throughput difference of 1,498 events/s (pointwise percentile bootstrap 95% interval [362, 2,852], 10,000 resamples). This small rate advantage belongs to the greedy policy in this workload; the mixed-workload study evaluates DQL's separate multi-objective benefit.

For remaining backlog $B(t)$, the primary endpoint is the mean normalized backlog over the recovery window,

$$\mathrm{AUC}_B = \frac{1}{T}\sum_{t=1}^{T}\frac{B(t-1) + B(t)}{2B(0)}, \qquad T = 120. \tag{49}$$

Lower AUC_B indicates less accumulated queued work. Twenty fresh paired seeds vary probe placement while holding the workload fixed. We use 10,000 paired bootstrap resamples and 100,000 sign-randomization draws. The registered criterion requires the upper 95% confidence bound of the DQL-minus-baseline AUC difference to be below zero and the one-sided p-value to be below 0.05. These intervals describe variation caused by probe placement within this model.

Table 10 shows that DQL raises mean throughput from 58,000 to 219,762 events/s, a factor of 3.79. It reduces backlog AUC from 0.95510 to 0.83759, an improvement of 12.30%. The paired AUC difference is −0.117504 (95% confidence interval $[-0.121210, -0.113345]$; one-sided $p \simeq 10^{-5}$). DQL leaves 65.97% of the initial backlog after 20 minutes, compared with 91.02% for scale-out only, or 27.52% less remaining backlog relative to that baseline. All 20 DQL episodes use all four pods by the end, with a mean of 7.75 connection migrations. The baseline retains all ten connections on the original pod.

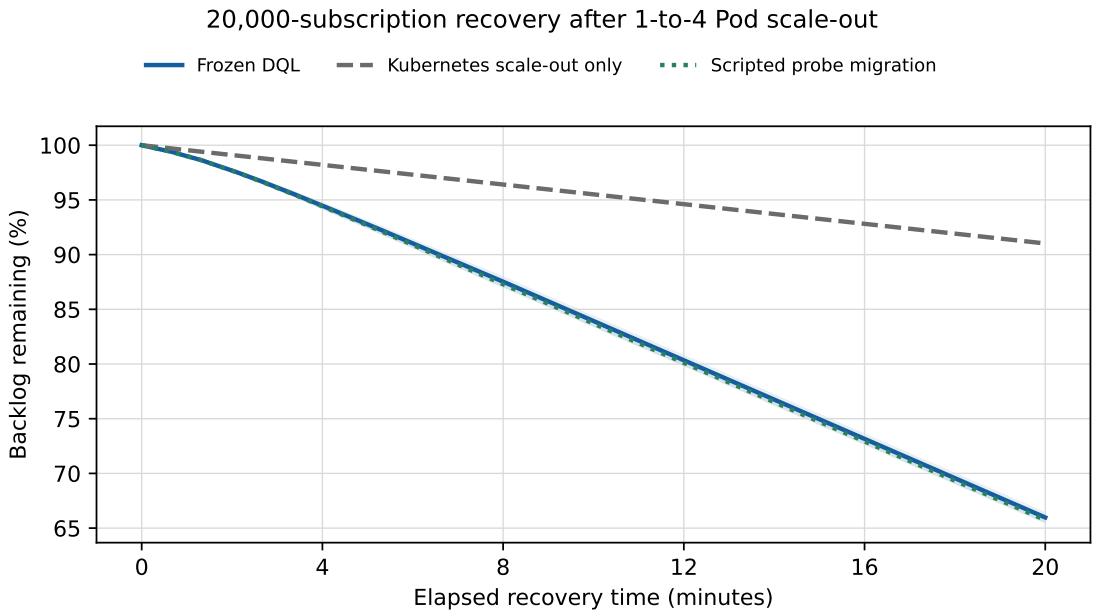


Fig. 13. Backlog recovery after adding three subscriber pods. All policies start with the same 775-million-event backlog. Curves are means over 20 paired seeds; shaded bands are descriptive 95% normal-approximation intervals. The area under each normalized curve defines the primary endpoint.

Figure 13 connects the endpoint to the recovery process. The steeper DQL curve follows successful probe promotions that place delivery work on additional pods. The nearby scripted-control curve shows that DQL reaches about the same service capacity as guarded scripted migration. Both improve on the unchanged placement, establishing the benefit of migration in this workload rather than a ranking above every controller with the same actions.

Figure 14 provides the mechanism evidence: the throughput increase occurs as more pods acquire subscription-bearing connections. Added pod capacity and client-directed reassociation jointly produce the gain. In the model's saturated regime, a single pod supplies approximately 58,000 events/s; four utilized pods supply approximately 232,000 events/s. The mean over the full recovery period is lower because probe discovery and cooldown take up part of the observation window. The maximum relative difference in queue accounting is 1.45×10^{-13} across the 60 episodes. The simulator represents reassociation as a discrete action; network delays and the live receipt protocol are outside this timing model.

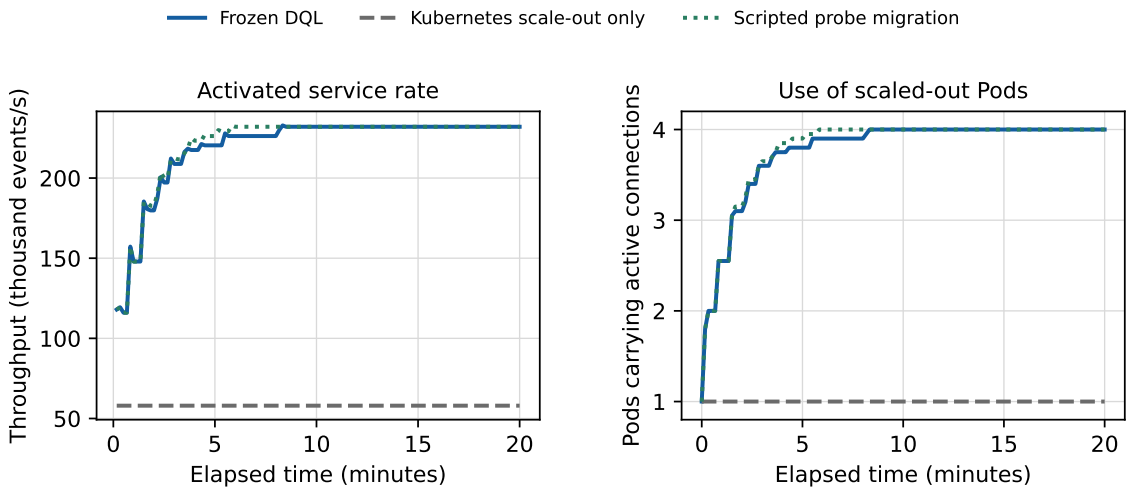


Fig. 14. Use of the added subscriber capacity. Left: mean delivered events per second over each measured interval. Right: mean number of pods carrying active client connections. The ten delivery connections are retained as a bounded pool while probe promotion replaces their endpoint assignments.

Figure 15 makes the seed-to-seed variation behind the recovery averages visible. Each AUC observation is the episode-level quantity in Eq. (49); each throughput observation is the episode's delivered-event total divided by its 1,200-second duration. The violin uses a Gaussian kernel with Scott's bandwidth, $h = s_{\text{sample}} n_{\text{ep}}^{-1/5}$ for sample standard deviation s_{sample} and $n_{\text{ep}} = 20$, and is drawn only between the smallest and largest observations. Its width is descriptive, not a confidence interval. The scale-out-only outcomes are constant because that policy never changes its initial placement. The adaptive distributions reflect variation in probe discovery and migration timing. Their separation from the baseline agrees with the paired AUC result above. The overlapping DQL and scripted distributions describe similar recovery performance for these two policies, without establishing DQL superiority over the guarded scripted control.

Independent ad-hoc migration. An exploratory extension retains all sixty original observations and adds eighty episodes: four pressure-unaware schedules evaluated on the same twenty paired seeds. Each attempt chooses a connection and target Pod uniformly, without consulting backlog, pressure, or the learned policy. A same-Pod draw remains a no-op. The schedules use fixed periods of 40 or 60 seconds, or a 60-second nominal period with ±30% or ±40% uniform timing jitter. All schedules first act at interval zero. The nominal jitter ranges are 42–78 and 36–84 seconds; rounding upward to the ten-second decision clock gives executable ranges of 50–80 and 40–90 seconds, respectively. These percentages modify the schedule, not the measured throughput. Rounding also changes the average attempt frequency, so jitter is not isolated from timing. In the following comparison, *Scripted* denotes the independent 60-second schedule with ±30% timing jitter; *guarded-greedy migration* denotes the original pressure-aware positive control. These are distinct non-learning policies.

Table 11 shows mean DQL throughput of 219,762 events/s, compared with 195,737 for the ±30% schedule and 195,640 for the ±40% schedule. The relative gains are 12.27% and 12.33%, with pointwise paired 95% bootstrap intervals of [9.38%, 15.55%] and [9.31%, 15.81%], respectively. Each interval uses 10,000 resamples of complete seed pairs and the

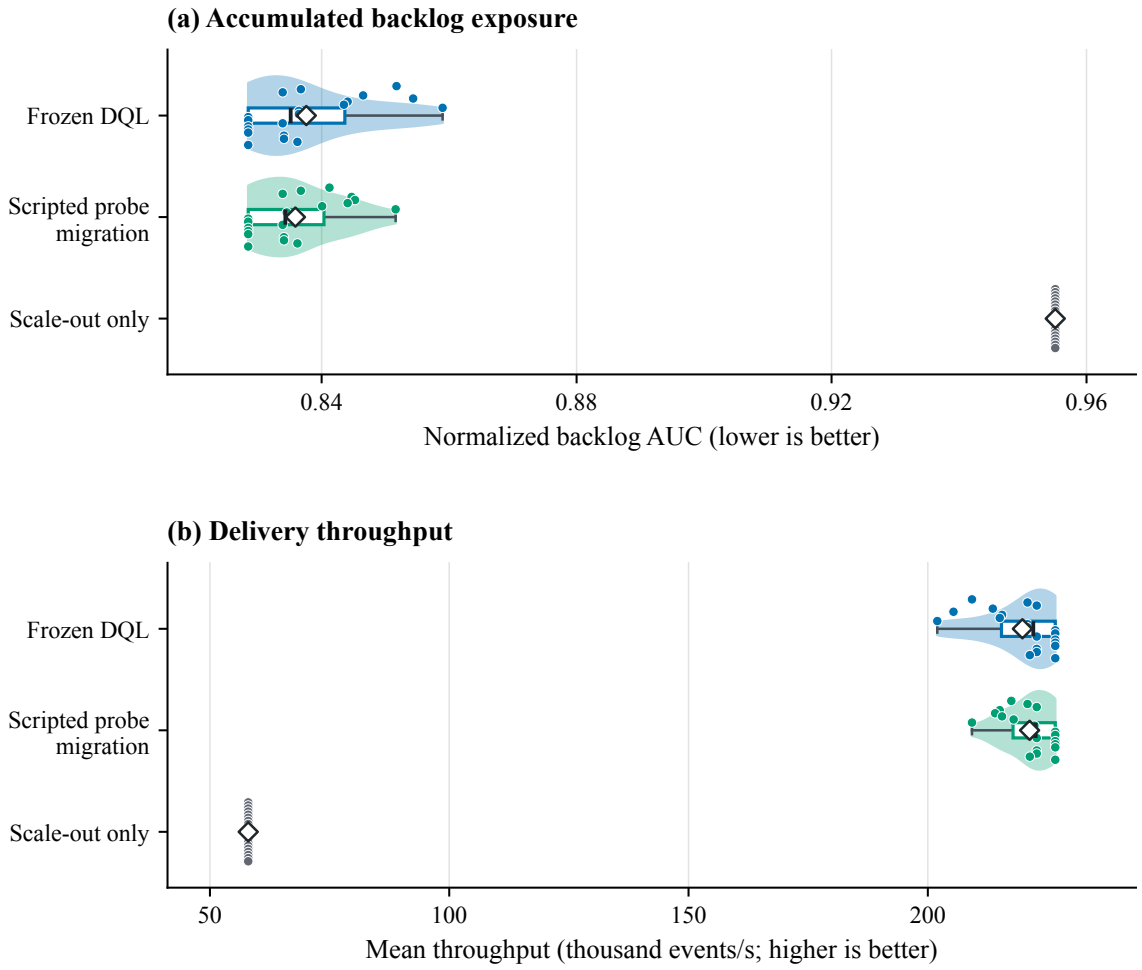


Fig. 15. Scale-out recovery distributions over 20 paired seeds per policy. Violin width shows a kernel-density estimate within the observed range; boxes show quartiles and medians, diamonds show means, and points show all episodes. Whiskers use the 1.5-interquartile-range rule. The constant scale-out-only baseline is shown without an artificial density. Lower normalized backlog AUC and higher throughput are preferred.

Table 11. Independent ad-hoc migration sensitivity in the scale-out simulator. Means over twenty paired seeds per policy; timing jitter changes the schedule, not measured values.

Policy	Events/s	Backlog AUC	Migrations	Final Pods
Frozen DQL	219,762	0.83759	7.75	4.00
Guarded-greedy migration	221,260	0.83589	7.75	4.00
Ad-hoc: fixed 40 s	205,295	0.84997	22.85	3.70
Ad-hoc: fixed 60 s	197,636	0.85906	15.40	3.90
Scripted	195,737	0.86064	14.70	3.85
Ad-hoc: 60 s, ±40%	195,640	0.86053	14.65	3.85
Scale-out only	58,000	0.95510	0.00	1.00

ratio of resampled mean rates. DQL uses 7.75 migrations per episode, compared with 14.70 and 14.65 for the two jittered schedules. The fixed 40-second and 60-second schedules yield gains of 7.05% and 11.20%, so the comparison covers both fixed and jittered schedules. This exploratory sensitivity analysis reuses the scenario seeds. Its scope is simulated migration scheduling, separate from independent holdout testing and live connection-TTL measurements.

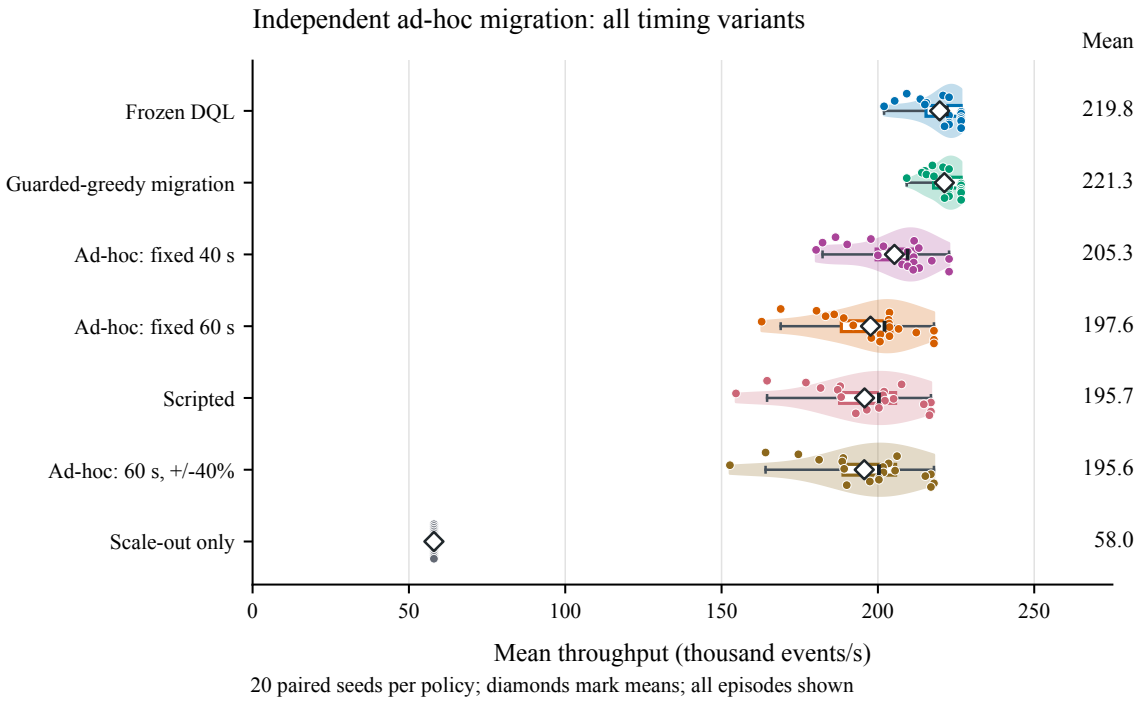


Fig. 16. Throughput under independent ad-hoc migration schedules. All twenty episodes per policy are shown. Violin density uses Scott's bandwidth within the observed range; boxes show quartiles and medians, whiskers use the 1.5-interquartile-range rule, and diamonds mark means. The constant scale-out-only control has no artificial density. All seven policies are retained, including the original guarded-greedy control.

Figure 16 shows the variation hidden by the averages. Random reassignment can revisit an already occupied Pod or move work away from a useful placement, whereas the guarded DQL controller admits pressure-improving candidates. The distributions and migration counts compare the complete controllers, including their admission and endpoint-selection rules. The nearby guarded-greedy distribution shows that the result compares complete policies, including their guards, rather than isolating the effect of learned action selection.

Figure 17 summarizes the same stressed workload, rather than importing the mixed-workload radar's averages. Its throughput coordinate divides each episode's mean rate by the four Pods' combined nominal capacity, $4 \times 100{,}000 = 400{,}000$ events/s. The other coordinates are the fraction of initial backlog removed, the final active-Pod count divided by four, and $10/(10 + M^{\mathrm{ep}})$ for an episode with M^{ep} migrations. The last coordinate decreases with intervention count; it is not a handoff-latency measurement. Each plotted point averages these per-episode coordinates across all twenty seeds. No observed-best or policy-specific denominator is used. The ±30% ad-hoc schedule appears alongside the original three controls; all timing variants remain in Table 11.

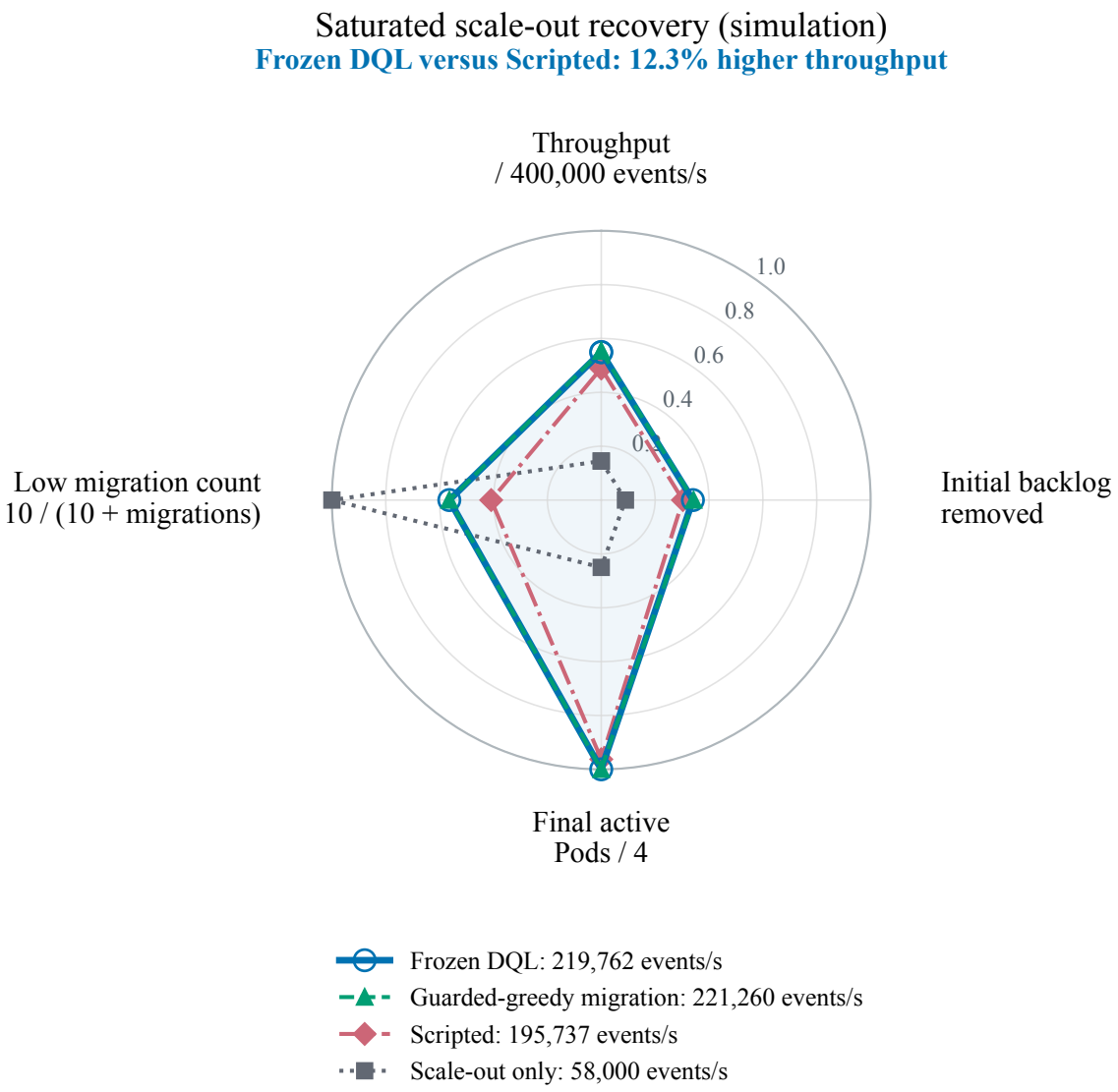


Fig. 17. Stress-scenario radar from all twenty paired simulator seeds per displayed policy. **Frozen DQL versus Scripted: 12.3% higher throughput**, with measured mean rates of 219,762 and 195,737 events/s, respectively. The guarded-greedy and scale-out-only controls remain visible at 221,260 and 58,000 events/s. All coordinates use fixed zero-to-one scales, with outward preferred; migration count exposes the intervention tradeoff. This is a descriptive profile, not an additional objective or statistical test.

The DQL throughput radius is 12.3% larger than the Scripted radius and 3.79 times the scale-out-only radius because all use the same capacity denominator. The guarded-greedy comparator has a nearly overlapping profile and slightly higher mean throughput. The result establishes the benefit of reaching added capacity, with similar performance from a pressure-aware non-learning control. Backlog removal and throughput are linked because the workload has no new

arrivals during drainage. They are not independent confirmations. The polygon's area is not a performance measure. These measurements remain distinct from the Oracle Cloud expiry and receipt-completion results in Section 6.6.5.

7.2 Supporting Study Design

The independent v3 holdout compares a frozen DQL policy with deterministic controls. The adverse-probe ablation holds that checkpoint fixed and varies the promotion guard, using separate 30-seed inference and five-seed scale cohorts. Expanded v4–v6 studies evaluate action coverage and contract changes; they are development diagnostics, not additional holdout performance results. Table 12 identifies these version boundaries. Appendix C retains their detailed outcomes.

Table 12. Checkpoint, action-contract, and execution scope. A policy trained under one action meaning is not evidence for another meaning.

Study	Policy / action contract	Scope of the evidence
v3 holdout and adverse-probe ablation	Frozen seed 101; legacy open-only scale-up	Reported rebalancing and multi-objective results; no cap-restoring claim
20,000-subscription scale-out	Same frozen checkpoint; fixed ten-connection pool	Endpoint migration; workload does not admit scaling actions
October loaded cluster trial	Same checkpoint; explicit rebalancing-only profile	Scaling actions disabled; three selective live migrations
v4–v5 development	Legacy scale-up; v5 corrects the scale-down guard	Action-coverage diagnostics; no selected holdout policy
v6 development	Versioned cap-restoring scale-up	Scripted action preflight and learner gates; no selected holdout policy

The shared frozen deployment file has SHA-256 prefix `3e68371d848eca85`; the reviewer evidence package retains the full hash and weights. A missing contract field identifies the legacy checkpoint. The current client permits it only for a compatible restricted action profile; full-action deployment requires the versioned contract.

7.3 Research Questions and Baselines

The evaluation addresses four questions. **RQ1** asks whether a frozen DQL policy improves the multi-objective return and connection distribution relative to a deterministic, two-dimensional heuristic. **RQ2** asks how the policy trades off the individual operating measurements alongside the scalar reward. **RQ3** asks whether independently trained candidates use every allowed action in its intended test scenario while respecting the fault guards. **RQ4** asks whether rejecting an adverse probe promotion reduces the amount of queued work over time compared with overriding that guard, blind promotion, static placement, and the guarded heuristic.

The v3 holdout compares three policies. *Static* retains the initial transport assignment. *Heuristic* applies the threshold-based subscription-count and backlog rules without value learning. *Frozen DQL* uses the checkpoint selected on the fixed development-validation bank, sets $\epsilon = 0$, disables training, and invokes the same action masks and execution guards as the deployed client. The heuristic is the primary baseline because it has the same rebalancing authority as DQL; the static policy is the no-rebalancing control for the initial connection assignment.

During each episode, the policy receives only client-deployable observations: local subscription queues, active-connection telemetry, interval throughput counters, and one retained probe candidate. Cluster-wide pod distributions

are calculated by the simulator only after an action for that interval has been chosen. They are evaluation outputs and are not inserted into the policy state. This maintains the partial-observability boundary formalized in Section 4.

7.4 Predeclared Protocol

Patterson et al. identify hyperparameter selection, baseline construction, and uncertainty reporting as central issues in empirical RL evaluation [39]. We therefore keep development selection separate from the locked holdout and report component metrics alongside return. Testing adaptation under changed execution conditions is also distinct from measuring performance in the training setting, as demonstrated by Biagiola and Tonella's plasticity analysis [4]. The operating and fault scenarios below test this distinction without treating successful training as evidence of universal adaptation.

Table 13 summarizes the evidence-generating runs. In v3, three candidate networks were trained for exactly 10,000 transitions under seeds 101, 211, and 307. Candidate selection used ten fixed validation seeds and required subscription-count rebalancing ($a^{\mathrm{reb}N}$), backlog rebalancing ($a^{\mathrm{reb}G}$), and full connection migration (a^{mig}) in at least 80% of admissible episodes in their intended scenarios. Candidate 0 (seed 101) passed this gate and was frozen before the holdout seed bank was read.

Table 13. Locked simulator study design. "Not opened" means that no performance episode was run on the reserved holdout seed bank.

Study stage	Training budget	Fixed evaluation	Selection requirement	Holdout status
v3 development	3×10,000 transitions	10 validation seeds; 4 scenarios	$a^{\mathrm{reb}N}$, $a^{\mathrm{reb}G}$, and a^{mig} each at $\geq$ 80% episode coverage	Not opened
v3 holdout	Frozen seed 101; no updates	10 new seeds; 4 scenarios; 3 policies; 120 episodes	No selection on holdout	Completed
v4 uniform replay	5×20,000 transitions	10 validation seeds; 7 operating and 4 fault scenarios	All five intervention actions at $\geq$ 80%, plus all safety guards	Not opened
v4 stratified replay	5×20,000 transitions	Same fixed v4 validation bank	Same all-action and safety gate	Not opened
v5 corrected guard	5×20,000 transitions	10 fresh validation seeds; same 11 scenarios	Same all-action and safety gate	Not opened
v6 scale-up preflight	No learner training	10 paired seeds; 500 intervals	Target metric and reward improve for every seed, plus safety gates	Not defined
v6 learner factorial	$4 \times 5 \times 20{,}000$ transitions	10 fresh validation seeds; same 11 scenarios	At least 4/5 candidates pass every action and safety gate	Not defined
Adverse-probe ablation	Frozen v3 checkpoint; no updates	30 paired inference seeds and 5 paired 20k scale seeds; 500 intervals	Lower paired backlog AUC for guarded DQL than forced promotion	Completed

The independent v3 holdout contains ten previously unused seeds for each of four scenarios and 500 decision intervals per episode. The connection-count skew scenario uses 1,000 subscriptions; the tiered-backlog scenario uses 1,200 subscriptions; and the client-slowdown and pod-pressure-skew scenarios each use 1,600 subscriptions. Every scenario has four modeled pods, four initial persistent connections, and a maximum of six connections. The tiered-backlog scenario assigns 25% of its queues to each initial backlog level 100,000, 50,000, 5,000, and zero. The client-slowdown scenario reduces acknowledgement capacity to 35% of its nominal value between intervals 125 and

375. The other two scenarios isolate initial subscription-count skew and the placement of three of four connections on one pod, respectively.

For policy π, paired episode j, and metric v, let $Y_{\pi,j,v}$ denote the episode mean. We define a direction-adjusted paired difference

$$D_{j,v} = d_v \left(Y_{\mathrm{DQL},j,v} - Y_{\mathrm{base},j,v} \right), \qquad d_v = \begin{cases} +1, & \text{higher is better,} \\ -1, & \text{lower is better.} \end{cases} \tag{50}$$

Thus, a positive $D_{j,v}$ always favors DQL. The relative benefit reported below is

$$\mathrm{RB}_v = 100 \, \frac{\overline{D}_v}{\left| \overline{Y}_{\mathrm{base},v} \right|}. \tag{51}$$

Overbars in these two equations denote means across paired episodes. The metric index v is distinct from the client-population size m. We resample the 40 paired differences 10,000 times to obtain percentile 95% confidence intervals. Two-sided paired sign-flip tests are exact for the scenario-level sets of ten pairs and use 100,000 Monte Carlo sign vectors for the pooled set of 40 pairs. Pooled p-values are adjusted by the Holm procedure across the complete family of 18 metric-by-baseline tests. The scenario box plots in Figure 19 describe quartiles and individual episode outcomes; the paired randomization tests provide the statistical comparison.

We evaluate DQL as a sequential controller through return, operational outcomes, paired effect sizes, and action coverage. Classifier accuracy and ROC–AUC apply to labelled prediction tasks. Applying them here would mainly measure agreement with a heuristic that supplied the correct-action labels, rather than the controller's operating performance.

7.5 Training Diagnostics and Frozen-Policy Selection

Figure 18 shows the archived v3 development traces. The upper panel reports the cumulative mean interval reward $J^{-1} \sum_{\ell=1}^{J} R_\ell^{\mathrm{train}}$, where J counts logged training transitions and R_ℓ^{train} is the recorded reward of transition ℓ. This is a mean, not accumulated reward, so its scale remains comparable across checkpoints. Candidate 0 recovered from the early high-exploration regime and ended with the highest development mean reward. The other candidates follow similar but distinct trajectories, which shows that the random seed has a clear effect on the learned values.

The cumulative mean TD loss rises for all three candidates after the early updates, with all archived values finite and bounded. This is a training diagnostic, not a convergence result. The target changes with the online network and the states visited during training, so TD loss measures target fit rather than model accuracy. Candidate selection used frozen-policy validation and action coverage. The independent holdout supplies the comparative performance results below.

7.6 Independent Holdout Results

Figure 19 shows that outcomes differ across scenarios. Frozen DQL has a higher return than the heuristic for connection-count skew, tiered backlog, and client slowdown. DQL and the heuristic are indistinguishable in the pod-pressure scenario because both select the same effective guarded response. DQL also matches the static policy under client slowdown: with client acknowledgement capacity as the bottleneck, additional connection movement leaves the service-rate bottleneck unchanged. These control cases identify when the policies provide equivalent outcomes.

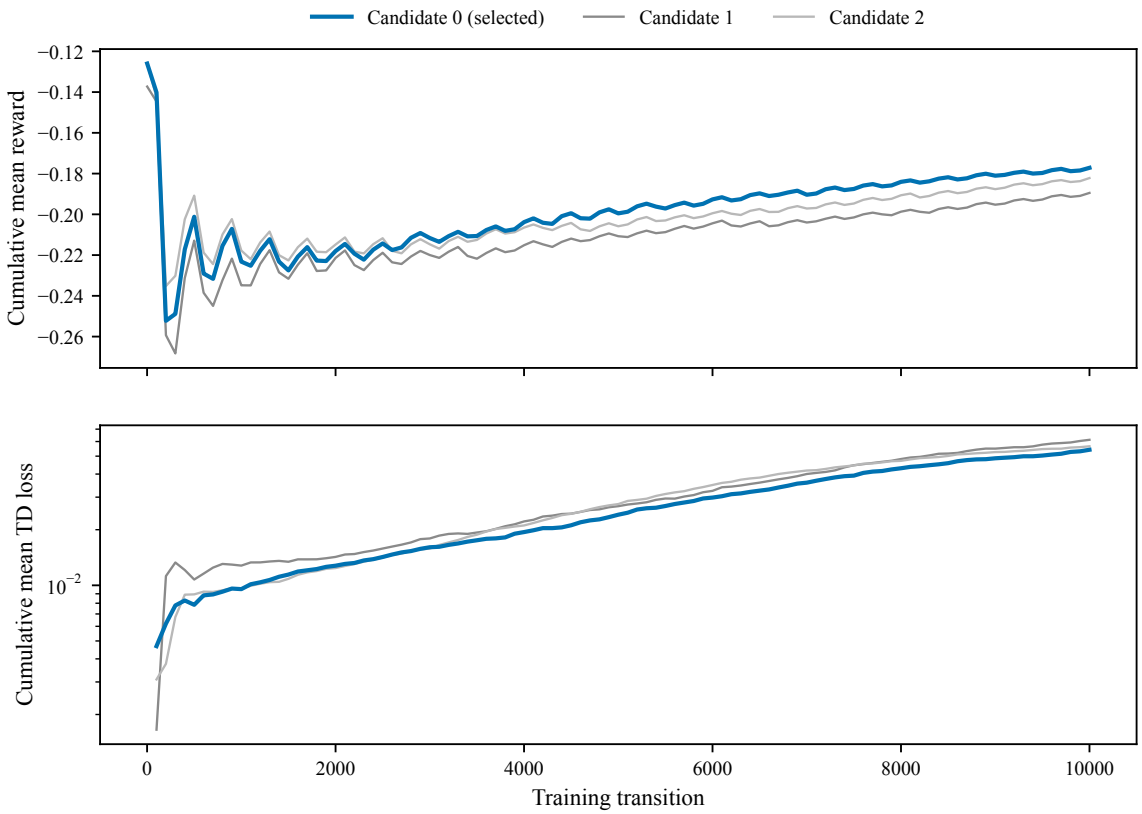


Fig. 18. Development traces for the three v3 candidates. Candidate 0 was selected by the predeclared validation rule and then frozen. The lower panel uses a logarithmic axis for the cumulative mean squared temporal-difference (TD) loss.

The boxes in Figure 19 describe the distribution of episode means, not the distribution of individual decision intervals. Horizontal point offsets separate the ten observations without changing their rewards. Collapsed boxes in the connection-count scenario reflect identical outcomes, not missing data. Separate panel scales make the smaller spread in the tiered-backlog case visible. The paired estimates below provide the statistical comparison, not the visual height of a box.

Across all 40 paired episodes, frozen DQL improves mean reward over the heuristic from −0.1609 to −0.1291. The mean paired difference is 0.0318 (bootstrap 95% CI [0.0191, 0.0455]), equivalent to a 19.8% relative benefit under Eq. (51); the paired standardized effect is $d_z = 0.729$, and the Holm-adjusted sign-flip p-value is below 0.001. Table 14 and Figure 20 show that this overall gain comes with tradeoffs.

The largest benefit is a 90.0% reduction in connection-level subscription variance, consistent with the design objective of moving subscriptions away from overloaded persistent connections. Subscription imbalance falls by 79.3%, combined connection imbalance by 48.3%, and mean pod pressure by 8.4%. Mean backlog ratio is 12.7% lower, but its interval crosses zero and the adjusted test is not significant ($p_H = .095$). Conversely, DQL has 45.4% higher connection-backlog variance, 34.0% higher normalized backlog imbalance, and 4.5% lower throughput than the heuristic. The scalar reward still favors DQL because its configured penalties place substantial weight on subscription pressure and combined

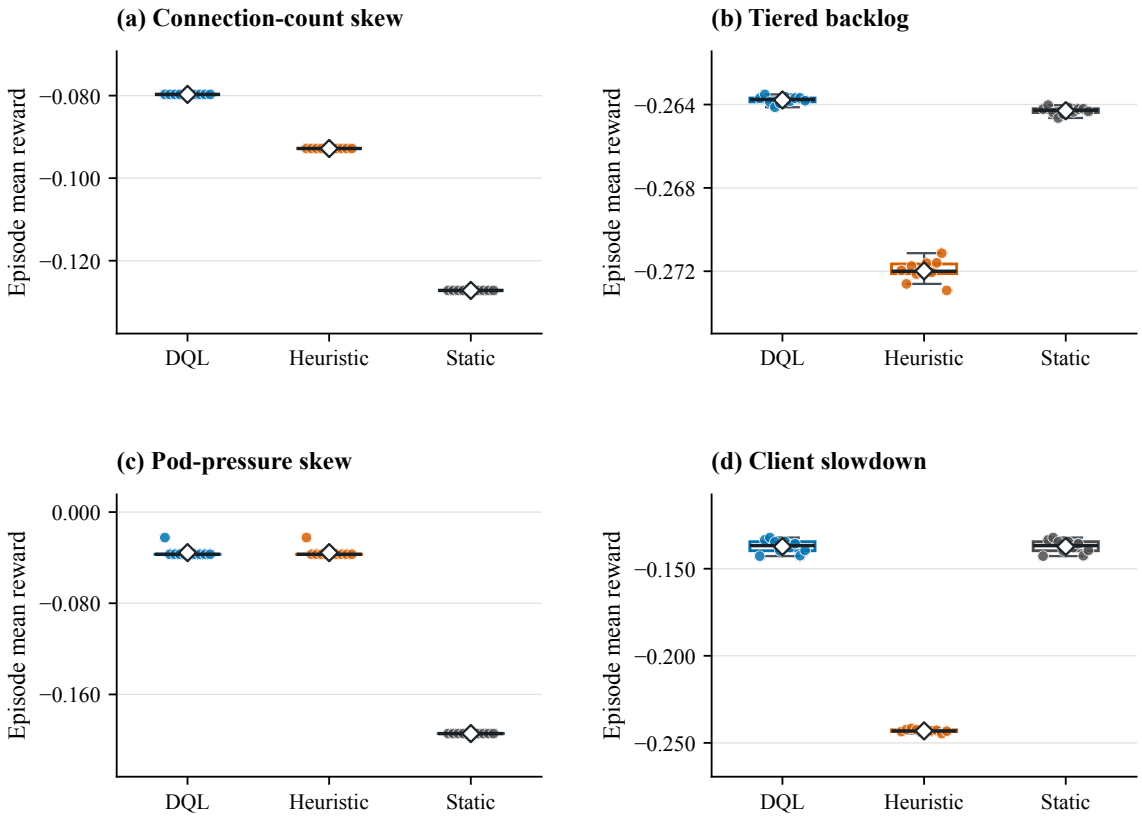


Fig. 19. Holdout reward distributions for frozen DQL, heuristic, and static policies. Each panel contains ten paired seeds per policy. Boxes span the first and third quartiles, the central line is the median, whiskers extend to observations within 1.5 interquartile ranges, and diamonds mark means. Every episode is shown as a point, including observations beyond whiskers. Panels use separate vertical scales; higher reward is better.

Table 14. Frozen DQL versus the heuristic on the independent v3 holdout. Positive benefit favors DQL. CIs are paired-bootstrap intervals transformed by Eq. (51); p_H is Holm-adjusted over 18 pooled metric-by-baseline tests.

Metric	DQL	Heuristic	Benefit [95% CI] (%)	p_H
Mean reward	−0.1291	−0.1609	19.8 [11.9, 28.3]	< .001
Backlog ratio	0.1749	0.2002	12.7 [−1.2, 27.1]	.095
Conn.-subscription variance	574.7	5768.7	90.0 [61.0, 120.8]	< .001
Conn.-backlog variance	2.280×10^{13}	1.568×10^{13}	−45.4 [−72.5, −22.6]	.022
Subscription imbalance	0.0898	0.4343	79.3 [59.1, 99.5]	< .001
Backlog imbalance	0.2194	0.1638	−34.0 [−56.4, −13.6]	.022
Combined conn. imbalance	0.1546	0.2991	48.3 [33.8, 63.8]	< .001
Pod pressure	0.1878	0.2051	8.4 [3.6, 13.6]	.022
Throughput ratio	0.2840	0.2973	−4.5 [−7.1, −2.2]	.022

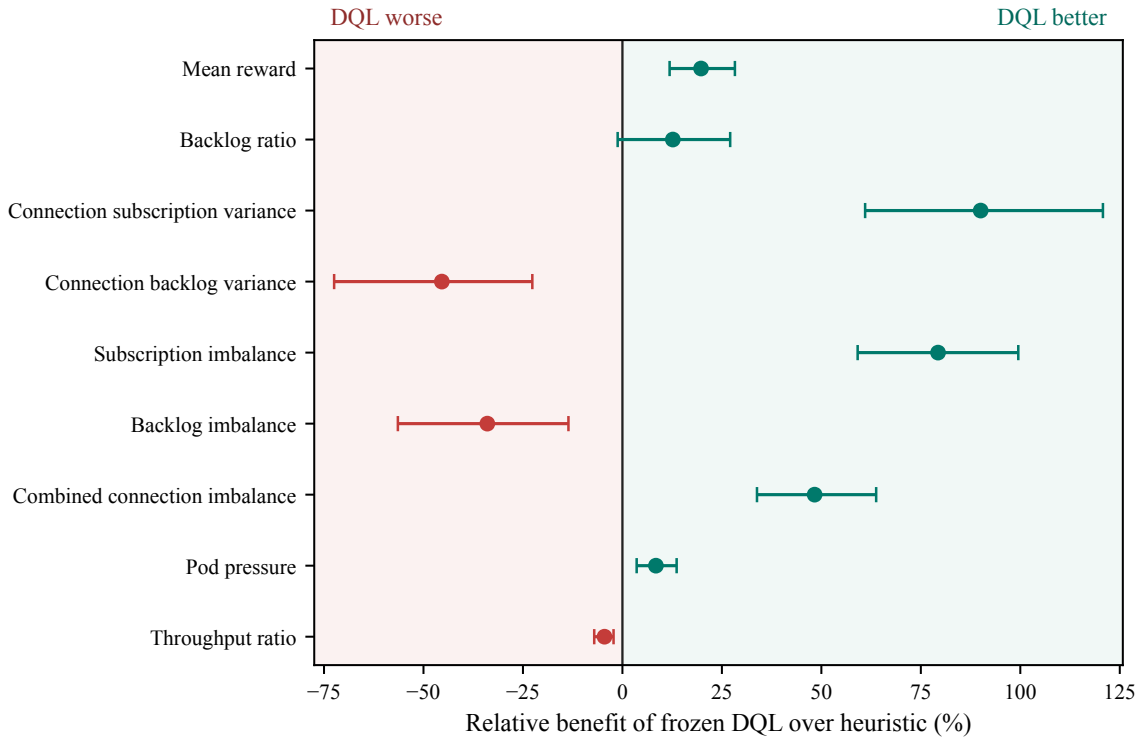


Fig. 20. Direction-adjusted relative benefit of frozen DQL over the heuristic across 40 paired holdout episodes. Points are mean effects and whiskers are paired-bootstrap 95% confidence intervals. Values right of zero favor DQL; values left of zero favor the heuristic.

imbalance. The holdout therefore identifies a multi-objective advantage for DQL and a queue-drainage advantage for the heuristic.

Figure 21 summarizes these tradeoffs without mixing raw variances with event rates. For policy π and a normalized cost metric x_v, the plotted coordinate is $1 - \bar{x}_{\pi,v}$; the throughput coordinate is $\bar{x}_{\pi,\text{throughput}}$. Here each mean covers the same 40 episodes, with ten episodes from each scenario. No observed minimum or maximum is used to stretch an axis. DQL's stronger subscription and combined balance coexist with the heuristic's stronger backlog balance and throughput. Polygon area has no statistical interpretation, and the combined-balance axis is derived from the two imbalance components rather than being an independent outcome. Uncertainty and paired comparisons remain those in Table 14 and Figure 20.

This pooled profile characterizes the mixed workloads; Pod saturation is evaluated separately. The connection-count-skew workload has no arrivals or initial backlog and therefore contributes zero throughput for all policies. In the pod-pressure workload, all policies serve the offered traffic with essentially zero residual backlog; their throughput is therefore equal despite different Pod pressure. The client-slowdown workload is limited by client acknowledgement capacity. Count balance and recovery of overloaded-Pod bandwidth are therefore separate endpoints. The saturated scale-out comparison is reported separately in Figures 13 and 15.

The distinction is operational, not merely graphical. Equal subscription counts can leave every persistent socket on the same saturated Pod, and moving idle subscriptions does not redistribute queued work. Conversely, moving

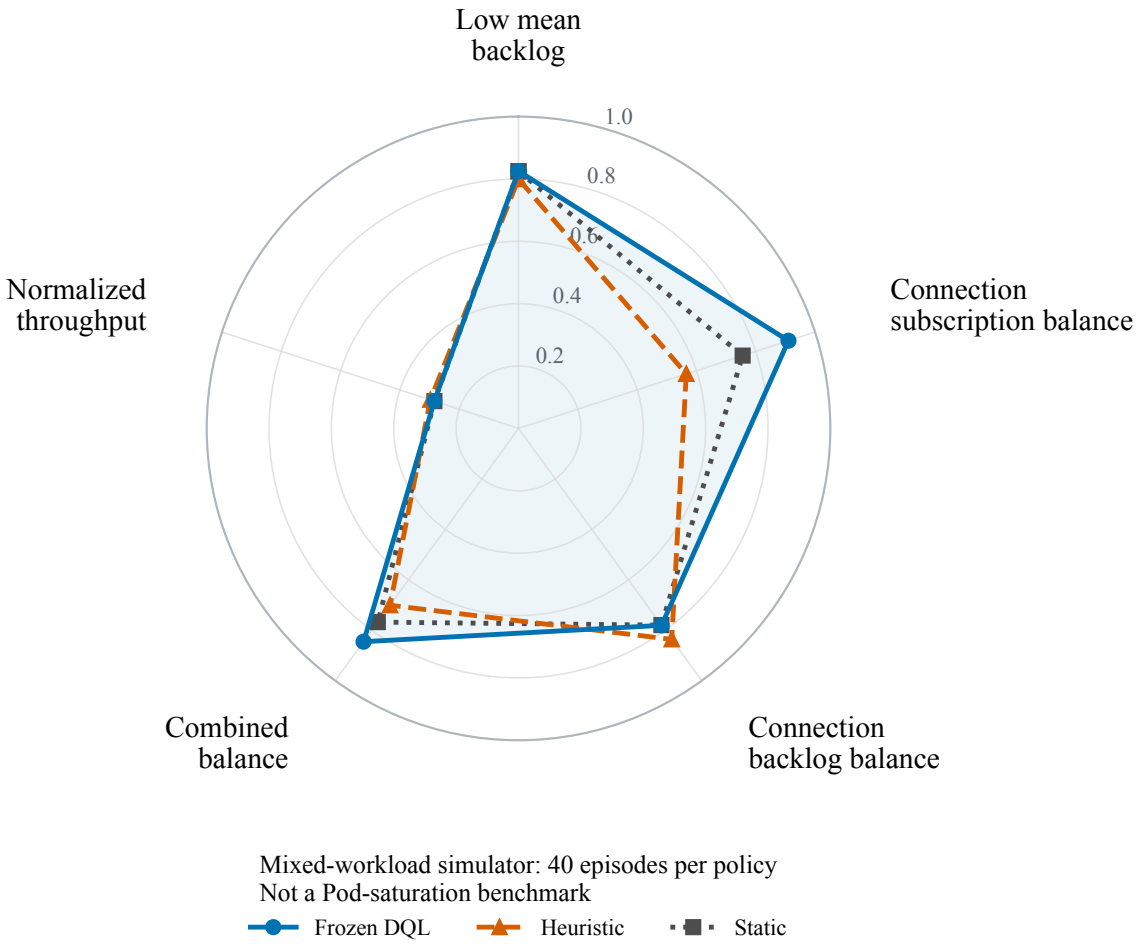


Fig. 21. Mixed-workload simulator profiles, not a Pod-saturation benchmark, across 40 paired holdout episodes per policy. Subscription balance is measured across client connections, not across all Pods. All axes use a fixed zero-to-one scale with larger values preferred. Backlog and imbalance costs are displayed as one minus their mean normalized value; throughput retains its recorded normalized value. The profiles are descriptive, not an additional weighted score or significance test.

backlogged subscriptions to a Pod with spare service capacity can raise aggregate throughput, provided client ACK capacity and shared Queue or network limits do not become the bottleneck. In the scale-out experiment, the initial connection counts and backlogs are already balanced ($V_u^N(0) = V_u^G(0) = 0$), yet all ten sockets terminate on one Pod. Migration across Pods raises measured simulator throughput from 58,000 to 219,762 events/s (3.79 times). Thus, the throughput-recovery evidence is the stressed placement comparison in Section 7.1. Connection-balance scores describe a different property. The Oracle Cloud preload pair in Section 6.6.5 measures expiry and ACK completion with the migration guard held at stay.

The comparison with static control isolates the benefit of rebalancing under concentrated load. In the 1,600-subscription pod-pressure scenario, three of four active connections initially terminate on the same pod. Static placement

therefore retains mean pod pressure 1.0125, whereas DQL promotes a guarded probe target and reduces it to 0.2176, a 78.5% reduction. The corresponding paired reward improvement is 0.1588 (95% bootstrap CI [0.1573, 0.1617]). In the connection-count-skew scenario, DQL reduces mean connection subscription-count variance from 6,250.0 to 2,298.7 (63.2%) and improves mean reward by 0.0475. Across all four scenarios, pooled mean reward improves by 0.0517 relative to static placement. The scenario plot also identifies cases where DQL and static placement are equal. The paired analysis assigns each measured gain to its workload, keeping the concentrated-load and tiered-backlog results separate.

7.7 Action-Contract Validation

The v3 policy passed the declared subscription-count, backlog-rebalancing, and migration coverage gates. The expanded v4–v6 studies tested all five interventions and fault guards. None passed every learned-action coverage gate, although the v6 candidates all passed the safety scenarios. These completed development studies do not supply additional holdout performance estimates. Appendix C retains their full results, including the scripted test of the corrected scale-up contract.

7.8 Adverse-Probe Promotion Ablation

The retained probe is a candidate endpoint, not an instruction to migrate. To isolate this distinction, we registered a paired mechanism experiment before the confirmatory run. In every episode, the probe was set to reach the pod with the highest initial pressure. The deployment mask evaluated the concrete source–target pair and rejected a^{mig} whenever its projected pressure improvement was below 0.01. The primary ablation then bypassed that mask for exactly one interval, moved the highest-backlog non-target connection to the adverse pod, and returned control to the same frozen DQL checkpoint. Thus, *Frozen DQL* and *DQL + forced promotion* differ only by that initial override and its subsequent state consequences. *Blind promotion* makes the same move and then stays; *Static* never changes the initial placement; and the guarded heuristic uses the same feasibility mask as DQL.

For total backlog $B(t)$ in the evaluated subscription population over T decision intervals, the primary endpoint is the normalized trapezoidal backlog area

$$\mathrm{AUC}_B = \frac{1}{T}\sum_{t=1}^{T}\frac{B(t-1)+B(t)}{2B(0)}. \tag{52}$$

It measures how much work remains queued over time. Lower values indicate earlier drainage. The statistical unit is the environment seed, not an interval. The primary cohort uses 30 previously unused paired seeds and 1,200 subscriptions.

Each episode runs for 500 intervals. Paired mean differences use 10,000 bootstrap resamples and one-sided sign randomization with 100,000 sign vectors; Holm adjustment covers the four named baselines. The action mask rejected the adverse migration in all 30 primary and all five scale-confirmation environments.

In the primary cohort, guarded DQL attains mean $\mathrm{AUC}_B = 0.66046$, compared with 0.68938 after the forced promotion. The paired difference is -0.028918 (95% CI $[-0.039444, -0.019073]$), a 4.19% reduction relative to the forced-promotion ablation. The raw one-sided randomization p-value is 1.0×10^{-5} and the Holm-adjusted value is 4.0×10^{-5}. Guarded DQL also reduces backlog AUC by 9.34% relative to blind promotion and by 1.41% relative to static placement. Its final backlog ratio is 0.35875, compared with 0.40936, 0.48584, and 0.36864, respectively. The throughput differences have the same direction: DQL delivers 43.02 million messages, 2.35 million more than the forced-promotion ablation and 5.91 million more than blind promotion.

The heuristic provides an important comparison. Its primary-cohort backlog AUC is 0.59104, so DQL is 11.75% higher (paired difference 0.069419, 95% CI [0.066977, 0.071826]). The action records explain the difference. Across 15,000

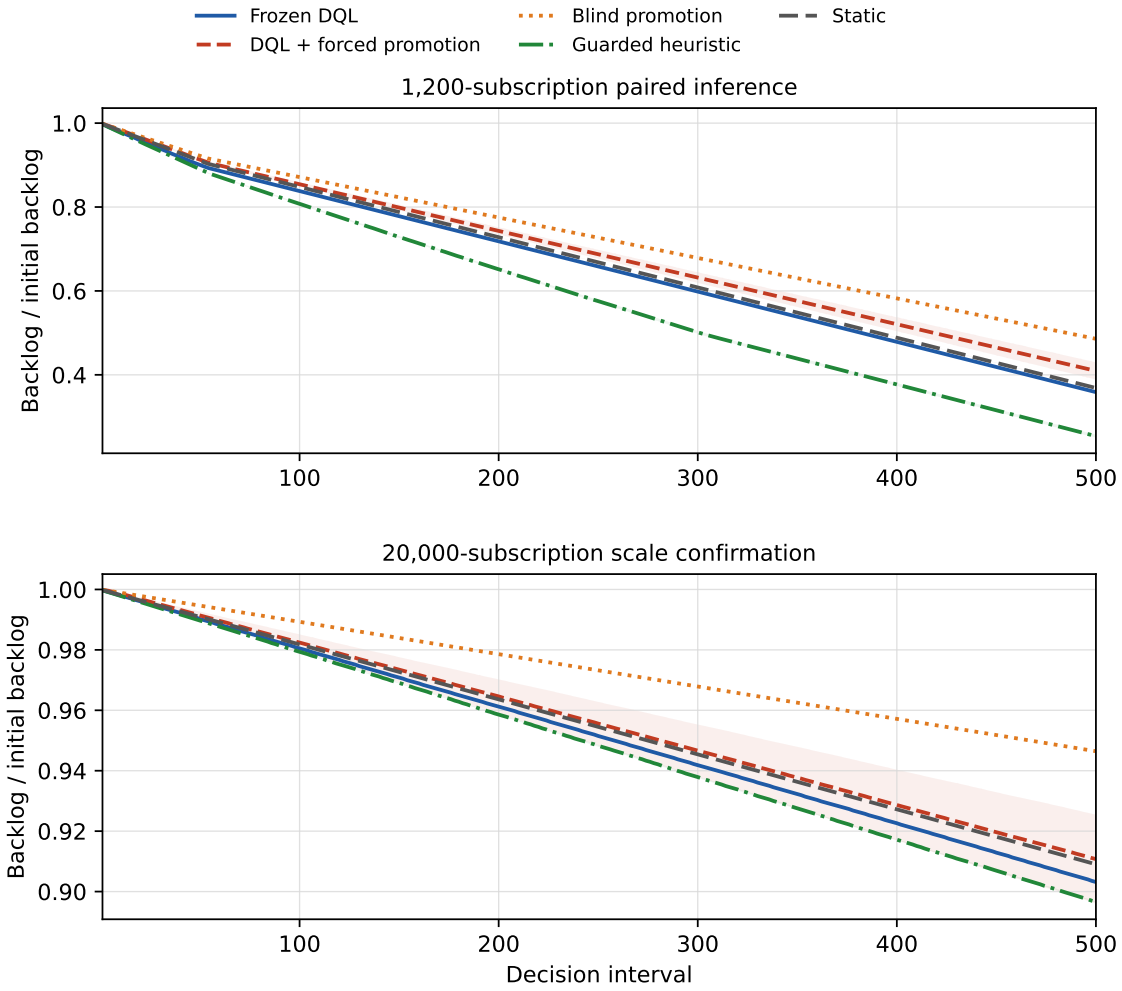


Fig. 22. Mean normalized backlog trajectories for the adverse-probe ablation. Shaded bands are descriptive 95% normal-approximation intervals across paired seeds. The upper panel contains the 30-seed inference cohort; the lower panel contains the five-seed 20,000-subscription scale confirmation. The forced-promotion policy resumes the same frozen DQL controller after the interval-zero override.

primary decision intervals, the heuristic executes 2,286 backlog-rebalancing actions, whereas DQL executes 175 backlog and 175 subscription-count moves and otherwise stays. In the 20k cohort, the heuristic uses all 625 interventions for backlog rebalancing. DQL splits 625 interventions almost evenly between 315 backlog and 310 subscription-count moves. The backlog-specific measure therefore favors the heuristic, which makes more backlog-focused moves. The ablation establishes the benefit of guarding probe promotion; the backlog-focused heuristic retains the stronger queue-drainage result.

The 20,000-subscription cohort retains the requested four equal groups: 5,000 subscriptions at 100,000 queued messages, 5,000 at 50,000, 5,000 at 5,000, and 5,000 at zero, totaling 775 million initial messages. Guarded DQL reduces

Table 15. Adverse-probe ablation outcomes. Backlog AUC and final backlog are normalized by initial backlog; lower is better. Throughput is cumulative over 500 intervals. The 20k cohort is descriptive ($n_{ep} = 5$).

Cohort	Policy	Backlog AUC	Final ratio	Throughput (M)	Mean pressure
1.2k inference	Frozen DQL	0.66046	0.35875	43.02	0.5873
1.2k inference	DQL + forced promotion	0.68938	0.40936	40.66	0.6100
1.2k inference	Blind promotion	0.72853	0.48584	37.11	0.5874
1.2k inference	Guarded heuristic	0.59104	0.25384	47.90	0.5941
1.2k inference	Static	0.66993	0.36864	42.56	0.5870
20k confirmation	Frozen DQL	0.95155	0.90316	97.05	0.7877
20k confirmation	DQL + forced promotion	0.95556	0.91076	91.16	0.7884
20k confirmation	Blind promotion	0.97323	0.94645	63.50	0.7875
20k confirmation	Guarded heuristic	0.94828	0.89658	102.15	0.7883
20k confirmation	Static	0.95452	0.90903	92.50	0.7875

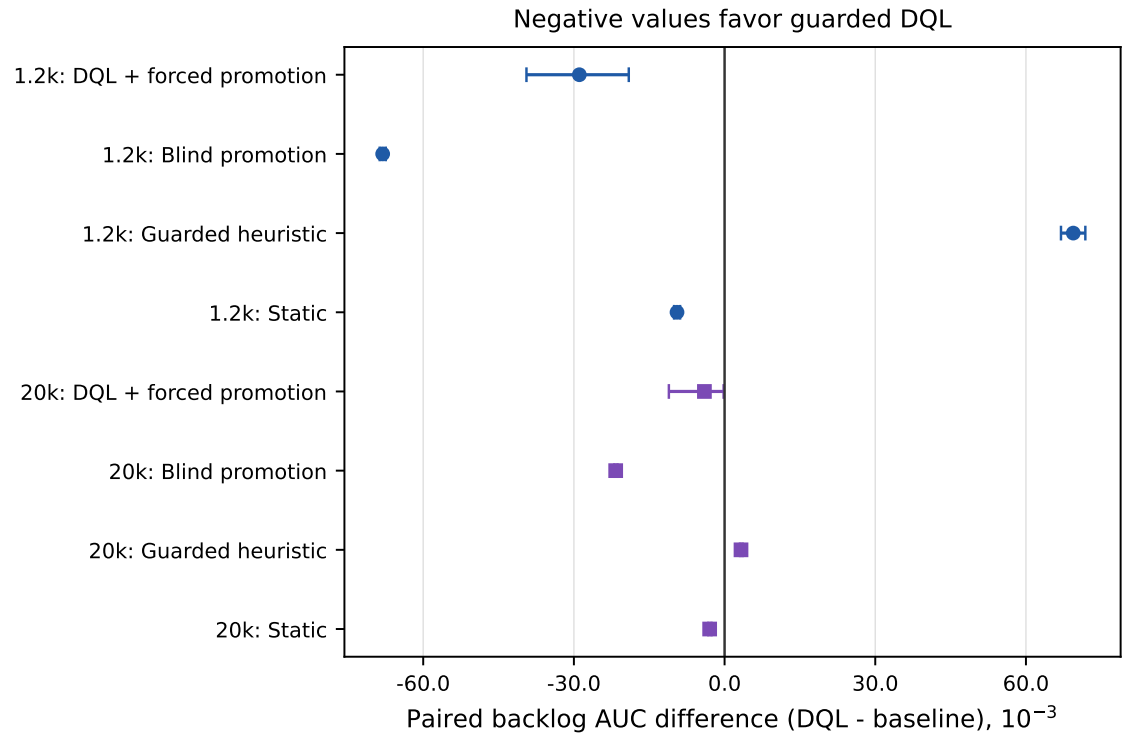


Fig. 23. Paired normalized backlog-AUC differences between guarded DQL and each baseline. Negative values favor DQL; whiskers are paired-bootstrap 95% confidence intervals. Squares denote the descriptive 20,000-subscription cohort.

backlog AUC by 0.42% relative to forced promotion, 2.23% relative to blind promotion, and 0.31% relative to static placement, while remaining 0.34% above the heuristic. The forced-promotion paired interval lies below zero, but its exact one-sided $p = 0.0625$ and Holm-adjusted $p = 0.125$ reflect the five-seed sample. We consequently treat this cohort

as scale confirmation, not as a second inferential test. The maximum normalized message-conservation residual over all 175 episodes is 1.031×10^{-14}; this verifies simulator accounting but does not establish loss-free production handoff.

7.9 Reproducibility and Study Scope

Reviewer artifact access. The accompanying supplementary archive, `reviewer-evidence.zip`, contains episode-level simulator CSVs, declared protocols, original aggregate analyses, the frozen 21–32–6 checkpoint, and identifier-free live summaries. Its README gives the file map and exact offline commands. A Python 3.9+ standard-library script verifies the checksum manifest, checks paired cohort coverage, recomputes policy means and selected effect sizes, and reconciles the loaded trial's reported subscription and ACK totals. A separate inference option evaluates the saved network without transport access or model updates. The archive supports reanalysis and checkpoint inspection; it does not expose corporate credentials, raw event payloads, private endpoint identifiers, or the proprietary service source. It is not a complete independent deployment rerun. Provenance records distinguish unchanged measurements from redacted local-path metadata.

The manuscript bundle contains the exact v3 aggregate CSV files, v3–v6 JSON protocols and summaries, the adverse-probe protocol and episode-level outputs, and the scripts that regenerate every figure in this section. The larger private evidence archive additionally records source-file SHA-256 hashes, frozen-checkpoint hashes, raw episode rows, and a recursive checksum manifest. The implementation test suite contains unit tests for state construction, action masks, replay, checkpoint persistence, migration state transitions, and fault guards, together with a Java STOMP fixture. At the source revision used for this manuscript, all 267 tests pass, including the four adverse-probe evaluator tests, cap-restoring scale-up tests, and legacy-policy rejection tests.

The recovery study includes 20,000 subscriptions and four modeled pods; the earlier four-scenario holdout contains up to 1,600 subscriptions. These are simulator measurements with explicit workload and service assumptions. The v4–v6 all-action studies ended at their development coverage gates and are reported as diagnostics, separate from holdout performance. The Oracle Cloud cluster experiment in Section 6 measures 20,000-subscription topology and ten live handoffs using existing queued events rather than the simulator's paired, controlled workloads. The backlog-recovery effect sizes therefore apply to the stated simulator. The Oracle Cloud measurements establish execution of the corresponding connection-replacement mechanism in Kubernetes. The additional loaded scale-out run in Section 6.5 records three selective DQL handoffs and 8,000 acknowledged STOMP deliveries for 1,000 subscriptions. It uses a fixed client-processing delay and bounded publication, and supplies functional handoff evidence rather than a matched throughput effect size.

8 STABILITY CONTROLS AND POLICY VALIDATION

Because the controller operates with partial and periodically refreshed statistics, stability is enforced at two levels. During learning, replay, target-network synchronization, bounded rewards, and epsilon decay limit the rate at which value estimates change. During execution, feasibility masks, minimum-improvement thresholds, cooldowns, and the receipt-gated handoff protocol prevent a high Q-value from bypassing required operating rules. These controls bound the available transitions. Their role is operational stability, distinct from a proof of learned-policy convergence.

Classical tabular Q-learning guarantees assume a stationary, fully observable Markov decision process and sufficient visitation of every state-action pair. The client instead receives delayed partial observations in an environment whose queue arrivals, acknowledgement capacity, and competing clients can change. Policy quality is therefore evaluated under the stated workloads, without a global-optimality guarantee. The finite v3 traces describe training behavior

rather than Bellman-loss convergence. The v4–v6 results distinguish training completion from action coverage. The validation protocol uses action-coverage, fault-safety, churn, and performance gates to assess each frozen policy.

9 DISCUSSION

The main systems finding is that subscriber capacity and client access to that capacity are separate. Adding Pods supplies service capacity; moving existing subscription-bearing connections makes it reachable by persistent clients. The guarded-greedy and DQL policies both achieve this in the scale-out model. Their nearby throughput results support the shared mechanism rather than requiring a claim that learning beats a strong greedy rule in every workload.

Policy choice depends on the operational objective. Greedy control gives a direct, inspectable response to measured pressure. DQL can account for delayed costs and several competing objectives. A subscription move changes future backlog distribution; migration trades immediate handoff cost for later capacity. Both selectors operate above the same bounded, checked actions.

QEdgeProxy selects service instances using client QoS feedback [14], while Yahya et al. adapt offloading through decentralized bandits and server-reward adjustment [48]. The present work focuses on subscription-aware persistent messaging clients under partial observability and treats connection-level rebalancing as a first-class control action alongside endpoint selection. This distinction matters operationally: endpoint selection chooses the serving Pod, while connection-level rebalancing corrects subscription skew within the client.

The holdout results show how learning compares with the heuristic. Both policies can perform the same actions, so they can reach the same capacity limit in some scenarios. DQL's measured advantage is concentrated in subscription distribution, combined connection imbalance, and avoiding a harmful heuristic intervention during client slowdown. The heuristic retains an advantage in throughput and backlog-distribution metrics. These results identify the operating objectives favored by each policy. The component-level analysis reports these tradeoffs alongside the scalar reward; policy selection uses development validation before the holdout is evaluated.

The distinction between a model's quality and the utility of the complete system also appears in Casimiro et al.'s probabilistic framework for adapting ML components [6]. In our setting, a favorable Q-value is therefore interpreted alongside backlog, throughput, and handoff outcomes. Gheibi and Weyns introduce a lifelong-learning layer to respond to changing adaptation options [20]. Our experiments evaluate a fixed policy under the stated workload and deployment conditions. Lifelong adaptation is outside this study's scope.

The loaded Oracle Cloud scale-out trial adds direct evidence for selective execution (Section 6.5). Every handoff preserved three unselected active sessions, while the full cohort reached 8,000 ACK receipts. The measured drain waits also show why opening a replacement socket is only one part of migration: client processing and acknowledgement completion can dominate the handoff duration. These measurements validate the learned decision-to-execution path under the declared load and processing limit.

The action-contract studies provide a further design lesson: a useful executor and a policy that reliably selects it are separate requirements. The cap-restoring scale-up preflight improved reward and maximum subscription count, but the expanded learner study did not pass every coverage gate. Appendix C retains these results. They explain why the live checkpoint uses a compatible, restricted rebalancing profile.

10 CONCLUSION

We presented a probe-guided, client-side framework for rebalancing persistent subscription connections without requiring cluster-wide client visibility. The mechanism combines per-subscription queues, active-connection telemetry,

and a load-balancer-assigned probe. Greedy rules or DQL select bounded actions, and a shared executor carries out receipt-gated handoff. The technical contribution is the link between partial observations, endpoint-preserving probe promotion, and selective subscription reassociation.

The primary experiment demonstrates recovery after a one-to-four pod capacity increase with 20,000 subscriptions and 775 million queued events. Guarded-greedy migration and frozen DQL deliver 221,260 and 219,762 events/s, respectively, compared with 58,000 events/s for unchanged placement. Frozen DQL uses all four pods, delivers 3.79 times the mean throughput, and leaves 27.52% less backlog after 20 minutes than retaining the ten original connection assignments. Its 12.30% reduction in backlog area satisfies the registered paired statistical criterion. The adverse-probe ablation complements this result: evaluating the candidate before promotion reduces backlog area by 4.19% relative to forcing one harmful migration.

These findings identify the operational benefit of the mechanism: scaled-out subscriber capacity becomes useful to existing subscriptions through client-directed connection reassociation, while combined subscription and backlog pressure determines whether a sampled target is suitable. The supporting studies characterize policy tradeoffs and action coverage. The mixed-workload holdout favors DQL in composite reward and subscription balance, while the heuristic favors throughput and backlog balance. These results support two evaluated policy options rather than universal DQL superiority or an untested DQL-tuned scripted policy. The isolated Oracle Cloud cluster qualification also demonstrates a 60% reduction in source subscriptions through three full-connection migrations, with all 2,000 subscriptions retained and 73,639 observed deliveries reconciled at closure. Its startup measurements explicitly capture activation while deliveries flow. A further loaded one-to-four-Pod trial retained all 1,000 subscriptions and four active connections through three DQL-selected handoffs. Each action preserved the other three sessions. The final client distribution was 500, 250, 250 and zero across the four available Pods, with 8,000 observed deliveries acknowledged and no unresolved client deliveries at closure. A separate Oracle Cloud preload pair measured twelve native TTL interruptions and 89.44% ACK-receipt completion, versus no interruptions and 100% completion in the TTL-disabled DQL arm (Section 6.6.5). That arm remained at guard-held *stay*: a 200-second minimum telemetry sweep exceeded the 180-second freshness deadline. The result measures expiry-related receipt completion under the telemetry constraint. The comparative throughput and backlog-recovery gains are simulator measurements under the stated service assumptions. The Oracle Cloud experiments establish live handoff execution, subscription retention, and observed delivery behavior. Together, these studies describe the mechanism's measured benefits and the operating conditions under which they were obtained.

A NOTATION REFERENCE

The tables collect the detailed definitions used in the model and experimental endpoints.

Table 16. Summary of Notations: System Sets and Pod Pressure

Symbol	Description
	Server-Level Placement Variables
$\mathcal{S} = \{1, \ldots, n\}$, $\mathcal{U} = \{1, \ldots, m\}$	Sets of Pod indices and client indices; n and m are their counts
t, T, Δ	Decision-epoch index, last epoch, and seconds per epoch; adjacent epochs are t and $t+1$
$\mathcal{K}_u(t)$, $C_u(t)$	Client u's subscription set and active delivery-connection set
$\mathcal{K}_{u,c}(t)$, $z_{u,k,c}(t)$	Subscriptions assigned to connection c and the binary indicator of that assignment
$Q_{u,k}(t)$, $b_{u,k}(t)$	Subscription queue and its backlog count, $b_{u,k}(t) = \lvert Q_{u,k}(t) \rvert$
$i_{u,c}(t)$, $O_u(t)$	Pod index reached by connection c and set of Pod indices observed by client u
p_i, $y_i(t)$	Pod name and reported tuple $(p_i, X_i(t), B_i(t), H_i)$; $y_i(t)$ is not the scalar Bellman target y_t
$\mathcal{R}_i(t)$	Set of server-side subscription queues currently served by pod i
$\tilde{b}_{i,r}(t)$	Backlog of server-side subscription queue $r \in \mathcal{R}_i(t)$
$X_i(t)$	Pod-level subscription count for pod i
$B_i(t)$	Cumulative backlog over all subscription queues served by pod i
H_i	Configured maximum processing threshold for Pod i in events/s, used as a normalization signal
X_{safe}	Subscription-count level after which pod reliability risk begins to increase
X_{hard}	Hard subscription-count reference used to normalize severe subscription pressure, with $X_{\text{hard}} > X_{\text{safe}}$
B_{ref}	Reference backlog used to normalize pod backlog pressure
$P_i^X(t)$	Normalized pod subscription-count pressure
$P_i^B(t)$	Normalized pod cumulative-backlog pressure
$P_i^H(t)$	Backlog pressure normalized by $H_i\Delta + \varepsilon_0$, then clipped at one
ε_0, $[x]_+$	Positive denominator offset in messages and positive part $\max\{x, 0\}$; distinct from exploration probability ϵ
$P_i^R(t)$	Nonlinear pod reliability-risk pressure caused by excessive subscription count
$P_i(t)$	Weighted pod pressure score used in the deployable reward
$\omega_X, \omega_B, \omega_H, \omega_R$	Nonnegative weights for subscription-count, backlog, threshold-relative backlog, and reliability-risk pressure terms

Table 17. Summary of Notations: DQL Observation and Reward Symbols

Symbol	Description
J_T	Offline finite-horizon system evaluation cost over epochs $0, \ldots, T$
$\Phi_X(t)$	Pod subscription-count variance across the cluster at time t
$M(t)$	Total number of migration or reconnect operations during epoch t
$\Lambda_B, \Lambda_X, \Lambda_C, \Lambda_M$	Nonnegative weights in the offline evaluation objective for backlog, pod balance, connection balance, and migration churn
$T_u(t)$	Effective client-side throughput from delivered, processed, and acknowledged messages
$R_u(t)$	Deployable reward observed by client u at time t
$Q_\theta(o, a)$	Online DQL action-value network used for action selection and gradient updates
$Q_{\theta^-}(o, a)$	Target DQL action-value network used only to compute Bellman targets
θ, θ^-	Online-network parameters and delayed target-network parameters
$\mathcal{D}, \xi_t$	Replay buffer and transition $(o_u(t), a_u(t), R_u(t), o_u(t+1))$
y_t, $\mathcal{L}(\theta)$	Scalar Bellman target and mean squared temporal-difference loss
α, K_{sync}	Online-network learning rate and target-copy interval in training updates
γ	DQL discount factor
ϵ	Exploration probability in the DQL epsilon-greedy action policy
δ	Required Q-value improvement of migration over stay; distinct from variance thresholds γ_N, γ_G
$\lambda_1, \ldots, \lambda_7$	Weight coefficients used in the deployable reward function
$o_u(t)$	Client observation vector used by the DQL policy
$o_u^B(t)$	Backlog observation branch containing normalized local backlog and backlog growth features
$o_u^N(t)$	Subscription-count observation branch containing normalized $N_{u,c}(t)$ and $V_u^N(t)$ features
$o_u^G(t)$	Backlog-load imbalance observation branch containing normalized $G_{u,c}(t)$ and $V_u^G(t)$ features
$o_u^C(t), o_u^P(t), o_u^D(t)$	Connection-count/throughput, pod-pressure, and decision-state observation branches
$\mathcal{A}$	Set of bounded DQL action templates
$\mathcal{A}_u^{\text{adm}}(t)$	Subset of action templates admitted by client u's guards at time t, used for action selection and next-state Bellman maximization
$i_u^{\text{probe}}(t)$	Pod endpoint bound to client u's currently retained, load-balancer-assigned probe socket
$a_u(t), a'$	Selected action at epoch t and candidate next-epoch action in Bellman maximization

Table 18. Summary of Notations: Connection Scaling and Local Counters

Symbol	Description
Connection Scaling Variables	
$C(t)$	Number of active persistent delivery connections for client u at time t, $C(t) = \lvert C_u(t)\rvert$
$C_{\max}$	Maximum allowable client connections
$N_{\text{sub}}(t)$	Total number of active subscriptions maintained by client u at time t, $N_{\text{sub}}(t) = \lvert\mathcal{K}_u(t)\rvert$
Θ_N	Maximum desired subscription count per connection used by the scaling rule
β	Backlog threshold per subscription
$b_{u,k}(t)$	Client-observed backlog of subscription queue k for client u
$C_{\text{sub}}(t)$	Required connections based on total subscription count
$C_k(t)$	Client connection-count requirement triggered by subscription k's backlog; not multiple owners for that queue
$C_{\text{back}}(t)$, $C^\star(t)$	Maximum backlog-based requirement and desired connection count after combining scaling requirements
c^{place}, $c^\star$	Connection chosen for new-subscription placement and connection evaluated for migration
$\widehat{N}_{u,c}(t)$, $\widehat{G}_{u,c}(t)$	Normalized subscription-count and backlog loads used by the placement score
$B_u(t)$, $\text{Grow}_u(t)$	Sum of client u's subscription backlogs and its positive increase since the previous epoch
$T_u^{\text{del}}(t)$, $T_u^{\text{proc}}(t)$, $T_u^{\text{ack}}(t)$	Delivered, processed, and acknowledged message rates in events/s; their minimum is $T_u(t)$
$\widehat{B}_u(t)$, $\widehat{\text{Grow}}_u(t)$, $\widehat{T}_u(t)$	Normalized backlog, backlog growth, and throughput under the experiment's stated scaling convention
$M_u(t)$, $\widehat{C}_u(t)$	Migration indicator or normalized count, and active connection fraction $C(t)/C_{\max}$

Table 19. Summary of Notations: Connection Rebalancing Symbols

Symbol	Description
Connection-Level Rebalancing Variables	
$N_{u,c}(t)$	Number of subscriptions of client u assigned to connection c at time t
$G_{u,c}(t)$	Cumulative backlog load on connection c, computed from the subscription queues assigned to that connection
c_h, c_l	Source high-load connection and target low-load connection used in a rebalancing move
$V_u^N(t)$	Connection-level subscription-count variance across client u's active persistent connections
$\bar{N}_u(t)$, $\bar{G}_u(t)$	Mean subscription count and mean cumulative backlog across client u's active connections
$V_u^G(t)$	Connection-level backlog-load variance across client u's active persistent connections

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Table 19. Summary of Notations: Connection Rebalancing Symbols (continued)

Symbol	Description
$\widehat{V}_u^N(t), \widehat{V}_u^G(t)$	Normalized subscription-count and backlog-load variance terms used in the local imbalance score
$V_{\text{ref}}^N, V_{\text{ref}}^G$	Positive variance references used to normalize $V_u^N(t)$ and $V_u^G(t)$
σ_{tol}^N	Tolerable standard deviation of subscription counts across a client's active connections
σ_{tol}^G	Tolerable standard deviation of cumulative backlog load across a client's active connections
Δ_N	Operator-chosen tolerable spread between a heavy and a light connection in subscription count
Δ_G	Operator-chosen tolerable spread between a heavy and a light connection in cumulative backlog messages
$\widehat{r}_u^{\text{ack}}$	Conservative client-observed ACK or message-processing rate for client u
τ_G	Acceptable additional drain-time skew used to derive a latency-driven backlog spread
ρ_N	Fraction of Θ_N used to set the default subscription-count imbalance tolerance
ρ_G	Fraction of $\beta\Theta_N$ used to set the default backlog-load imbalance tolerance
$V_u(t)$	Weighted local connection-imbalance score for client u
η_N, η_G	Nonnegative weights for subscription-count imbalance and backlog-load imbalance, with $\eta_N + \eta_G = 1$
γ_N, γ_G	Thresholds triggering subscription-count and backlog-based connection rebalancing

Table 20. Summary of Notations: Handoff and Experimental Endpoints

Symbol	Description
$\mathcal{K}_{\text{move}}, s_b, s_f$	Selected subscription set, barrier timestamp after source quiescence, and source telemetry-frame timestamp
$I_{c_h,k}(s_f)$, $\text{state}_{c_h,k}(s_f)$	Source-reported in-flight count and delivery state for subscription k at frame time s_f
c_p, i_h, i_l	Probe socket and Pod indices reached by c_h and c_l in the architecture figure
$t_k^{\text{act}}, t_k^{\text{send}}, t_k^{\text{ready}}$	Activation latency and submission/receipt timestamps in seconds; $t_k^{\text{act}} = t_k^{\text{ready}} - t_k^{\text{send}}$
$A^{\text{act}}(s)$	Number of receipt-confirmed activations by elapsed time s in seconds
$t_c^{\text{open}}, t_c^{\text{exp}}, T_{\text{TTL}}, J_c$	Connection opening and nominal expiry timestamps, TTL in seconds, and integer jitter in minutes
d_c, h_c, r_c	Disconnect, replacement-CONNECTED, and full binding-restoration timestamps in seconds
$T_c^{\text{conn}}, T_c^{\text{restore}}$	Transport reconnect and full restoration durations, measured from d_c
$T_u^{\text{handoff}}, T_u^{\text{drain}}$	Measured total handoff and source-drain durations for client u in seconds
$C_0, I_c(s), F(s)$	Initial trial connection pool, unrestored-connection indicator, and affected fraction at elapsed time s

Continued on next page

Table 20. Summary of Notations: Handoff and Experimental Endpoints (continued)

Symbol	Description
$D_{\text{obs}}, A_{\text{conf}}$	Recorded delivery attempts and attempts with confirmed ACK receipts
$B(t)$, AUC_B	Total backlog in the evaluated population and its normalized trapezoidal area over T intervals
$Y_{\pi,j,v}$, d_v, $D_{j,v}$	Episode mean for policy π, episode j, metric v; benefit direction; and signed paired difference
RB_v, p_H, d_z	Relative benefit in percent, Holm-adjusted p-value, and paired standardized mean difference
J, R_ℓ^{train}	Number of logged training transitions and the reward recorded for transition ℓ
M^{ep}, n_{ep}	Migration count in one episode and number of sampled episodes, distinct from client count m and Pod count n
$W_1, b_1, W_2, b_2, h(o)$	Weights and biases of the 21–32–6 Q-network, and its single ReLU hidden representation
$\mathbf{q}_\theta(o)$	Vector of online Q-values, one per action in $\mathcal{A}$, with $o = o_u(t)$

B EVALUATED FEATURES AND POLICY SETTINGS

The frozen network consumes the ordered 21-element vector in Table 21. For nonnegative x and positive reference r, the implementation uses $\text{clip}(x/r) = \min(1, x/r)$ and $\text{lognorm}(x; r) = \min(1, \log(1+x)/\log(1+r))$. These are numerical scaling rules, not drain-time estimates. A pressure feature can exceed one through the quadratic reliability term even though its count, backlog, and capacity components are clipped.

Table 21. Exact input order of the evaluated 21–32–6 network. Indices are one-based; "mean" and "maximum" refer to active client connections.

Indices	Features, in order
1–2	Log-normalized client backlog; log-normalized positive backlog growth
3–5	Mean and maximum of $\text{clip}(N_{u,c}/\Theta_N)$; $\widehat{V}_u^N$
6–9	Mean and maximum of $\text{lognorm}(G_{u,c}; B_{\text{ref}})$; $\widehat{V}_u^G$; combined imbalance V_u
10–11	$\lvert C_u \rvert / C_{\max}$ clipped at one; client throughput divided by the declared throughput reference and clipped at one
12	Active-Pod pressure, weighted by this client's subscription counts on each observed Pod; unweighted mean if the count is zero
13–15	Probe pressure; clipped probe subscription count relative to X_{safe}; clipped probe backlog relative to B_{ref}
16–18	Probe-present indicator; probe endpoint absent from active endpoints; cooldown-ready indicator
19–21	Maximum active telemetry/backlog age ratio; probe age ratio; previous-action-was-migration indicator

Pod-wide values are deduplicated when several sockets reach one Pod. Client backlog sums each known subscription once. When per-subscription values are absent, the builder uses the active-connection backlog aggregates. Age ratios are clipped at one. Missing probe data sets its pressure and age features to one and its present indicator to zero; the independent validity checks prohibit migration. Coverage and freshness checks are execution gates, not additional neural inputs.

For the simulator adapter, B_{ref} is the declared initial tiered backlog when it is positive, otherwise $\max(1, \beta|\mathcal{K}_u|)$. The growth reference equals B_{ref}; $V_{\text{ref}}^N = (0.2\Theta_N)^2$ and $V_{\text{ref}}^G = (0.2B_{\text{ref}})^2$. The safe and hard subscription thresholds are $2\Theta_N$ and $\max(2\Theta_N + 1, \text{round}(2.5\Theta_N))$. The throughput reference is three times the configured base client ACK rate.

These constants are scenario inputs, not fitted from holdout outcomes. The service-model pressure used to generate simulator throughput is distinct from this observation-builder pressure, as described in Section 7.1.

The October loaded deployment uses $\Theta_N = 250$, $C_{\max} = 5$, $B_{\text{ref}} = 100{,}000$, $V^N_{\text{ref}} = 2{,}500$, $V^G_{\text{ref}} = 400{,}000{,}000$, $X_{\text{safe}} = 800$, and $X_{\text{hard}} = 1{,}000$. Its throughput reference is 1,000 events/s, telemetry deadline 180 seconds, minimum backlog coverage 0.8, and decision interval 10 seconds. The four active connections fit within the configured cap. Both scaling actions are disabled by the deployment profile. Migration requires projected pressure improvement above 0.01 and Q-value margin above $\delta = 0.05$. These live normalization constants differ from the simulator's; the live trial evaluates execution, not a matched cross-domain reward gain.

The mixed-workload *Heuristic* checks admitted actions in this order: subscription variance above 0.45; backlog variance above 0.45; normalized backlog above 0.55 for scale-up; active-minus-probe pressure above 0.18 for migration; connection utilization below 0.25 for scale-down; otherwise stay. The scale-out *guarded-greedy* selector instead requests migration whenever its shared guard admits it. The *Scripted* timing controls use their declared random schedules. None receives the DQL network's action values.

C ACTION-CONTRACT DEVELOPMENT DIAGNOSTICS

The v3 protocol validated the three actions central to the claimed rebalancing mechanism, but it did not require scale-up and scale-down. We therefore declared a stricter v4 gate before running a new study. Each of $a^{\text{reb}N}$, $a^{\text{reb}G}$, $a^{\text{scale+}}$, $a^{\text{scale}-}$, and a^{mig} had to be selected in at least 80% of admissible episodes in its own fixed validation scenario. Every candidate also had to pass four probe and telemetry fault guards. Five seeds were trained for 20,000 transitions with uniform replay and again with action-stratified replay, giving 100,000 exact training transitions in each development run.

All ten candidates passed the scripted scenario-feasibility checks and every fault-safety contract. No row in Figure 24 passes all five policy-coverage conditions. Uniform-replay seed 401 exercises scale-down but misses backlog rebalancing; the other uniform candidates miss scale-down. Every action-stratified candidate misses scale-down, and several also miss subscription rebalancing or scale-up. Action-stratified replay left gaps in frozen-policy action coverage. Under the declared stopping rule, both v4 studies ended at the development gate and retained their checkpoints for diagnosis. Their results concern action coverage rather than new holdout performance estimates.

V5 then changed only the scale-down admissibility rule and used fresh candidate and validation seeds. The historical target-metric checks passed, including a 49.6% connection-cost reduction for scale-down. All five frozen candidates passed every fault guard and selected $a^{\text{reb}G}$, $a^{\text{scale+}}$, and a^{mig} in all ten intended validation episodes. However, each selected $a^{\text{reb}N}$ and $a^{\text{scale}-}$ in zero of ten intended episodes. For the diagnostic candidate with the best reward, we evaluated each missed action for one step from the same state. Both improved reward, but its Q-values ranked *stay* higher. A later audit showed that the old $a^{\text{scale+}}$ action reduced its declared maximum-subscription metric by 21.78% but lowered mean scalar reward by 0.04946 relative to static placement. The v5 diagnostics therefore distinguish missed one-step reward gains from a scale-up implementation that conflicted with its reward objective. The coverage gate was not met, and the study ended without a holdout evaluation.

V6 changed only the scale-up contract. The revised $a^{\text{scale+}}$ opens one connection and performs bounded, receipt-gated $a^{\text{reb}N}$ moves until the subscription target is restored or no eligible move remains. In the locked ten-seed preflight, the option improved mean interval reward by 0.04350 with a paired bootstrap 95% interval of [0.03559, 0.05114]. It reduced the mean maximum subscriptions per connection by 38.824, with interval [-38.824, -38.824]. Every episode restored the

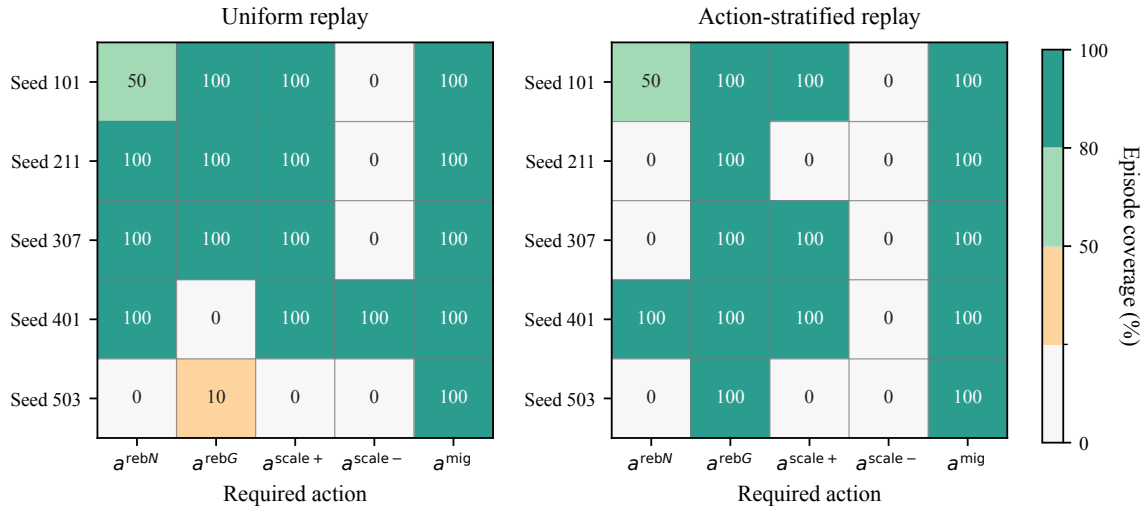


Fig. 24. Fixed-validation episode coverage for required actions in the two v4 development runs. A cell gives the percentage of admissible episodes in which the frozen candidate selected the indicated action; the predeclared pass threshold was 80% for every column in one row.

Table 22. V6 development-gate results. Each action and safety entry is the number of passing candidates out of five. TM denotes Target-Max bootstrap; DDQL denotes Double-DQL. An arm required at least four candidates that each passed all action and safety gates.

Arm	Learner/loss	Eligible	a^{rebN}	a^{rebG}	$a^{\mathrm{scale+}}$	$a^{\mathrm{scale-}}$	a^{mig}	Safety
A	TM/squared	0	0	1	0	5	5	5
B	DDQL/squared	0	2	2	0	5	5	5
C	TM/Huber	0	3	5	1	5	5	5
D	DDQL/Huber	0	4	5	0	5	5	5

target using exactly 50 moves, without an executor no-op or scale-down reversal. This scripted preflight verifies the revised action contract in the simulator; learned selection is evaluated in the study below.

The locked v6 study then tested each combination of Target-Max or Double-DQL bootstrapping with squared or Huber temporal-difference loss. Each arm trained five fresh candidates for 20,000 transitions and evaluated them on ten fresh development-validation seeds. Table 22 reports the number of candidates passing each action gate. Every candidate passed all four safety scenarios, but no candidate passed all five action gates. The predeclared requirement of at least four fully eligible candidates was therefore unmet in every arm. No policy was selected and no holdout was defined or executed. Squared and Huber loss magnitudes are not ranked against each other because the two losses have different numerical scales.

The cap-restoring executor improves the fixed reward, while learned selection varies across candidates. The diagnostics distinguish the corrected action contract from policy coverage. They leave the individual effects of value estimation, visited states, replay sampling, and credit across several moves unresolved.

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